

1-HACKBERRY MANUAL

User manual for assembling and customizing HACKberry



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Introduction

01 The world that HACKberry errs and



[1] What is HACKberry?

HACKberry is an electrical prosthesis designed by exiii Co., Ltd. and announced in 2015. The housing is manufactured using 3D printing, and all design data is available on the web as open source. Since September 2016, it has been taken over by Mission ARM Japan, a non-profit organization, and improvement and outreach activities are still ongoing.

An electric prosthetic hand is an electrical prosthetic hand that is activated by electrical signals from the muscles. An electromyoelectric sensor is placed on the surface of the skin, and the user's intent is read from the measured signal pattern, and the motor incorporated in the prosthetic hand moves accordingly. The user has a small

With just a little effort, you can move your prosthetic arm with the same intuition as a healthy person.

The development of electric prosthetic hands has a long history and was already commercially available in the 60s. However, it is difficult to lower the price in the traditional manufacturing and distribution system, and it is still very expensive, at more than 1.5 million yen. Therefore, only some people in developed countries can get electric prosthetic arms.

By combining 3D printing with open source, HACKberry aims to make electric prosthetic hands an "easy choice." 3D printing has no upfront costs, such as mold fees, and you can start creating on the spot by accessing open-source data through WEB. HACKberry



Expanding into the world of HACKberry

In this way, we have created a system that reduces the costs related to manufacturing and distribution, and if you pay about 50,000 yen in materials in costs, you can manufacture your own electric prosthetic hand anywhere in the world.

Although it has not yet reached the practical quality of commercial products, several people around the world are starting to manufacture HACKberry.

[2] Vision of H A C K b e r r y

HACKberry's vision is a "chance choice". This is about two aspects: "price" and "design". As mentioned before, the current electric prosthetic hand is very expensive, and if the price does not go down, it will not become more popular. However, it is important not only to lower the price, but also to expand the range of designs. Our hobbies and lifestyles are very diverse. Until now, prosthetic hands were usually designed to look like real hands as wellness equipment, but if you think of prosthetic hands as a part of fashion, such as shoes, clothes, glasses, etc., the possibilities of prosthetic hand design are also expanded. HACKberry is an example of this, with a sleek design that doesn't dare to be skin colored.

In addition, for prostheses to be a "casual choice", we must focus not only on the "user", but also on the "person who makes them". The prosthetic hand user and the creator are inextricably linked. To increase the chances of using a prosthetic arm, we must increase the chances of making a prosthetic arm. And to increase the chances of making a prosthetic arm, I think it's important to convey the "joy of doing." For example, we aspire to a world where prosthetic hands of various designs can be created one after the other with creativity,

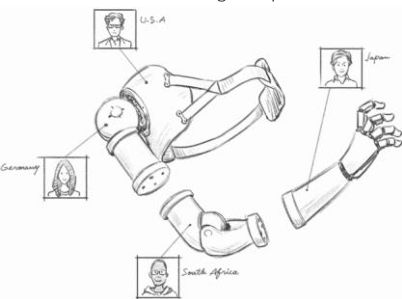
imagining the lifestyle of those who wear prosthetic arms, just like when they cook.

It was under this vision that the vision of H A C K b e r r y y y y y a la vision of H A C K e r r y was named H A C K B E R R Y is the English name for the enoki tree, a broadleaf tree in the elm family. Enoki trees grow their branches widely and produce many leaves and fruits. H A C K b e r r y too,

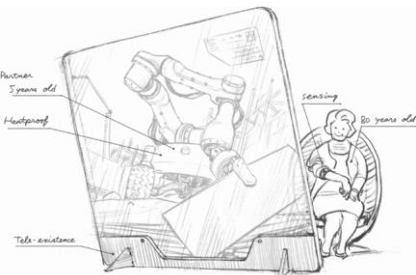
With the idea that "improving (HACK) will expand the branches and leaves of enjoyment, reach the hands of users, and bear fruit (berry)", I would like to grow it into a large tree that carries various prosthetic hands around the world.



Customizing the prosthetic hand



Developed by the open source community



Applications other than prosthetic arms



Pinch HACKberry leaves

Introduction

02 Specifications

[1] How H ACKberry Works

First, the standard specifications of the H ACKberry are presented in the table on the next page. The dimensions and weight correspond to the hand and do not include the cup part.

The sensor used in HACKberry is a simple pressure sensor. The sensor is made by overlaying an inexpensive distance sensor called a photoreflector with a 10mm thick hollow sponge inside. By placing the sensor on the user's arm (where the muscles are located) and fixing it around the arm with a non-elastic band, the movement of the user's arm muscles can be physically read. (This technology is based on that developed by Professor Masahiro Yoshikawa of the Osaka Institute of Technology.)

The angle of the three servo motors varies depending on the reading value of the simple pressure sensor. The information is processed by the microcontroller. When the user applies force to the arm, the fingertips close, and when the force is released, the fingertips open. Also, the stronger the force you exert, the faster your fingertips close. Open source programs have this option, but it is also possible to rewrite the program to make it different (for example, reverse the opening and closing of the finger).

[2] Design Features of H ACKberry

With today's science and technology, it is difficult to perfectly reproduce the innate and skillful hand, and when designing a prosthetic hand, it is necessary to choose between functions. Thinking about practicality in everyday life, H ACKberry has the following features:

By adopting a linking mechanism such as a page diagram, we have both the ability to "pinch" small objects and the ability to "hold" large objects. In this linking mechanism, when an object hits the middle finger, the rotational force of the base joint is sent to the tip joint, so that a single motor can passively alternate between pinching small objects and grasping large objects.

(2) Ease of assembly

Prosthetic arms are more complex than general electrical appliances.

I'm doing it. It is also used in various environments, so there is a high risk of failure. Furthermore, HACKberry assumes that it will be assembled by a third party as open source. For these reasons, it is important that the H ACKberry is easy to assemble and reassemble. All H ACKberry components are available worldwide. In addition, a gear is used to drive the knuckles without the need for wires. Wires tend to express smooth movements similar to tendons, but they are also prone to loosening and require frequent maintenance. On the other hand, the gears move strongly, but they do not loosen and are easy to repair even if they fail.

(3) Wrist mechanism

Considering the comfort of daily life, the hand Not only is it ahead, but the wrist movement cannot be underestimated. The HACKberry wrist can be adjusted at a manual angle. By pressing the button located in the wrist area, the wrist can be rotated or tilted. You can also put on and take off the hand and wrist area, and adjust the placement every 90 degrees. When combined, they can reproduce the three degrees of freedom that the wrist has: pronation/supination, palmar flexion/dorsiflexion, and radial/ulnar flexion. For more information, see "3.HACKberry

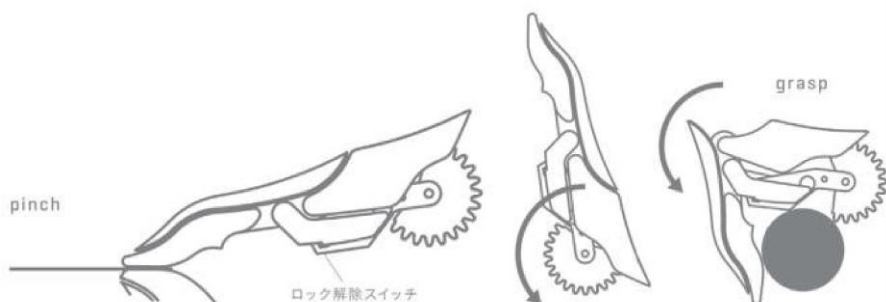
(1) 「Balancing "pinch" and "grab"

Let's enjoy it more" (p145).

When a person picks up something in their hand, they can skillfully adjust the posture of their fingers and wrist to match the shape of the object with that of the object. However, in the case of a prosthetic hand, the number of motors is limited, so it cannot be freely adjusted like a natural hand. Therefore, H ACKberry is as follows

Dimensions	225 x 150 x 60 (mm) *When thumbs are parallel
Weight	450 to 500g *Varies by material
Sensa	Simple Pressure Sensor x1
Engines	Servo Motor (Large) x1 *Index Finger Flexion Servo Motor (Small) x2 *Thumb Induction & Absorption, Middle Finger/Medicine/Pinky Flexion.
Battery	Li-ion電池 (7.2V, 2200mAh)
Mai Kong	Mk1: Arduino micro Mk2: Arduino nano互換品

HACKberry Specifications



HACKberry's index finger binding mechanism

Introduction

03 License Forms

If you wish to use HACKberry as open source, you must comply with the following terms and conditions set forth by the EXII Shares Association.

=====

HACKberry Open Source Project Policy

ixy Corporation (hereinafter referred to as "the Company"). In this policy, the source code and data, etc., related to "HACKberry" (hereinafter referred to as the "Project") will be disclosed. The content of the Use Items and their terms of use will be determined.

1. Purpose

1.1. The purpose of this project is to ensure that the Company provides all users of this project (hereinafter referred to as "users"). The purpose of this policy is to promote the research, development, and dissemination of prosthetic hands by disseminating source code and data, and licensing the use of our inventions.

1.2. The use of source code and data published under this Policy or inventions licensed under this Policy for the purposes specified in the following points is prohibited.

1.2.1. Use for military purposes

1.2.2. Use for purposes contrary to public order and morality

1.2.3. Use them for the purpose of discrediting us or HACKberry

1.2.4. Use for purposes that the Company believes are likely to fall under any of the items listed above

2. Software Source Code

The source code of the software related to HACKberry is published under the following license: If you wish to use the source code under conditions different from those of each license (commercial use, etc.), please contact us at the following address.

info@exiii.jp

Source Code License

2.1. Arduino sketch (means the source code for writing on the Arduino Micro integrated into the prosthetic arm.) GNU General Public License version 3

3. Data in hardware

Each hardware data on HACKberry is published under the following licenses or conditions: If you wish to use the data under different conditions (such as commercial use) of each license, please contact us at the following address. info@exiii.jp

Data Licensing

3.1. 3D data (refers to the shape data of each part used in the prosthetic arm).) Creative Commons BY-NC-SA 4.0 International License

3.2. Hand board data (refers to the circuit board used to connect microcontrollers, motors, sensors, and batteries). Creative Commons BY-NC-SA 4.0 International License

3.3. Sensor board data (means circuit board for attaching a reflective light sensor (photoreflexor) to detect muscle lumps.) Creative Commons BY-NC-SA 4.0 International License

3.4. Battery board data (i.e. the circuit board integrated into the battery box).) Creative Commons License by - NC-SA 4.0 International

4.1. The Company grants the User a royalty-free, unrestricted, and non-exclusive license (excluding sublicenses) for the Company to implement any patented or patent-pending inventions (excluding sublicenses) to the extent necessary to create a prosthetic hand using the source code or data under the licenses set forth in 2 and 3 and not in violation of 1.2. I will.

5. Trademarks and Content

5.2. The following trademarks are the ones we apply for in Japan. Trademark: HACKberry
Application No.: 2016-042844, 2016-042845

5.4. With respect to the source code and data published by the User under this Policy, the User may not be able to obtain information about the source code or data published by the User pursuant to this Policy or the forums <http://exiii-hackberry.com/forums/> from third parties, including but not limited to the Company, and the User hereby agrees to indemnify and hold the Company harmless from any and all claims or damages, including attorneys' fees, that may be asserted against or incurred by the Company or its affiliates in connection with the User's use of the forums.

5.6. If you wish to use the marks or content, please contact us at the following address.

6. Priority of individual licenses

7. Legal notice

7.2. We are not responsible for any dispute or damage of any kind between you or between you and any third party in connection with the use of the source code and data provided by us or the interpretation and application of each license set forth in this Policy.

7.3. We do not guarantee that the source code and data will always be available on this website. The Company shall not be liable for any damage suffered by users or third parties due to the inability to provide source code and data on this website, regardless of the reason for the maintenance or inspection of this site's system.

8. Changes in the method of provision and content

8.1. The Company may modify the method or content of the provision of the source code or data specified in this Policy or the content of the license of the invention at any time without prior notice to the User (hereinafter referred to as "Changes, etc."). You can.

8.2. The Company shall not be liable for any dispute or damage arising from the User or a third party in connection with the changes in the preceding paragraph.

9. Changes to this policy

9.1. We may change the content of this Policy at any time without notice to the User.

9.2. The User shall be deemed to have given valid and irrevocable consent to the revised Policy by continuing to obtain or using the source code or data provided by the Company or by continuing to use the Company's inventions following any changes to this Policy. However, this does not apply to the content of the license that applies to the source code or data obtained by the user under the policy prior to the change in that policy.

10. Applicable law and jurisdiction

10.1. This policy is written in Japanese and is governed by the laws of Japan.

10.2. Any dispute arising out of or relating to this Policy shall be subject to the exclusive jurisdiction of the Tokyo District Court or the Tokyo Summary Court in the first instance, depending on the amount of the claim.

====

These are the terms that should be followed when using H A C K b e r r y as open source. In addition, in relation to H A C K B E R R R Y , E X I I I - H A C K B E R R R Y . Content uploaded to the forums and G l u b o n c o m is not limited to program code, but also includes bug reports, opinions, ideas, and other content, whether commercial or non-commercial. It is used for the development of "H A C K b e r r y" and C r e a t i v e

In addition to the C M m o n s license, it can be distributed under a closed-source license established by the company itself.

Introduction

04 Guarantees and Liabilities

H A C K b e r r and has made it easier for anyone in the world to build their own electrical prosthetic hand with the help of 3D printers and open source. The reason H A C K b e r r y is offered in this form is because we want to increase the opportunities for those who use prosthetic hands to try electric prosthetic hands and for those who make prosthetic hands to know the fun that comes with it.

However, H A C B E R R Y does not guarantee that it is 10% safe like commercial electrical prosthetic devices, nor is it sold as a finished product or quality guaranteed. When using H A C K b e r r y as a prosthetic arm outside the scope of research and testing, please fully understand the performance and accuracy of H A C K B E R R Y and act at your own risk. H A C K B E R R Y shall not be liable for any damages or damages arising out of or in connection with any dispute arising from the use of H A C K B E R R Y.

2. Let's build it

1. ¿Qué HACKberry montas? P016
2. How to get parts P019
3. Before you set up P022
4. Basic welding P026
5. Uploading a program P052
6. Mounting the battery box connector P058
7. Thumb Mount P074
8. Index finger assembly P080
9. Three-finger assembly P088
10. Voltage adjustment and initial servo motor position adjustment .. P092
11. Palm Assembly P094
12. Wrist Mount P122
13. sensor suite..... P130
14. Initial Position Setup P134

Mounting

01 What hack to set up

There are several settings in HACKberry. First, decide which HACKberry you want to build.

A Right hand



The most basic type of HACKberry. When used as a prosthetic arm, a socket is made that fits the actual arm by requesting a prosthetic fabric. Apart from this image, you will need an external battery box to share the sensor and power supply to operate the HACKberry.

B Right hand (with dummy socket)



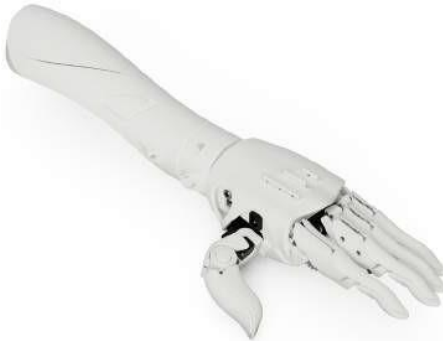
A is a guy with a temporary cup 3D printed on the HACKberry. The battery is stored inside the socket, so you don't need to have an external case. In addition to this image, you will need a sensor to operate HACKberry.

C Left hand



It is the specification for left-handers of the most basic type, H A C K b e r r r. When used as a prosthetic arm, a socket is made that fits the actual arm by requesting a prosthetic fabric. Basically, there is no difference with respect to the right hand. Apart from this image, you will need an external battery box to share the sensor and power supply to operate the H ACKberry.

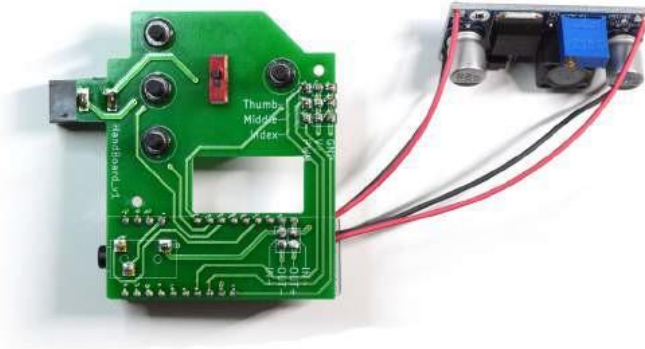
D Left hand (with dummy base)



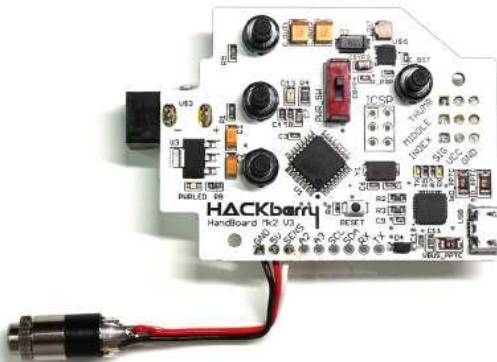
This is a guy with a temporary connector 3D printed on the HACKberry in C. The battery is stored inside the socket, so you don't need to have an external case. In addition to this image, you will need a sensor to operate HACKberry.

In addition, there are two types of substrates for H ACKberry: Mk1 and Mk2. HACKberry works on both boards, but M k2 is easier to implement and cheaper. At the same time, the board size is small, and ports are provided for feature expansion, making it highly customizable.

- **H ACKberry handboard Mk1**



Hand Board • H ACKberry Mk2



Mounting

02 How to Get Parts

[1] 3D Printed Parts Printing

Data on 3D printed parts is available on the Internet. Open the link below and click on DATA (GITHUB) on the page that appears.

<http://exiii-hackberry.com/>



Githubのサイトが開きます。

<https://github.com/mission-arm/HACKberry>

そこから

HACKberry_3Dmodel / STL / ...

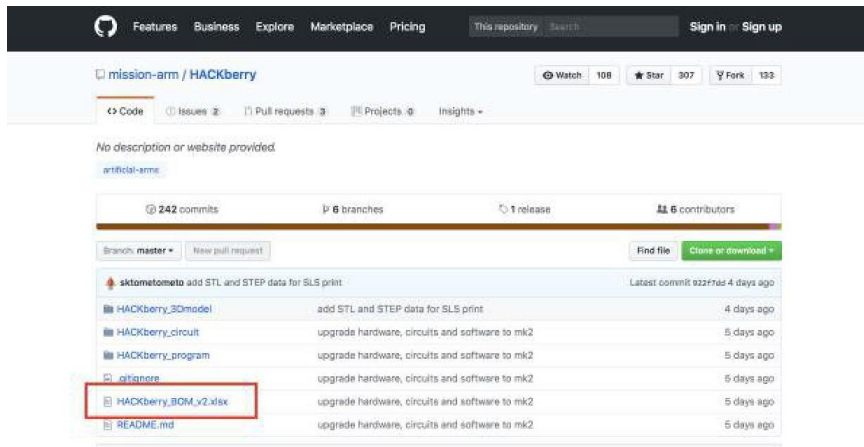
to download the desired STL data.

Commit Message	Author	Time
sktometometo add STL and STEP data for SLS print	sktometometo	Latest commit 922f7dd 4 days ago
..	..	..
C upgrade hardware, circuits and software to mk2	sktometometo	5 days ago
ForSLSPrint add STL and STEP data for SLS print	sktometometo	4 days ago
L upgrade hardware, circuits and software to mk2	sktometometo	5 days ago
R upgrade hardware, circuits and software to mk2	sktometometo	5 days ago

[2] How to Buy Mechanical and Electronic Parts That Aren't 3D Printed

First, please refer to the section above and visit the Github page. <https://github.com/mission-arm/HACKberry>

Download the "HACKberry_BOM_v2.xlsx" from that page.
(The BOM version may be updated in the future.)



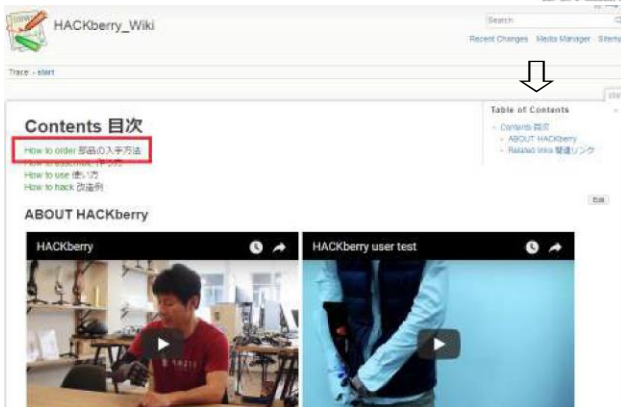
Open the downloaded file using Excel or compatible software.

Unit	Part Name	Description	Quantity	US delivery	US substitution	JP delivery	JP substitution
Hand	S91-5UJ-1d4	Steel ball -Spherical diameter:1/4"	2	Amazon.com		Amazon.jp	
	flatScrewM2L6Phi3.3T1.3	Tapping screw M2L6 -Shaft diameter:2 -Head diameter:3.3 -Length:6	25			willco	
	ScrewM3-.89tbs	Tapping screw M3L8 -Shaft diameter:3 -Head diameter:5.5 -Length:8	6			willco	
	E308MO	Servo motor(small)	2	Emax	Emax	KK HOBBY	Emax
	HD3001H8	Servo motor(large)	1	rcMarT		rcMarT	Akizuki
	RXFP050	Resettable Fuses	3	Myusai		Akizuki	
	UsmBearing	Bearing -Bore diameter:4 -Outside diameter: -Width	1	sp8tada		dMT	
	flatScrewM2L6Phi3.3T1.3	Tapping screw M2L10 -Shaft diameter:2 -Head diameter:3.5 -Length:10	25				

You can open the buying site from the link on the right of the file screen that opens, so check the part name and the number of parts needed before purchasing parts.

For information on how to obtain a HACKberry board, please contact Mission ARM Japan.
(You can make your own without going through Mission ARM Japan, or you can create it using a PCB fabrication service like S eed Studio.) For more information, go to U R L below and click T U T O I A L to see the page that opens.

<http://exiii-hackberry.com/>



Mounting

03 Before Riding

Notes

- Please follow the order in the manual.
- The following explanation is basically about the right hand. If you're riding your left hand, think of it backwards.
- Depending on the version, there may be slight differences between the image in this book and the parts you have.

-Essential tools-

- | | | |
|----------------------|--|--------------|
| 1. Welder and trowel | 2. Thread welding | 3. PC |
| 4. Cable micro USB | 5. Nipper | 6. Conductor |
| 7. Vise | 8. Taladro ($\phi 1.6$ mm, $\phi 2.2$ mm) | 9. Medida |
| 10. Tester | 11. Vinyl tape | |



-Useful tools and equipment if you have them—

12. Needle-nose pliers

13. Tweezers

14. Cable decapabler

15. Heat Shrink Tubing



⑫



⑬



⑭



⑮

-General purpose parts—

1. Screw (M2L10)

2. Screw (M2L6)

3. Screw (M3L8)

4. Bearings

5. Washing machine

6. Torsion spring

7. Muelle helicoidal

8. Tension spring

9. Spacer

10. Shaft (10mm)

11. Shaft (15mm)

12. Steel Ball

①



②



③



④



⑤



⑥



⑦



⑧



⑨



⑩



⑪



⑫



-How to read part numbers-

Learn about naming conventions for 3D printed parts. The letters are separated by a hyphen (-).

The first element is an important classification.

R	Right side
L	Parts of the left
C	Common parts left and right

The second point is the classification of the units.

So	Intake unit
T	Thumb unit
I	Index finger unit
Or	Three-finger units (middle, ring, and pinky)
H	Palm Unit
W	Wrist unit
B	Battery Housing Unit
SE	Sensor Unit

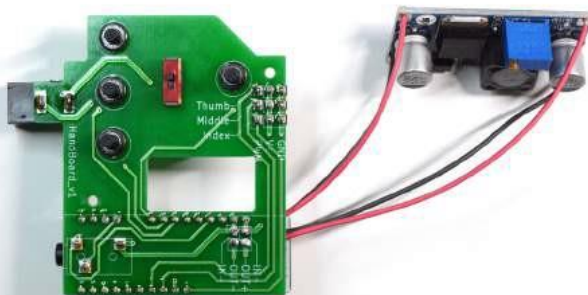
The third element is the number assigned within that unit.

For example, if it's R-SO-01, it's part number one of the right plug unit.

Mounting

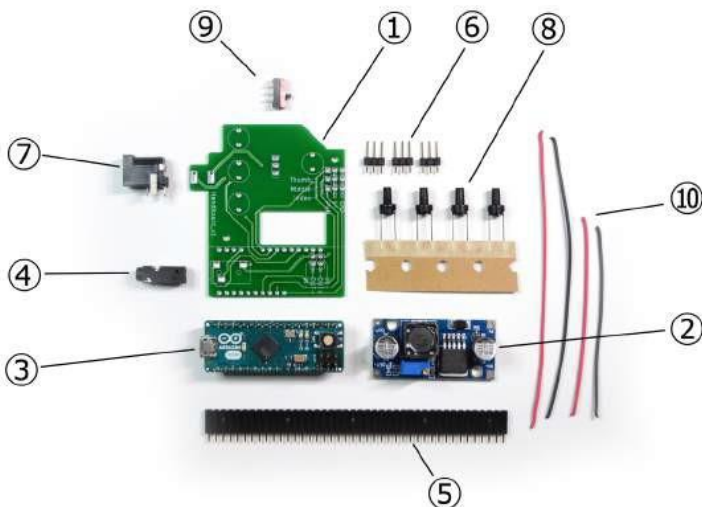
04 Soldering the board

HACKberry Mk1 Soldering Hand Plate



-Used parts—

- | | | |
|------------------------------|-----------------------|-----------------------|
| 1 . HACKberry Mk1 Hand Board | 2 . D C D C Converter | 3 . Arduino Micr
o |
| 4 . Stereo jack | 5 . Pin Socket | 6 . Pin Header |
| 7 . DC Connector | 8 . Change of feel | 9 . Slide switch |
| 10 . Lead Wire | | |



*When welding, be careful not to solder the electrical element (component) firmly or while floating or at an angle.

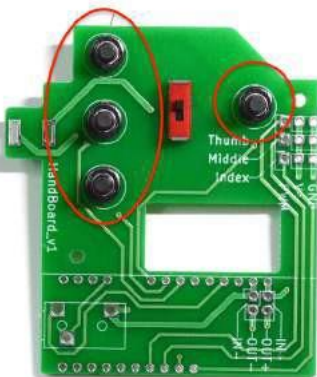
*Small electrical elements on the board can be damaged by static electricity. Before welding, use a pad to remove static electricity as much as possible.

*Touching the electrical element with wet hands will cause a short circuit and damage. Clean moisture thoroughly before welding.

[1] Solder the slide switch to the Omote side of the board. **The surface of the board is the surface where the white letters are printed. In addition, the slide switch has no polarity (direction).**

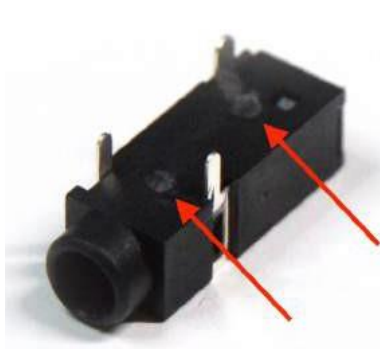


[2] Solder the touch switch to the Omote side of the plate. Cut the wire sticking out the back with scissors. As a rule, do not protrude more than 2 mm from the board.



Bac

[3] Then, weld the backside. If you're soldering a board for your right hand, cut the bump at the bottom of the stereo connector with a cutter before making the circuit.

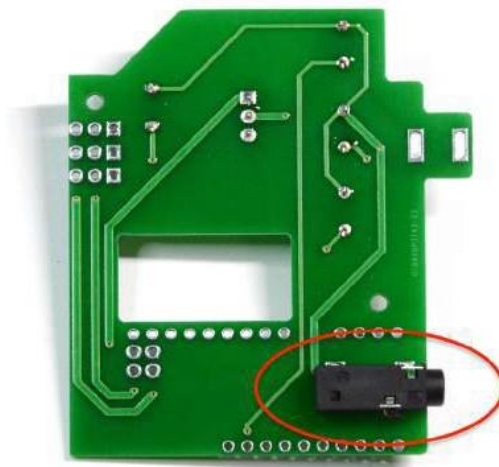


Cut bump

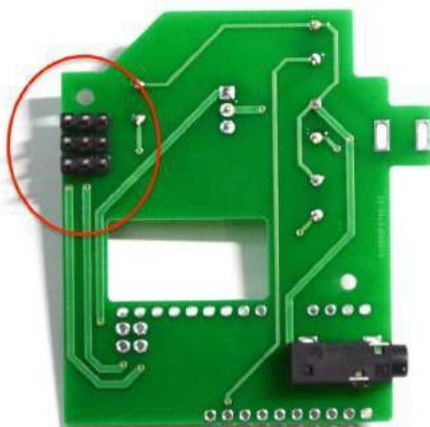
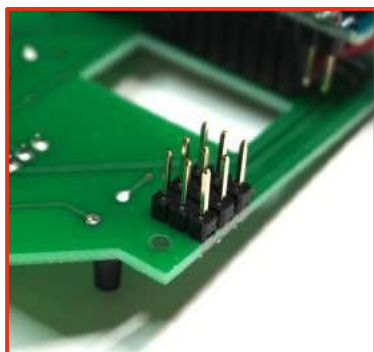


What is not cut with bumps?

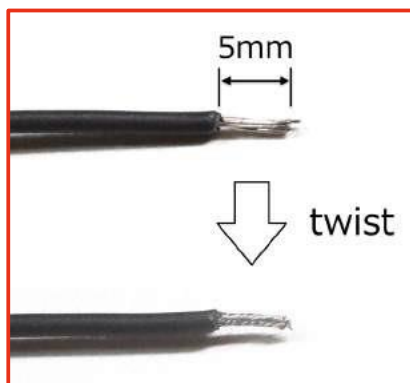
[4] Next, solder the stereo jack to the photo position.



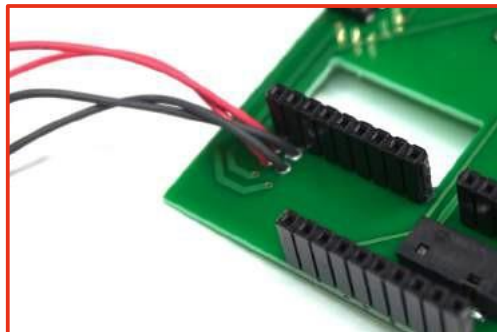
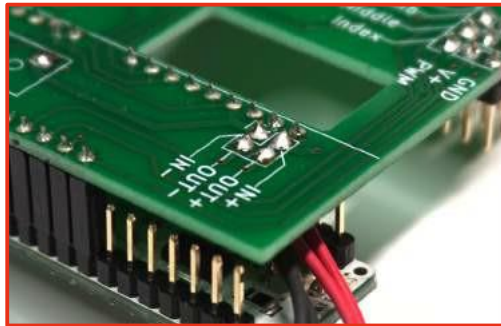
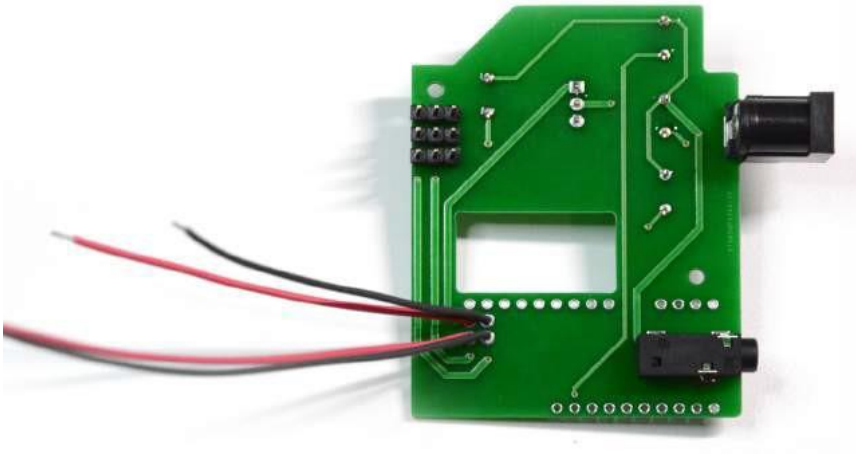
[5] The pin connector is soldered by inserting the short metal part into the plate. **Be careful not to tilt the pin.**



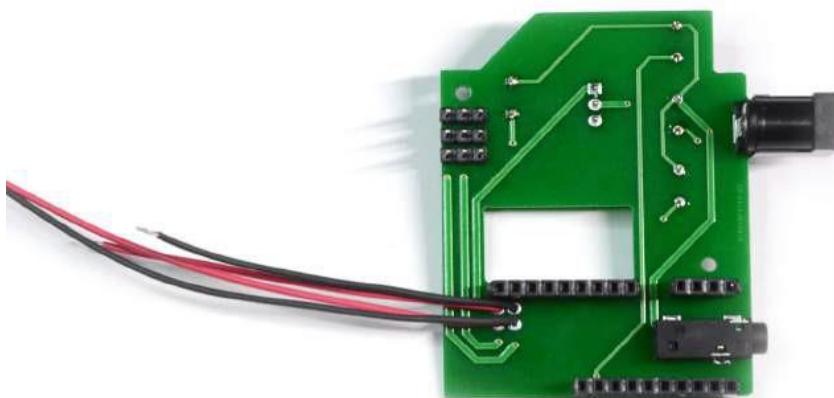
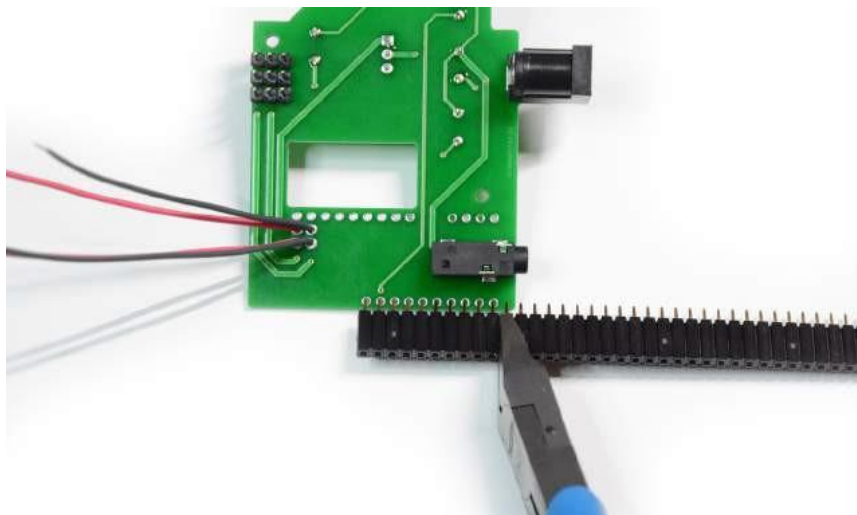
[6] Peel the 5mm tip sheath off the branch wire using a stripper, etc., and twist the wire together. If you don't have a stripper, pinch the case with tweezers and peel off the sheath being careful not to cut the core.



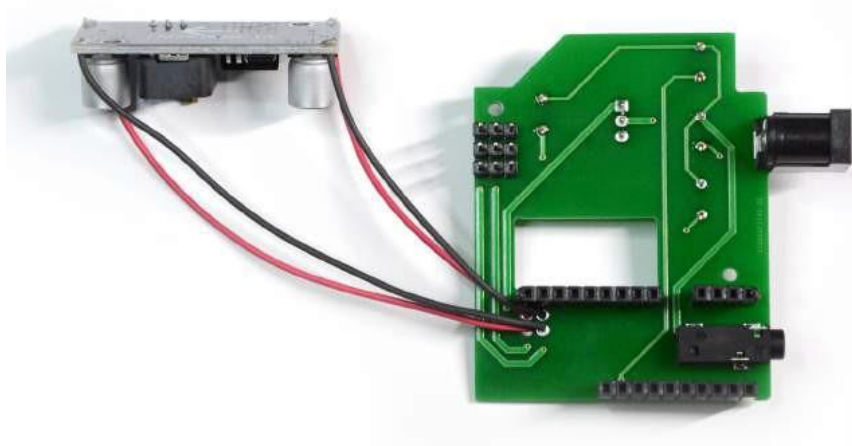
[7] The longest is the OUTPUT and the shortest is the INPUT, and the red is + and the black is connected to the $-$, respectively. The D-C connector is also soldered.



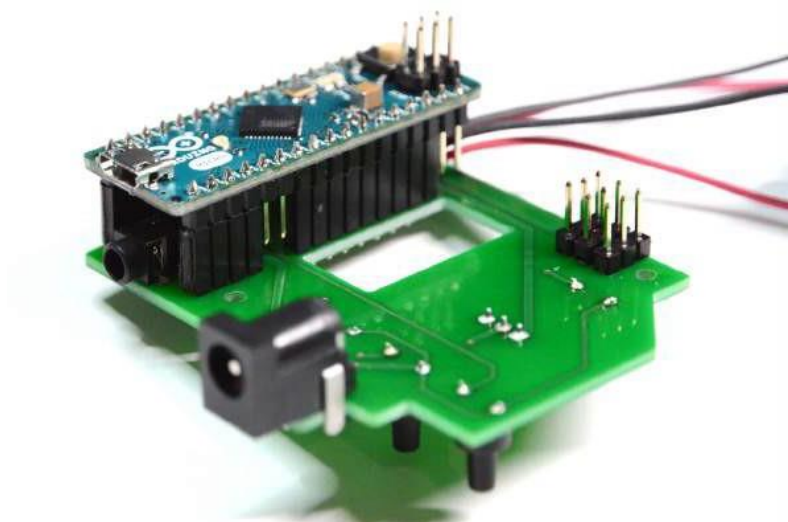
【8】 Cut the pin bushing with a clamp to fit the holes in the plate, and then



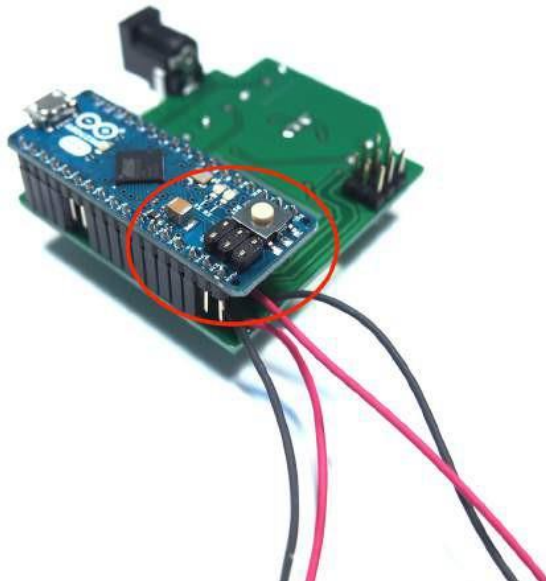
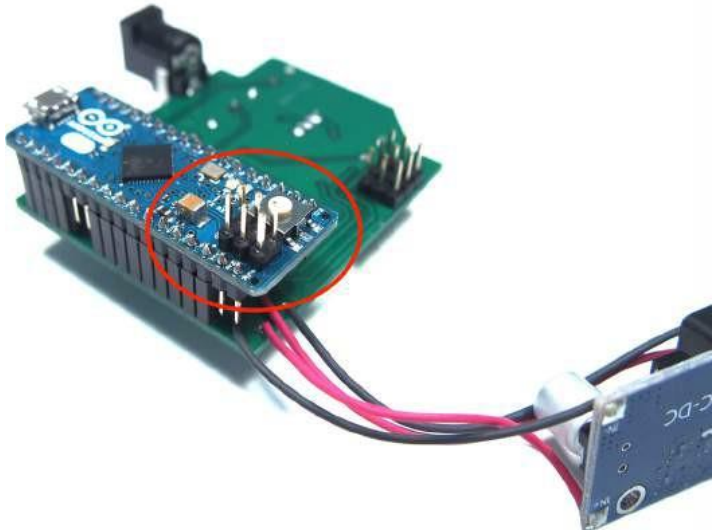
【9】DCDCコンバーターとHACKbrryhandboardMk1のOUT+,OUT-,IN+,IN-を対応さ
I will attach it.



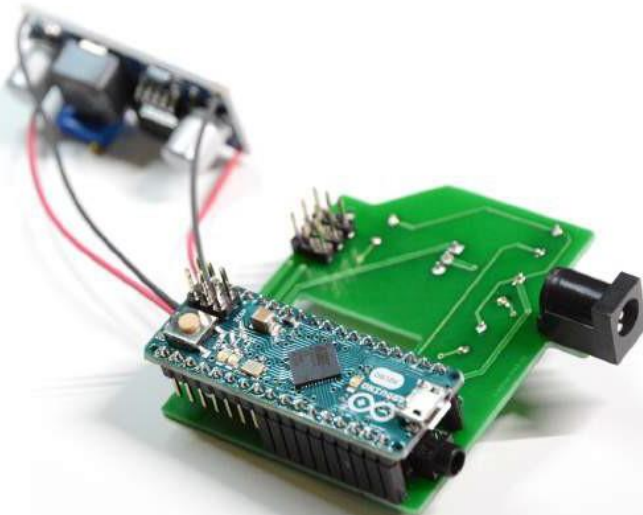
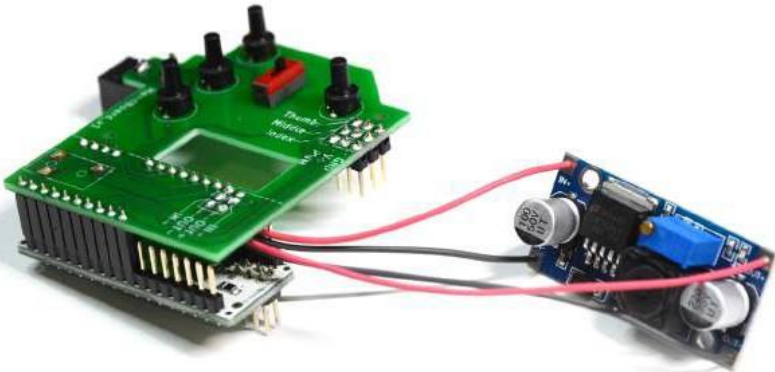
【10】 A Micro rduino is fully inserted into the pin connector so as not to lose the hole.



[1 1] For boards on the left, cut the pins into the red frame of the diagram with tweezers as shown in



[1 2] This completes the HACKberry Mk1 handheld board.



[1 3] Next, insert the poly switch into the servo motor.

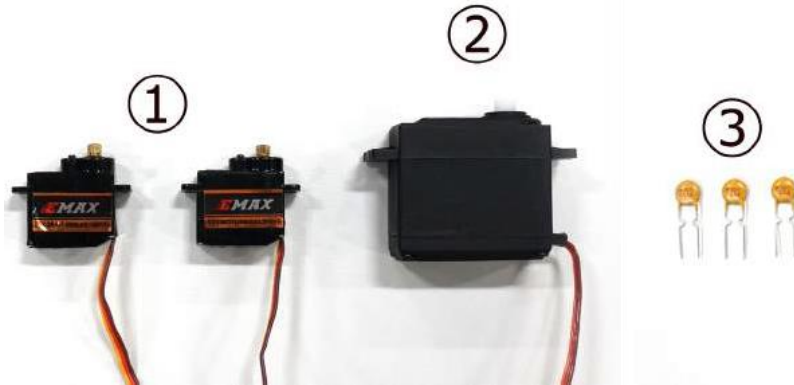
**It is possible to move the servo motor even if this process is skipped, but by installing a poly switch it is possible to avoid damage when the motor is overloaded.*

**This work is not necessary because the Mk2 board already has a polyswitch installed on the board itself.*

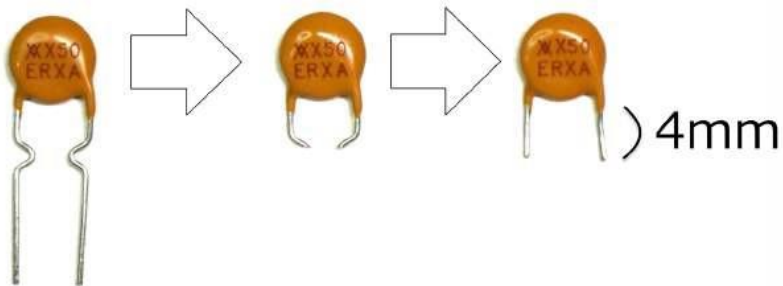
1. Small servo motor

2. Servomotor grande

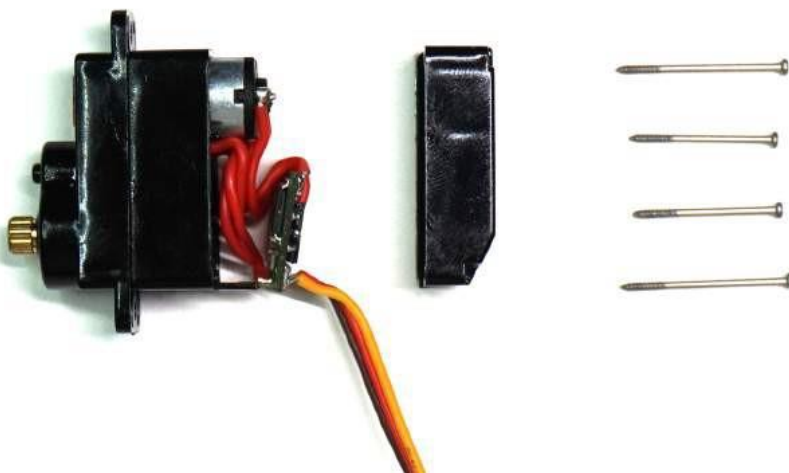
3. Polyswitch



[1 4] First, cut the polyswitch branch wire about 4mm and straighten it with fine-tipped tweezers. This work is done in 3 pieces.



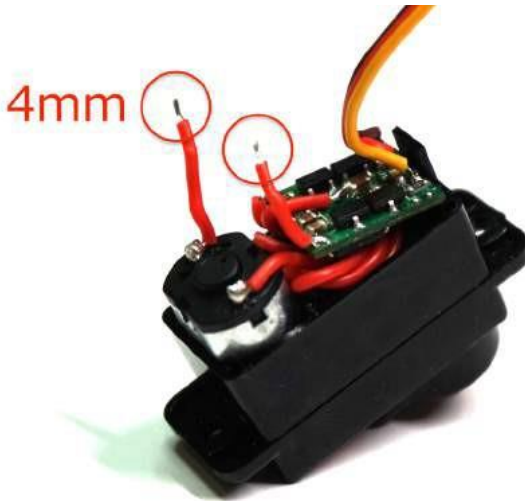
【15】 Loosen the small screw of the servo motor and remove the cap. There is a manufacturer's seal on the side, which you can remove or cut at the joints of the pieces.



【16】 Cut one of the two wires soldered to the inner DC motor with the blue wire in the photo.



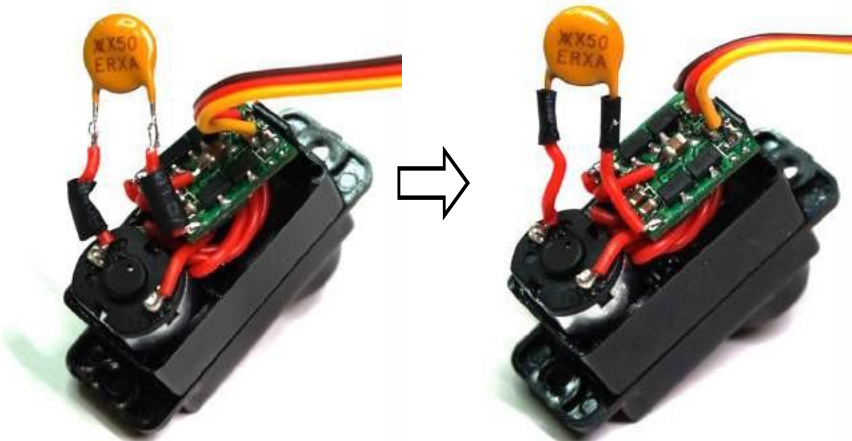
[17] Peel the sheath off the cut wire with a stripper or other means about 4 mm and twist the main wire.



[18] Solder the polyurethane switch to the cut wire and cover it with vinyl tape. **It's dangerous to expose the metal parts, so cover the entire metal part well.**

**If you have a heat shrink tube, it is safer to use a heat shrink tube.*

**Polyconnectors do not have polarity.*



【19】 Bend the cable as shown, close the lid, and tighten the screws. **At this time, don't pinch the cord**
Please be careful.



【2 0】 Unscrew the large screw from the servo motor and remove the cap. A rubber gasket is removed. However, it is not necessary to close the lid at the end.



[2 1] Cut one of the two wires soldered to the inner DC motor as shown in the photo, peel off the sheath about 4mm using a wire stripper, and twist the center wire together.



【2 2】 Solder the polyswitch to the cut wire and cover it with vinyl tape. **Exposed metal parts**
It is dangerous to do so, so please firmly cover the entire metal part.

**If you have a heat shrink tube, it is safer to use a heat shrink tube.*

**Polyconnectors do not have polarity.*



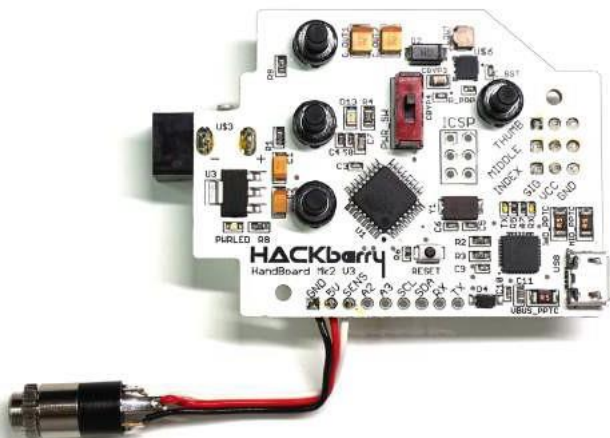
【2 3】 Close the lid and tighten the screws. The polyswitch is already inserted.



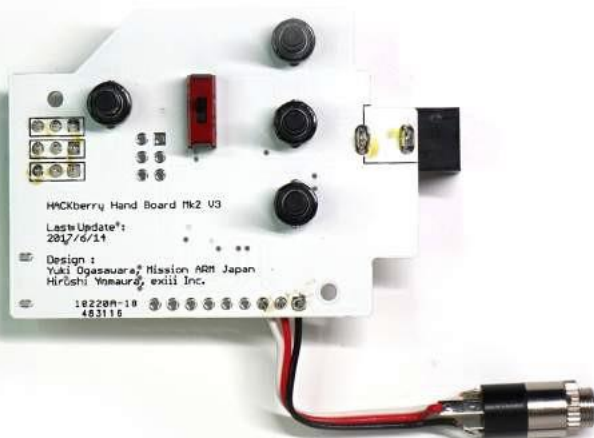
Mk2 Plate Welding

The HACKberry Mk2 handboard is a control board that is shared between the left and right, but the **front and back are inverted depending on whether you use it on the left or right**. Please be careful not to make mistakes, as there is a description of the left-hand welding method after the right-hand welding method.

Mk2 Board Completion Image (Right Specification)



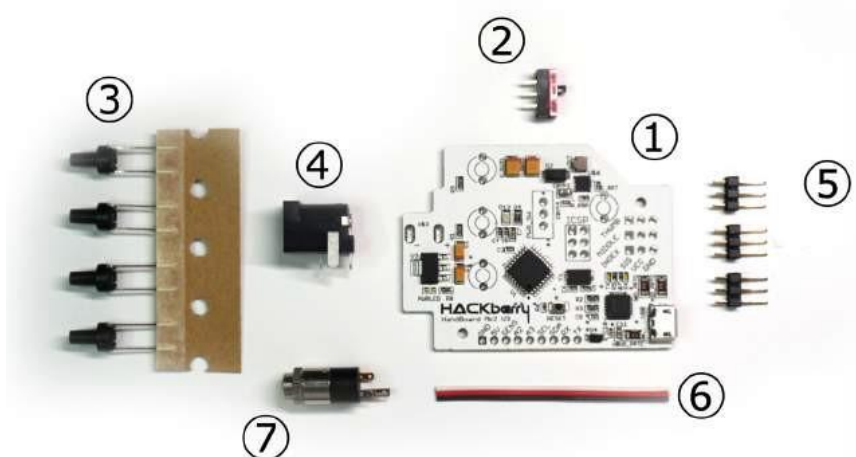
Mk2 Board Completion Image (Left Specification)



**Appearance may differ slightly from the previous photo depending on the version of the plate.*

-Used parts—

- | | | |
|-----------------------------|--------------------|--------------------|
| 1. HACKberry Mk2 Hand Table | 2 . Slide switches | 3 . Touch Switches |
| 4. DC Jack | 5. Pin Header | 6. Ribbon Cable |
| 7. Stereo Jack. | | |



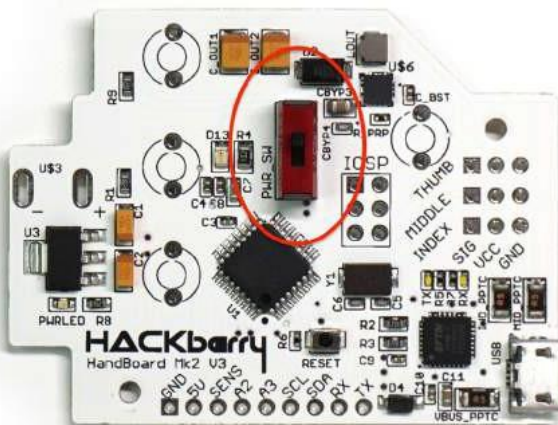
*When welding, be careful not to solder the electrical element (component) firmly or while floating or at an angle.

*Small electrical elements on the board can be damaged by static electricity. Before welding, use a pad to remove static electricity as much as possible.

*Touching the electrical element with wet hands will cause a short circuit and damage. Clean moisture thoroughly before welding.

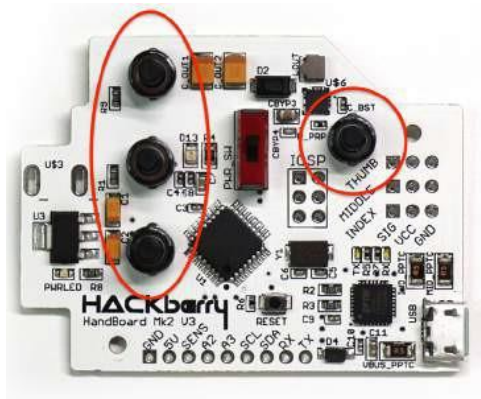
Welding when used on the right hand

[1] Solder the slide switch to the Omote side of the board. **The plaque is printed with the H A C K b e r logo on the side.** In addition, the slide switch is not polar.



[2] Solder the touch switch to the Omote side of the plate. Cut the lead wire sticking out from the Ura side at the root. As a rule, cut to within 2mm of the board's surface.

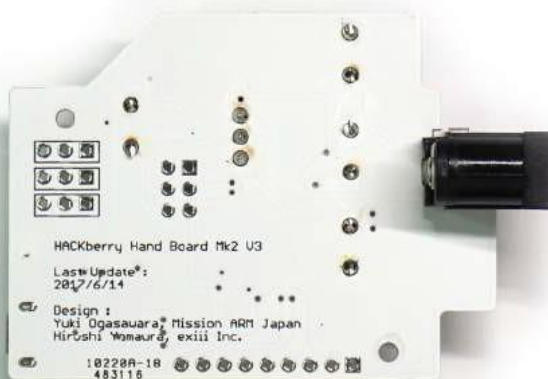
Omote side



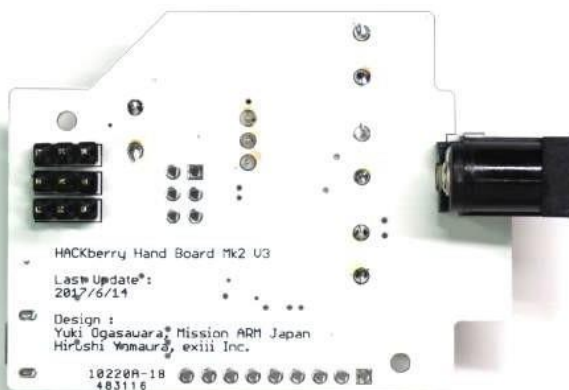
Ura's side



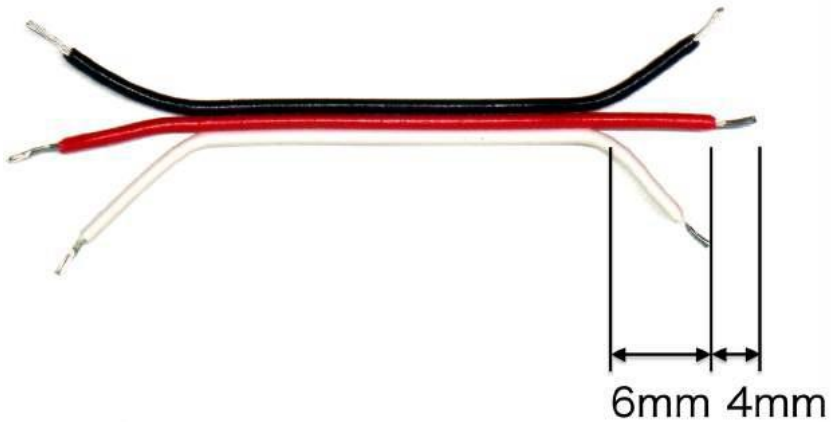
【3】 Solder the D-C connector as shown.



【4】 The pin connector is soldered by inserting the short metal part into the plate. **Be careful because it tends to tilt.**



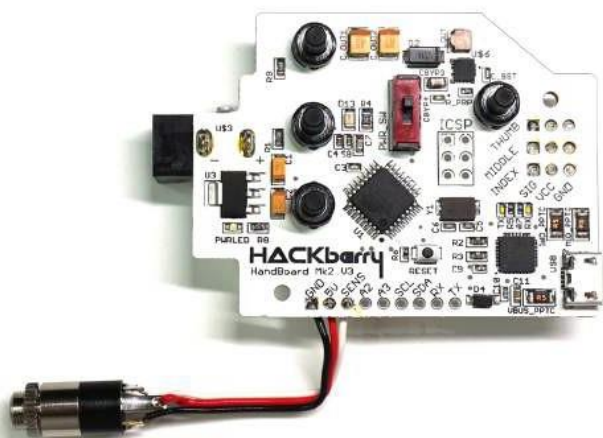
[5] Divide each end of the ribbon wire by about 10 mm each, then peel off the film by about 4 mm each using a stripper. The core cable is twisted by hand and assembled.



[6] Solder the ribbon wire to the terminal of the stereo connector. At that point, **solder the wires in white on the top right terminal, red on the bottom terminal, and black on the top left terminal so you don't make mistakes.**

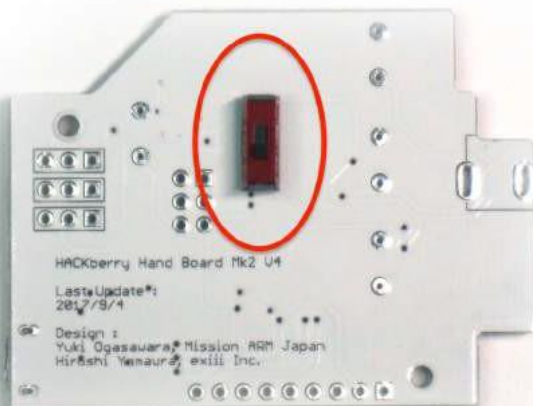


[7] Solder the other side of the ribbon wire to the board. **The white wire is SENS, the red 5V and the black one is GND** . This concludes the soldering of the HACKberry Mk2 handplate.



Welding when used on the left hand

[1] Solder the slide switch to the ULA side of the board. The plate is **the Ura side without the HACKberry logo printed on it**. In addition, the slide switch has no polarity (direction).

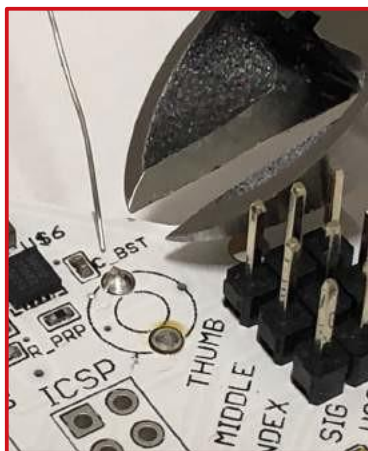


[2] Solder the touch switch to the Omote side of the plate. Cut the lead wire sticking out from the Ura side at the root. As a rule, cut to within 2mm of the board's surface.

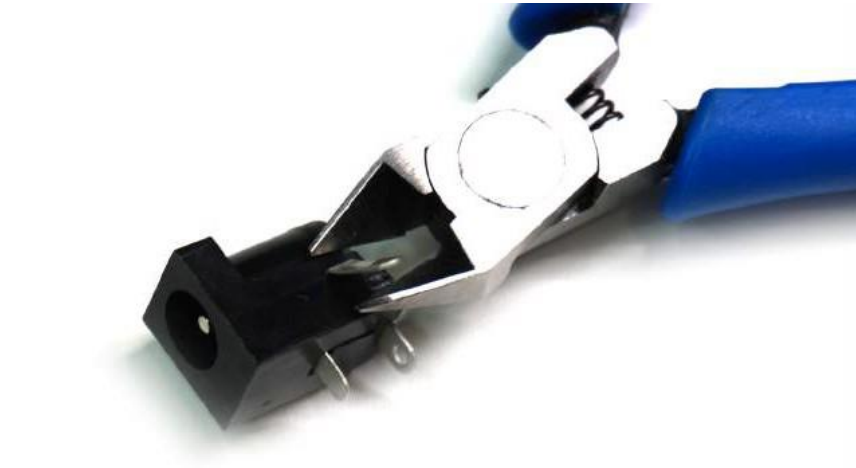
Ura's side



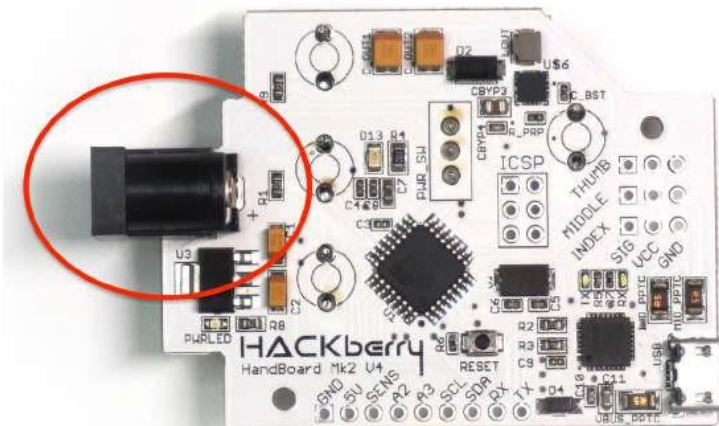
Omote side



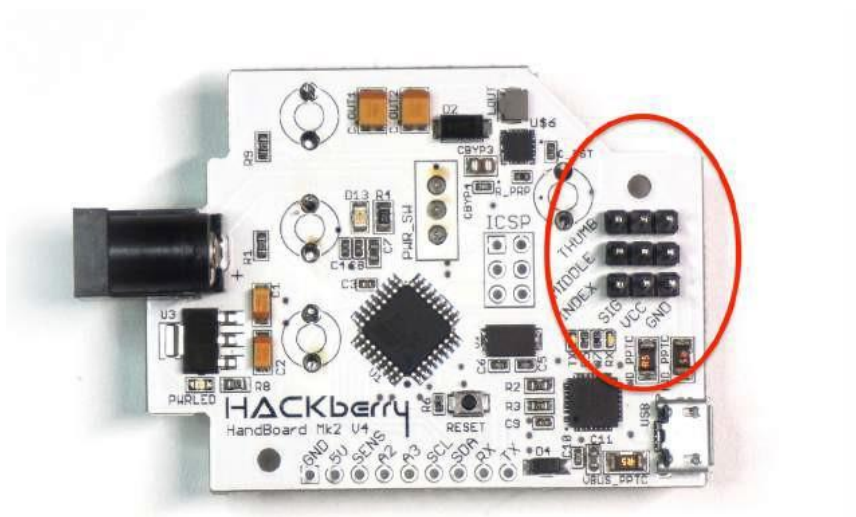
[3] Then weld the side of the omote side. Cut the metal terminal coming out of the side of the D-C connector I will.



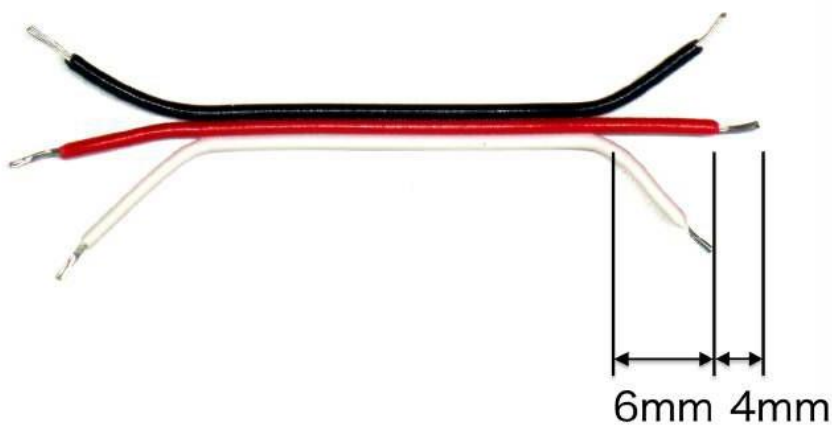
[4] Solder the D-C connector as shown.



[5] The short-pin connector is inserted into the board and soldered. **Be careful, as it's easy to lean**



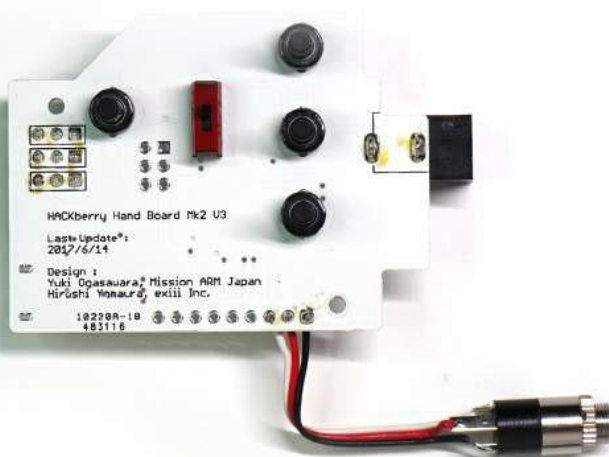
[6] Divide each end of the ribbon wire by about 10 mm each, and then peel off the film by about 4 mm each using a stripper. The core cable is twisted by hand and assembled.



【7】 Solder the ribbon wire to the terminals of the stereo jack. At this time, **the upper right**
Solder the red wire to the child and the black wire to the top left terminal so you don't make mistakes.



【8】 Solder the other side of the ribbon wire to the board. **The white wire is SENS, the red 5V and the black one is GND** . This completes the soldering of the HACKberry Mk2 handplate.



05 Uploading the program

Plate Type and How to Change the Program in the Case of Left and

This is how to upload a program using Wikis.

[1] Enter the official Arduino website <https://www.arduino.cc/en/Main/Software#> and download the latest version of the Arduino IDE software.

**The design and version of the website may differ from the time this book was written.*



Download the Arduino IDE



ARDUINO 1.8.4

The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. It runs on Windows, Mac OS X, and Linux. The environment is written in Java and based on Processing and other open-source software.

This software can be used with any Arduino board. Refer to the [Getting Started](#) page for installation instructions.

Windows installer
Windows ZIP file for non-admin install

Windows app 

Mac OS X 10.7 Lion or newer

Linux 32 bits
Linux 64 bits
Linux ARM

[Release Notes](#)
[Source Code](#)
[Checksums \(sha512\)](#)

ARDUINO SOFTWARE
HOURLY BUILDS

LAST UPDATE
29 August 2017 11:12:38 GMT

Download a preview of the incoming release with the most updated features and bugfixes.

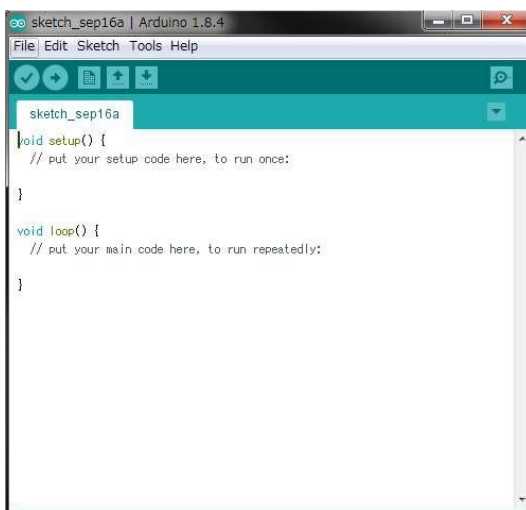
Windows
Mac OS X (Mac OS X Lion or later)
Linux 32 bit, Linux 64 bit, Linux ARM

ARDUINO 1.0.6 / 1.5.x / 1.6.x
PREVIOUS RELEASES

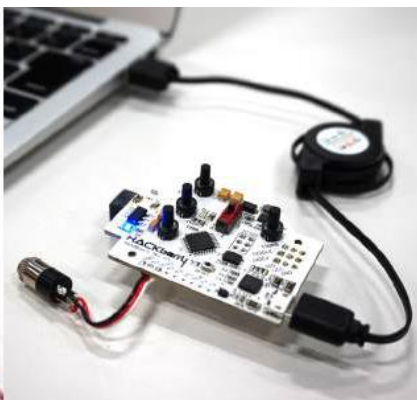
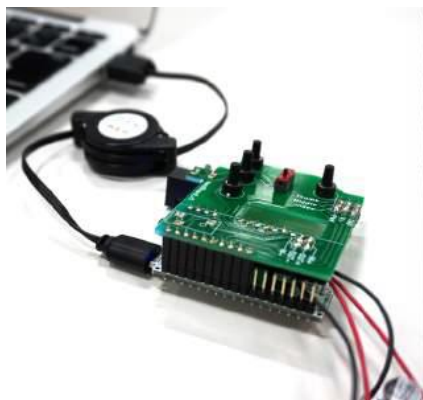
Download the previous version of the current release, the classic Arduino 1.0.x, or the Arduino 1.5.x Beta version.

All the Arduino IDE versions are also available for download. The Arduino IDE can be used on Windows, Linux (both 32 and 64 bits), and Mac OS X.

【2】 Run the Arduino installation file and follow the instructions to install Arduino on your computer. Once the installation is successful, launch the Arduino IDE.

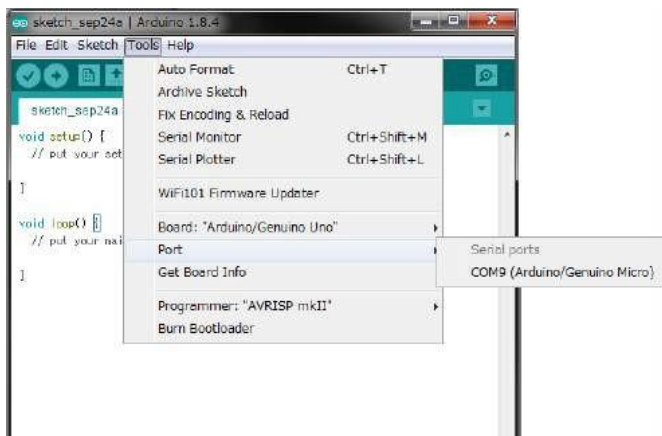


【3】 For MK1 board, connect the Arduino Micro to the PC, and for MK2 board, connect the Mk2 board to the PC with a micro USB cable.



【4】 The driver installation will start automatically. Once the installation is complete, a new COM port will appear in the Ports section of the Arduino IDE Tools tab, select it.

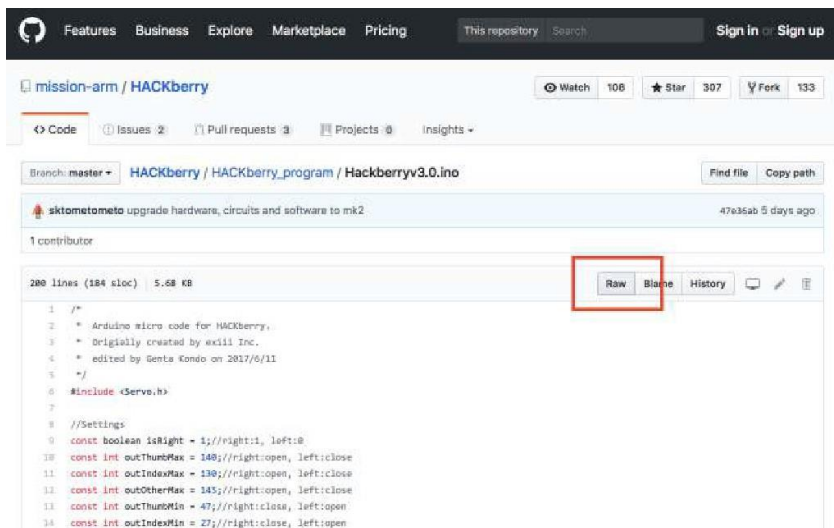
**If it doesn't appear, disconnect the micro USB cable from the Arduino Micro or Mk2 board and plug it back in.*



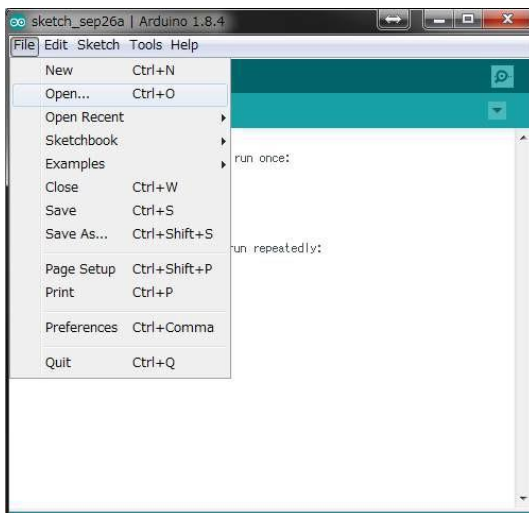
【5】 GithubのHACKberryリポジトリの [HACKberry/HACKberry_program](#) /Hackberrylv3.0.0.0.ino にアクセスし、手元にHACKberry handboard Mk1があれば"[Hackberrylv3.0.0](#)"を、

If you have a HACKberry handboard Mk2, open "[HackberryMk2v1.0](#)", right-click on the "R AW" button and select "Name and save as link".

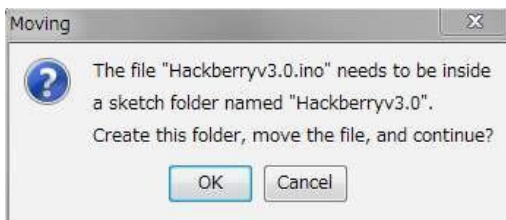
**The version of the program may be different from the time it was written.*



[6] An IDE "Open File →" and select the file you just downloaded to open it.



[7] The following message may appear in the center, but click OK to continue.



[8] Next, select the appropriate microcontroller board from "Tools → Microcontroller Board". Finally, in the top left Press the "➡" button to upload the program.

**For the microcontroller board, select Arduino Micro if you use an Mk1 board or Arduino Nano if you use a Mk2 board .*

A screenshot of the Arduino IDE interface. The title bar shows 'Hackberryv3.0 | Arduino 1.8.4'. The menu bar includes 'File', 'Edit', 'Sketch', 'Tools', and 'Help'. Below the menu bar is a toolbar with icons for opening, saving, and uploading. The main text area displays the code for 'Hackberryv3.0'. The code includes comments about the file's origin and date, an include statement for 'Servo.h', and various settings and pin definitions for a servo motor. The code is as follows:

```
/*
 * Arduino-micro code for Hackberryv3.0
 * Originally created by exiii Inc.
 * edited by Genta Kondo on 2017/8/11
 */
#include <Servo.h>

//Settings
const boolean isRight = 0; //right:1, left:0

//For right hand, find optimal values of ThumbMin, IndexMax and OtherMax first
//For left hand, find optimal values of ThumbMax, IndexMin and OtherMin first.
//Then, calculate the remaining values by following rules.
//Difference of ThumbMin and ThumbMax is 88
//Difference of IndexMin and IndexMax is 117
//Difference of OtherMin and OtherMax is 55

const int outThumbMax = 150; //right:open, left:close
const int outIndexMax = 150; //right:open, left:close
const int outOtherMax = 150; //right:open, left:close
```


Mounting

06 Mounting the Battery Housing and Plug

Mounting the battery housing

This section shows how to mount a battery case for HACKberry that does not contain a dummy connector. If you're building a HACKberry with a dummy socket, start on page 62.



-Used parts—

1. C-B-01

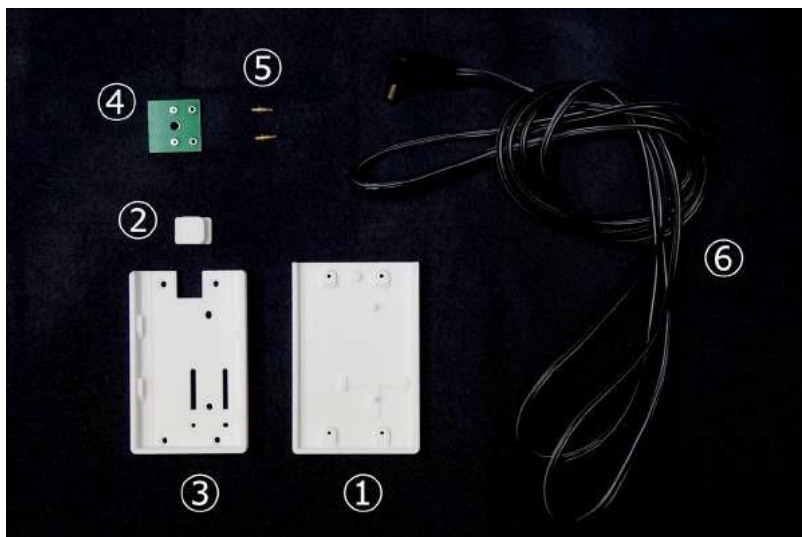
2. C-B-02

3. C-B-03

4. Battery Plate

5. Battery Probe

6. DC plug



-General parts used-

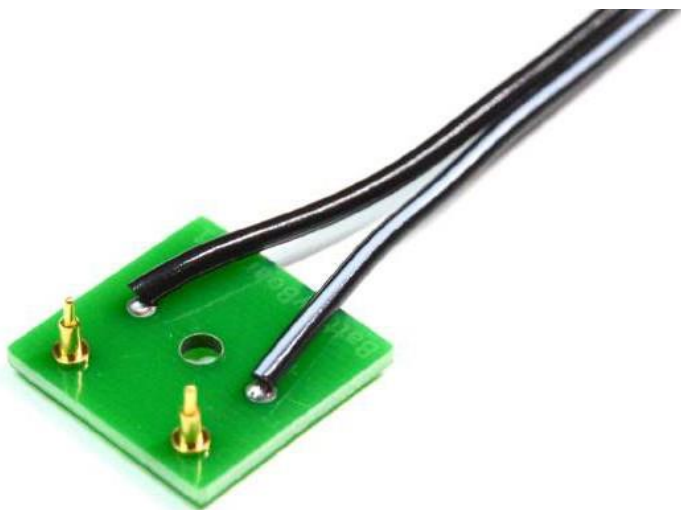
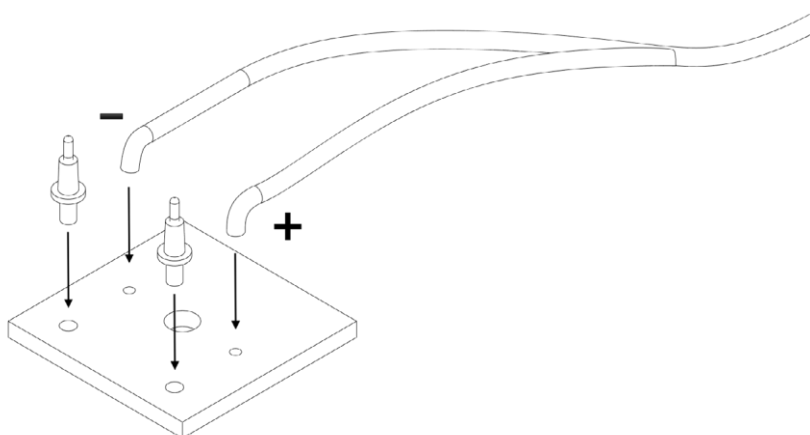


x4

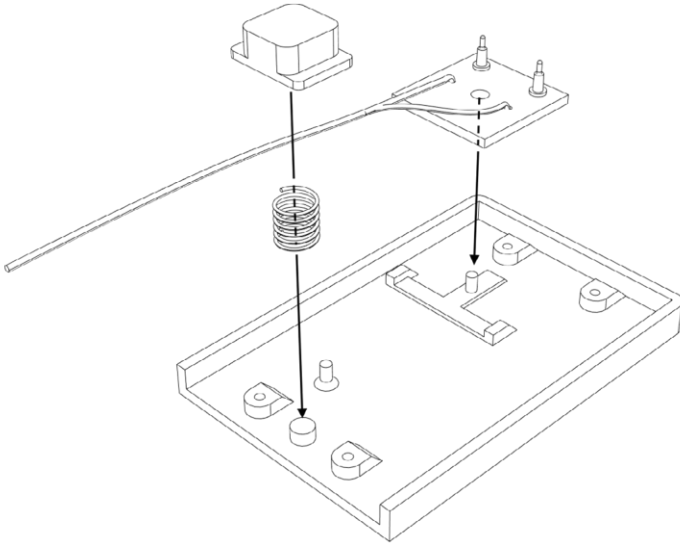


x1

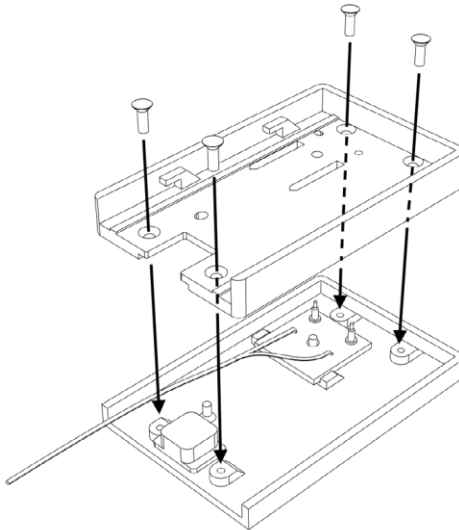
【1】 Soldering the battery probe and DC plug to the battery plate as shown in the figure below. At Make sure to solder the wire with the a+ line.



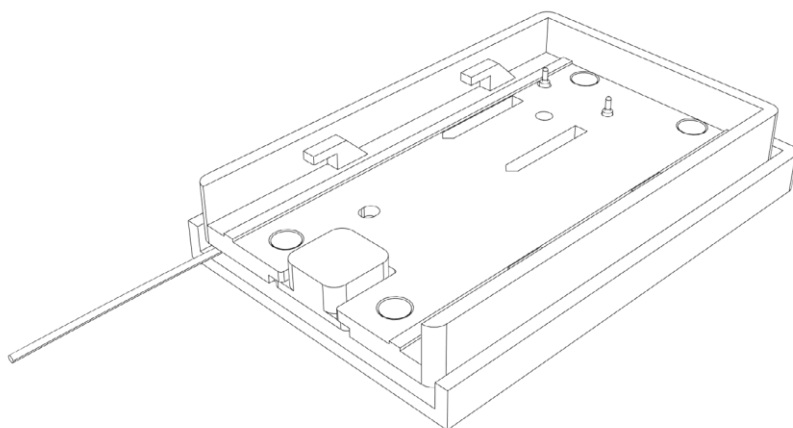
[2] Place the coil spring and C-B-02 on top of the C-B-01



[3] Attach the C-B-03 with 4 screws (M2L6) as shown below.



【4】 Now the battery box assembly is complete.

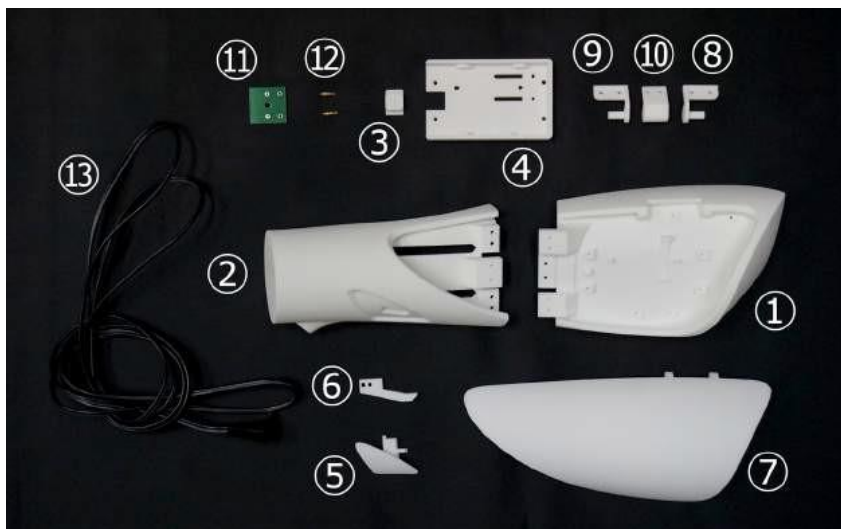


Assembling the fictitious plinth



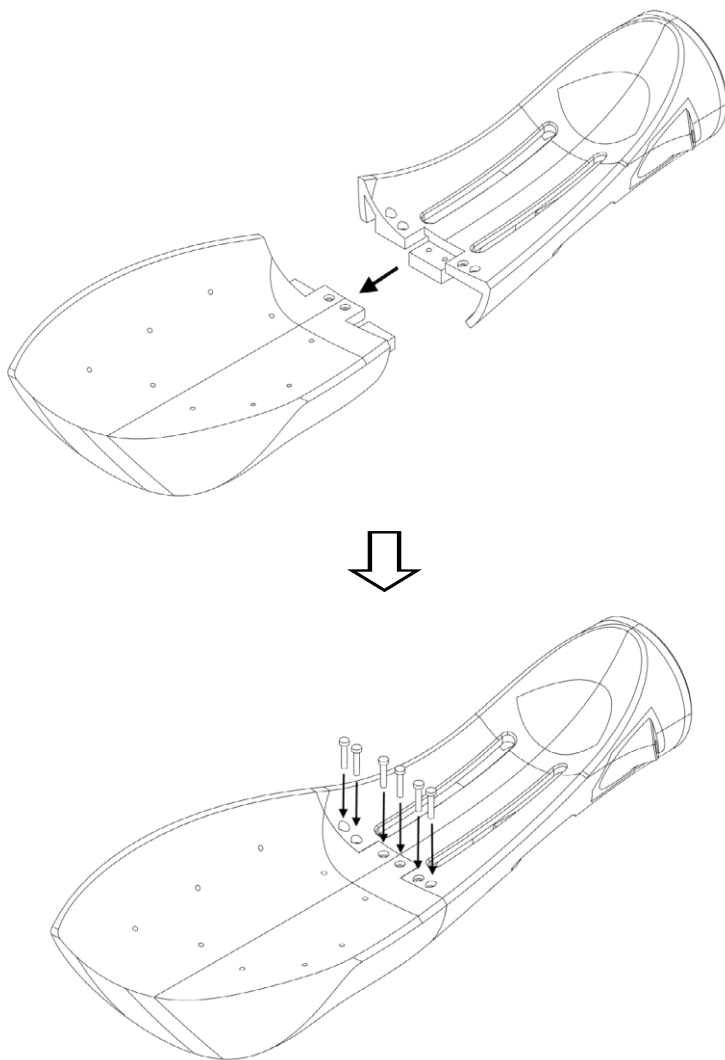
-Used parts—

- | | | |
|--------------|-------------------|-------------------|
| 1. R-S O-01 | 2. R-S O-02 | 3. R-S O-03 |
| 4. R-S O-04 | 5. R-S O-05 | 6. R-S O-06 |
| 7. R-S O-07 | 8. R-S O-08 | 9. R-S O-09 |
| 10. R-S O-10 | 11. Battery Plate | 12. Battery Probe |
| 13. DC plug | | |

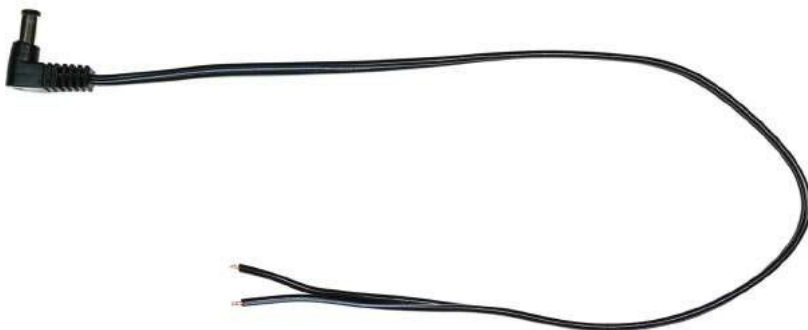


-General parts used- x7 x10 x1 x1

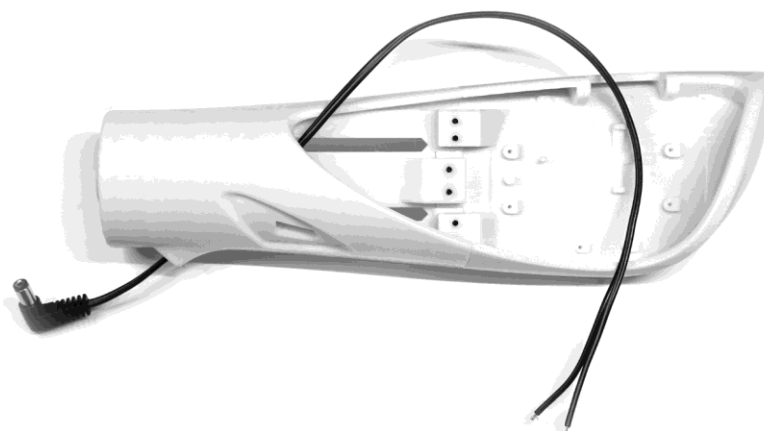
【1】 Fixes R-SO-01 and R-SO-02 with 6 screws (M2L10) as shown in the figure below.



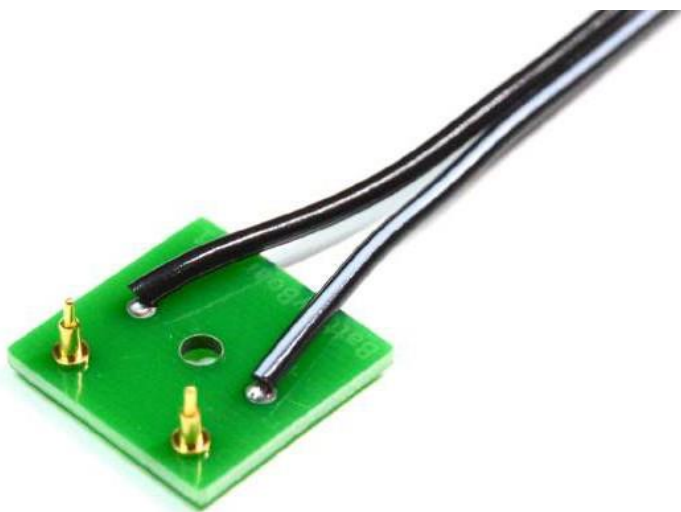
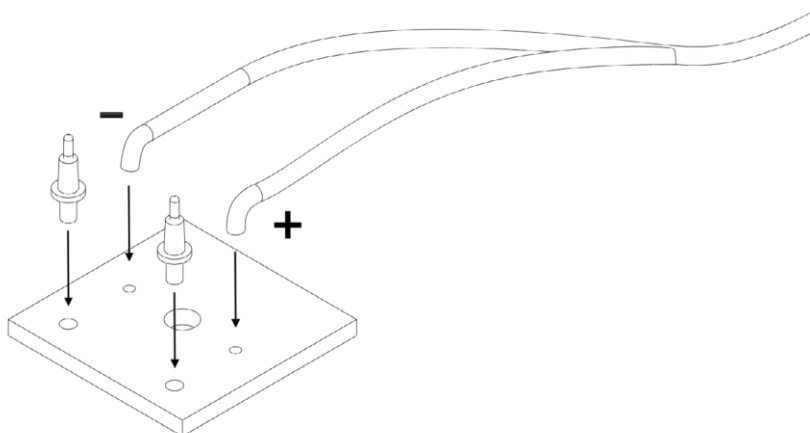
[2] Cut the plug from D&C to 33cm, peel off the sleeve from the cut end 4mm, and twist the inner core cable together. If you don't have a pickle on hand, use a tipper to remove the sheath without cutting the center wire inside.



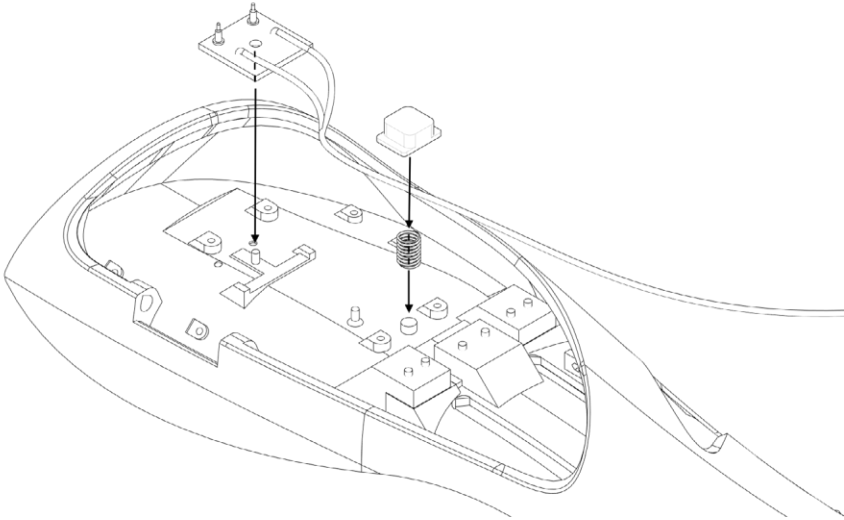
[3] Thread the wire through the hole in R-SO-02 as shown in the figure below.



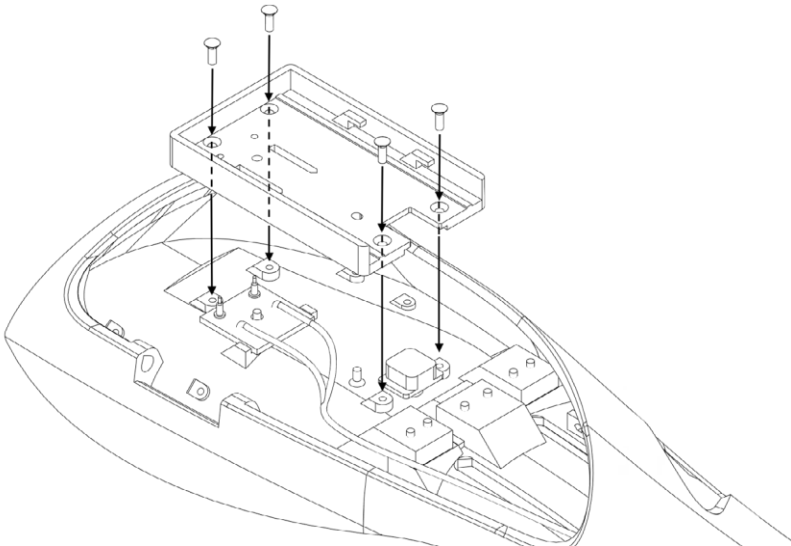
【4】 Soldering the battery probe and DC plug to the battery plate as shown in the picture. At this point, make sure to solder the wire with the white line to +.



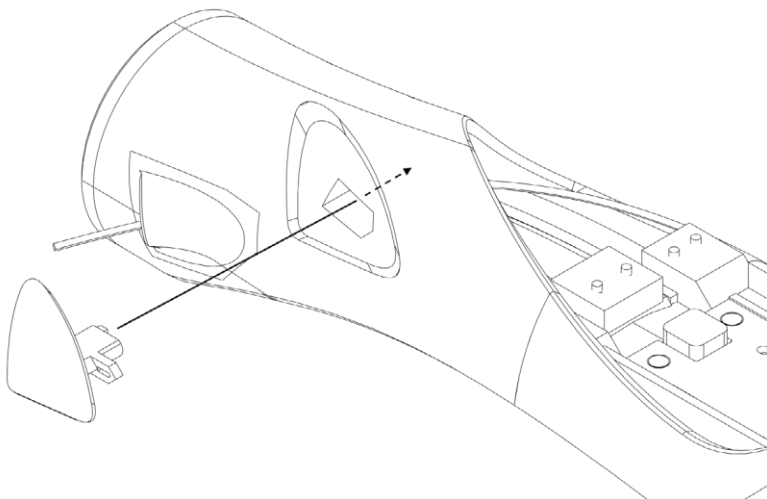
[5] Attach the coil spring and R-SO-03 as shown in the figure below.



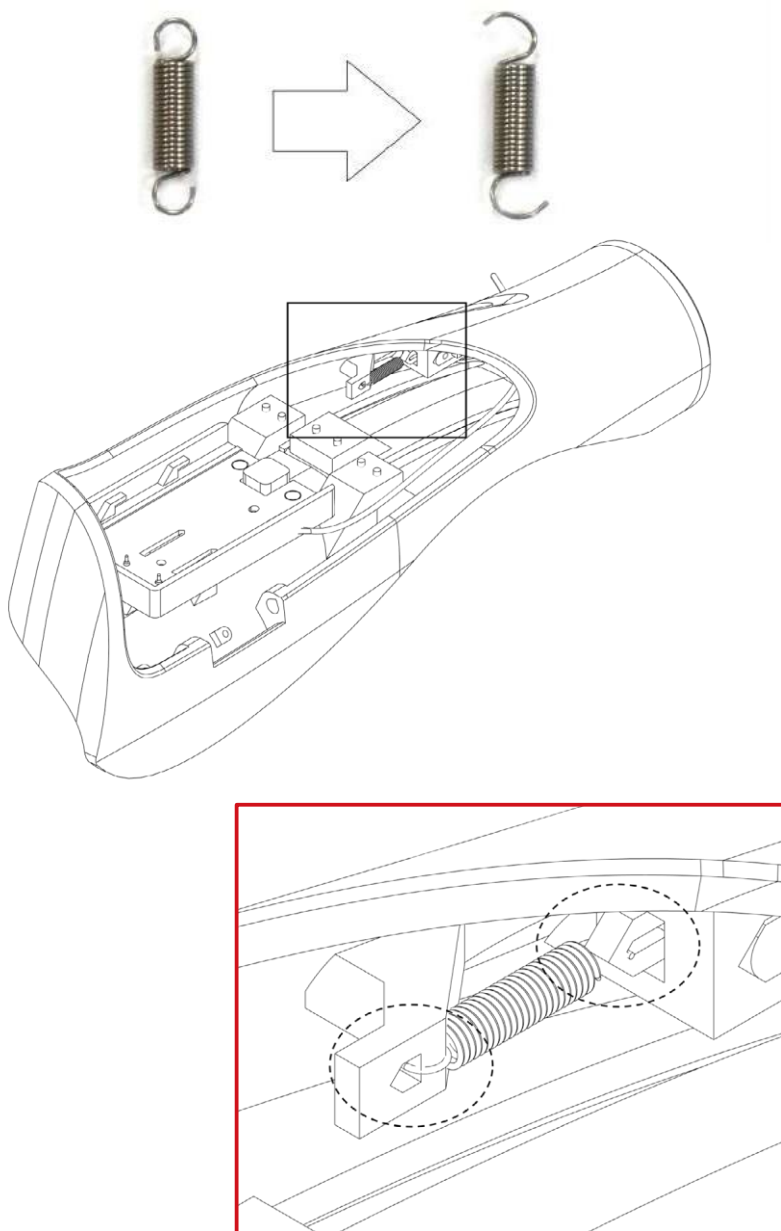
[6] Fix the R-SO-04 with 4 screws (M2L6) as shown in the figure below.



[7] Couple R-SO-05 to R-SO-02 as shown in the

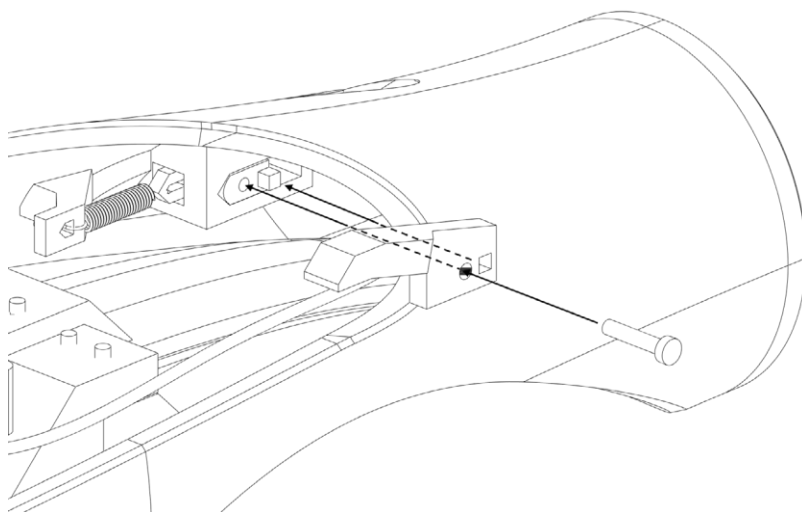


[8] Install the two ends of the tension spring unfolded as

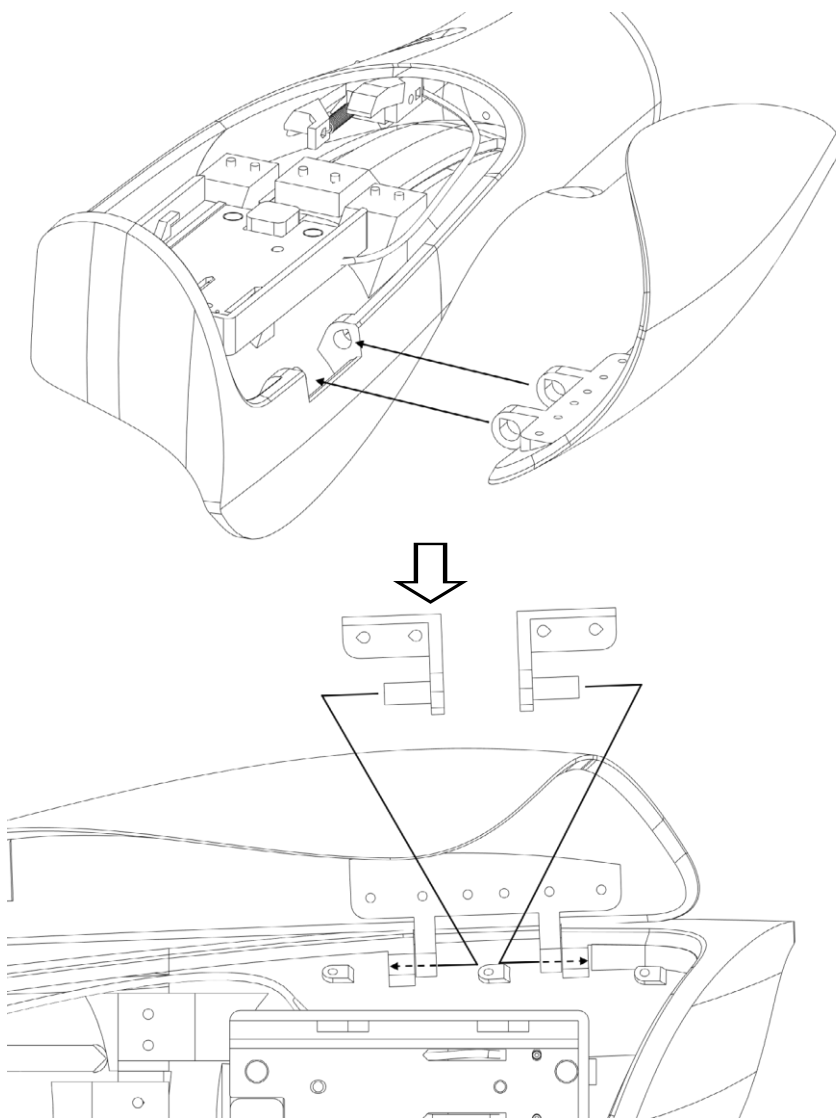


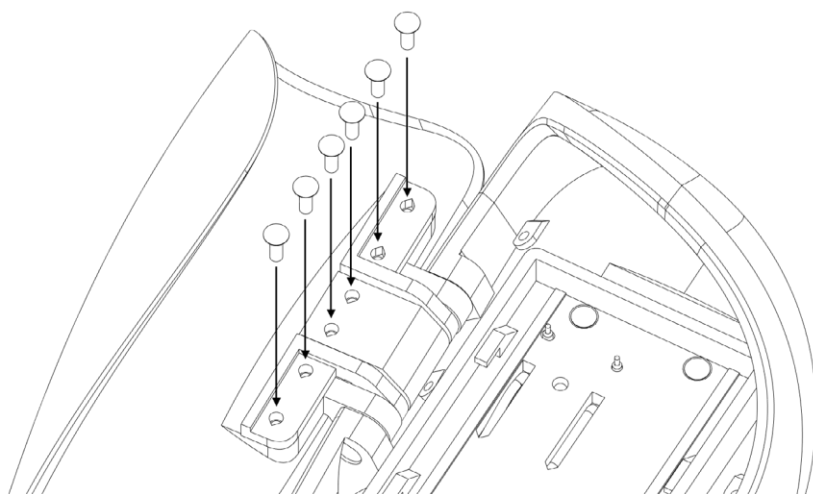
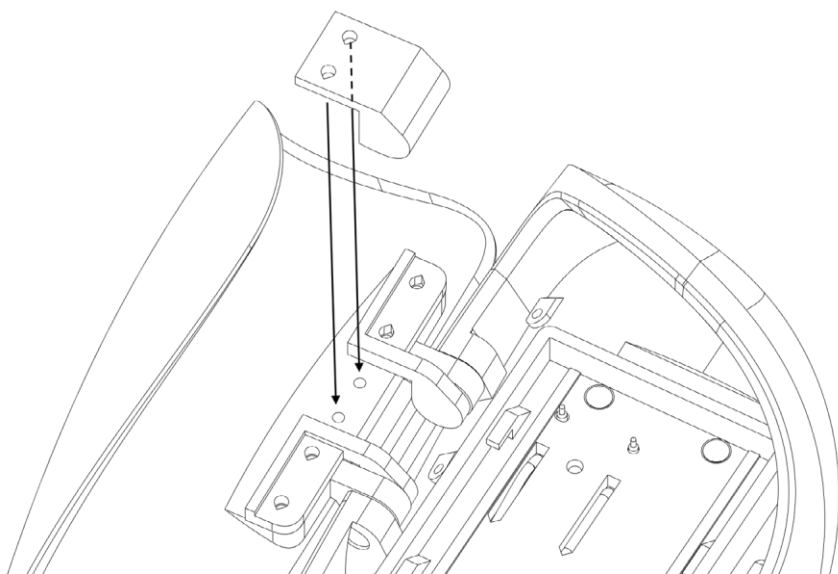
Enlarged view

【9】 Use the screw (M2L10) to fix the R-SO-06 as shown in

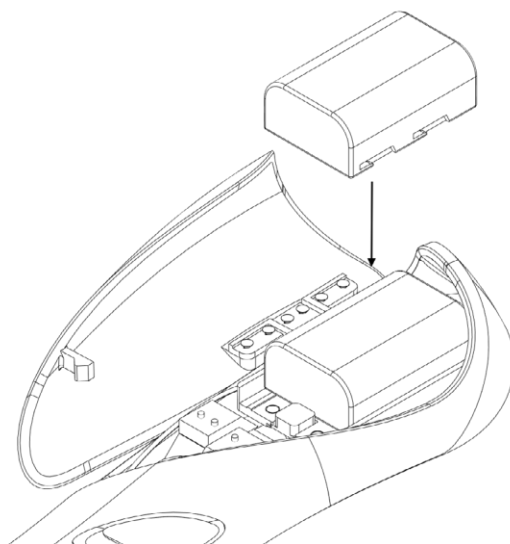


[1 0] Secures R-SO-01 and R-SO-07 with 6 screws (M2L6) up to R-SO-08 and R-SO-09.

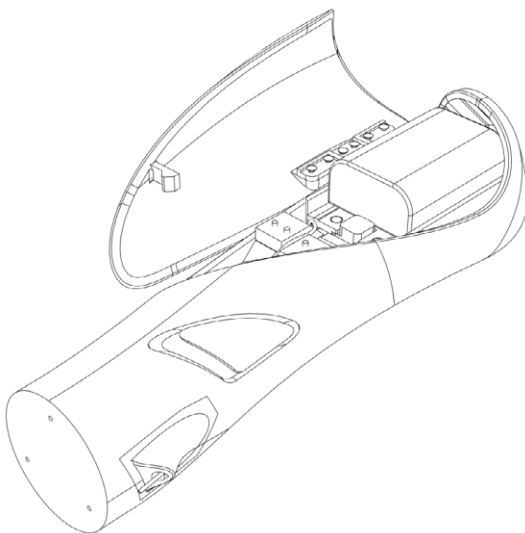




[1 1] Install the battery.



[1 2] This completes the socket assembly.



Mounting

07 Thumb Set



-Used parts—

1. R-T-01

2. R-T-02

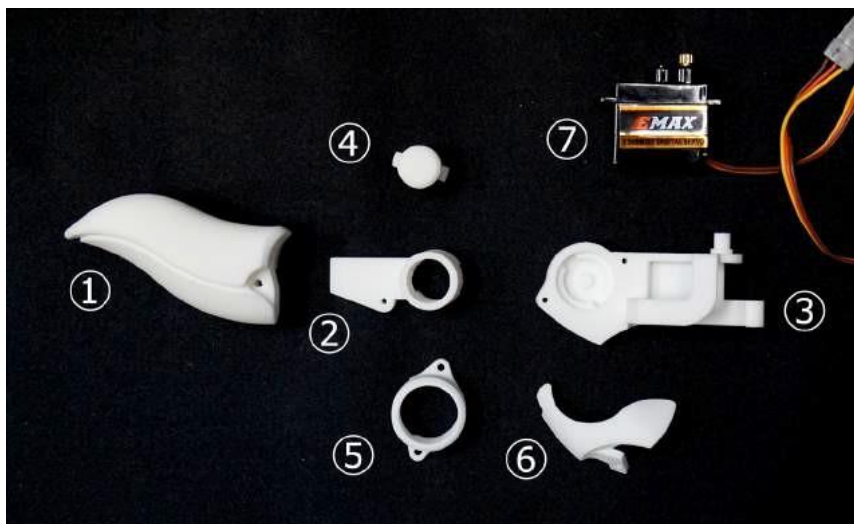
3. R-T-03

4. R-T-04

5. R-T-05

6. R-T-06

7 . Small servo motor



-General parts used-



x5

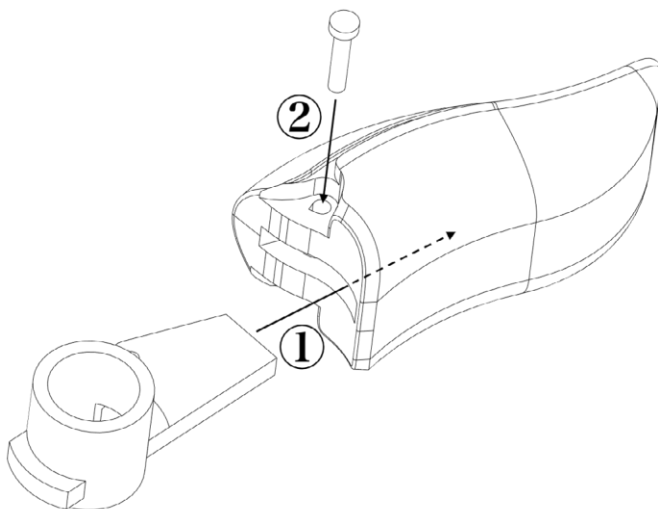


x1

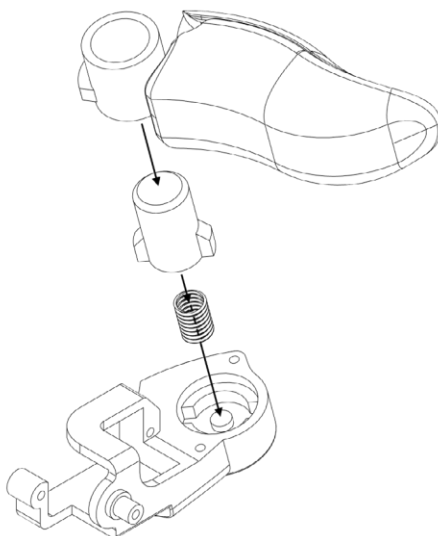


x1

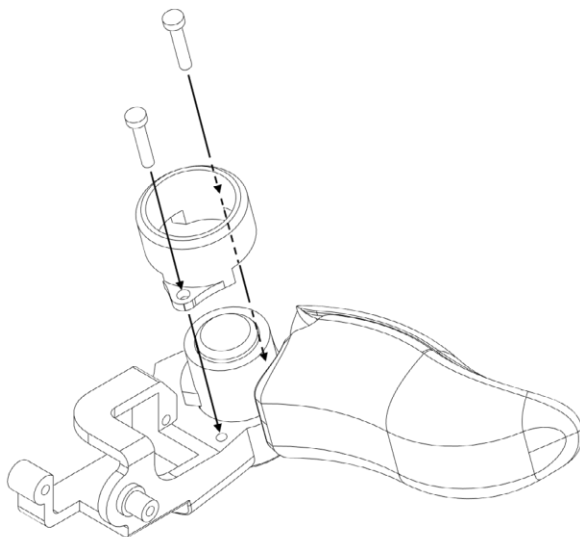
[1] Screw R-T-01 to R-T-02 (M2L10).



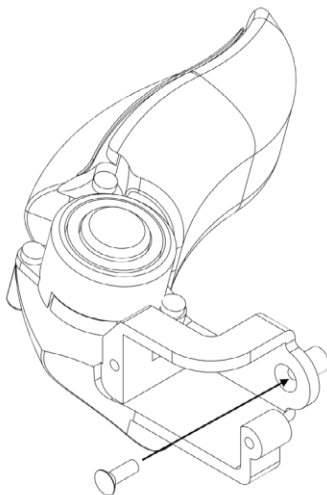
[2] Place the coil spring in R-T-03 as shown in the figure below, align R-T-04 with the groove, and position it so that it covers the coil spring. R-T-02 passing.



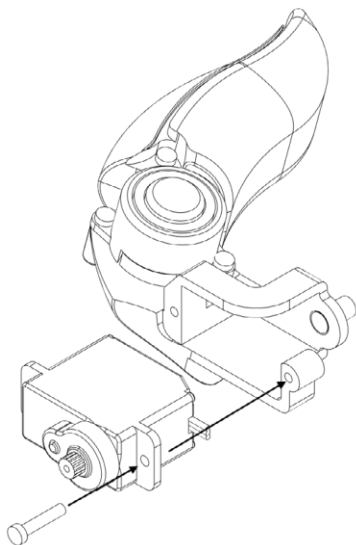
【3】 R-T-05 and fixing with two screws (M2L10).



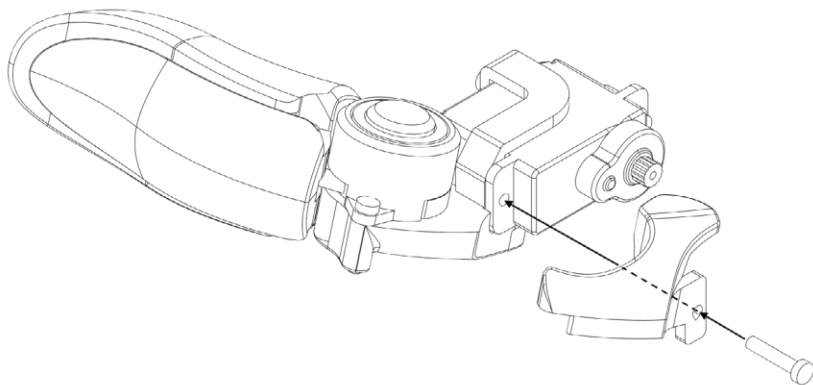
【4】 Attach the screw (M2L6) to the R-T-03 as shown in the figure below.



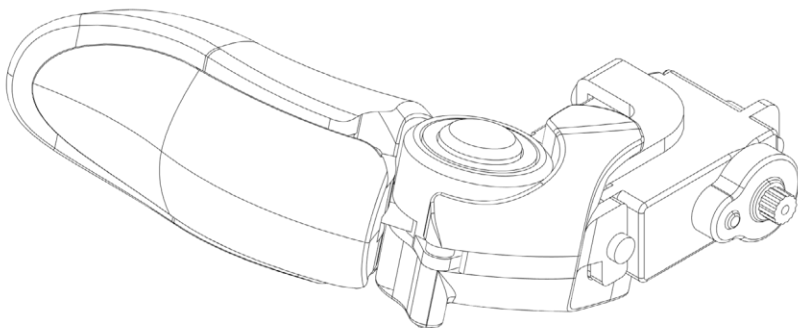
【5】 Fix the small servo motor with screws (M2L10).



【6】 Then fix the R-T-06 with screws (M2L10).



[7] This is the end of assembling the thumb.



Mounting

08 Mounting the Index Finger



-Used parts—

1. R-I-01

2. R-I-02

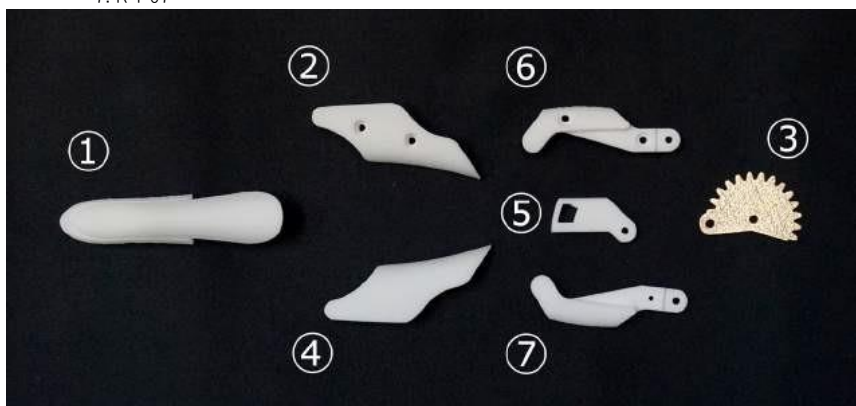
3. R-I-03

4. R-I-04

5. R-I-05

6. R-I-06

7. R-I-07

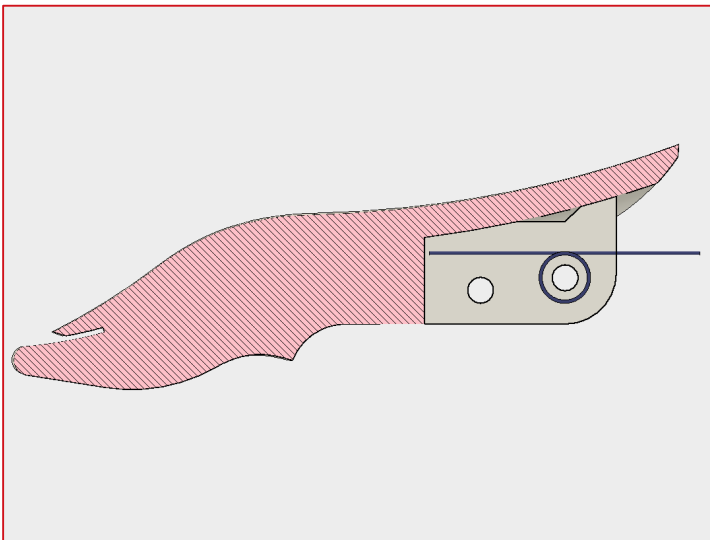
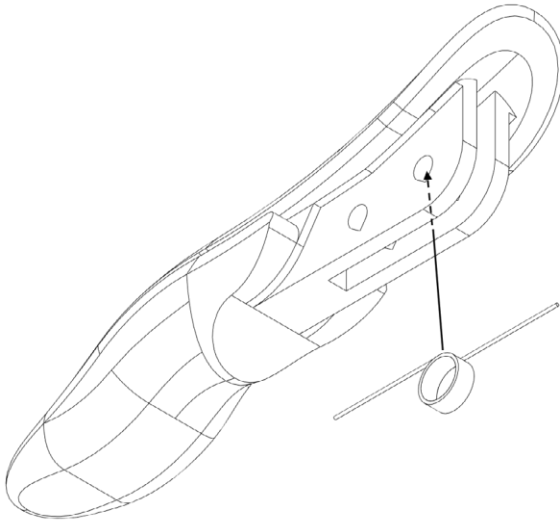


**The R-I-03 in the photo is 3D printed using metal materials to increase strength. Although it is not as strong or durable, it is possible to use resin materials like other parts.*

-General parts used-

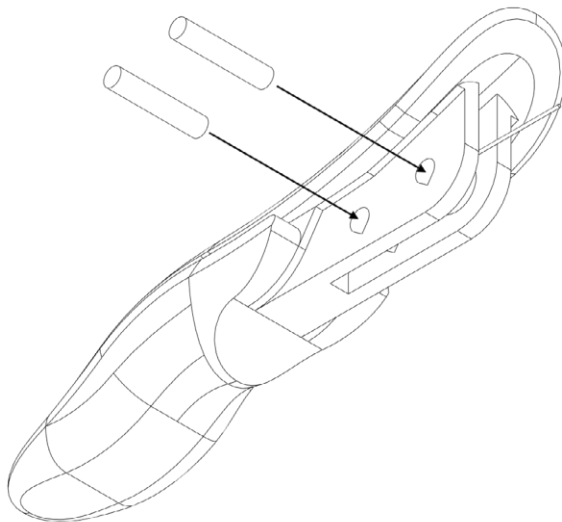


[1] Insert the torsion spring into the R-l-01 as shown in the figure



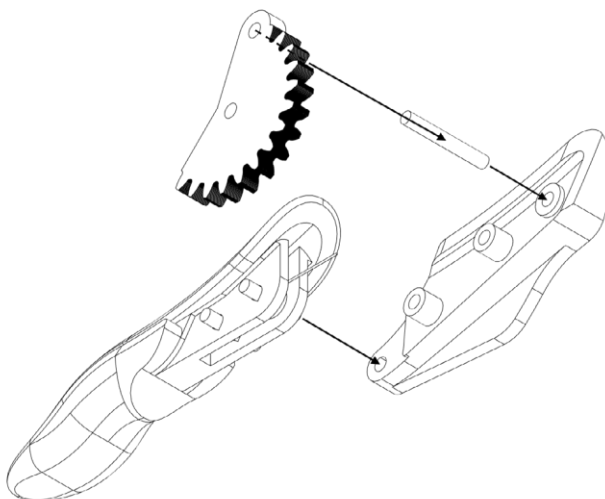
Cross view

[2] Insert two shafts (10 mm) into it as shown in the figure below. **At this time, the axis on the right side**
Be careful when passing the spring circle.

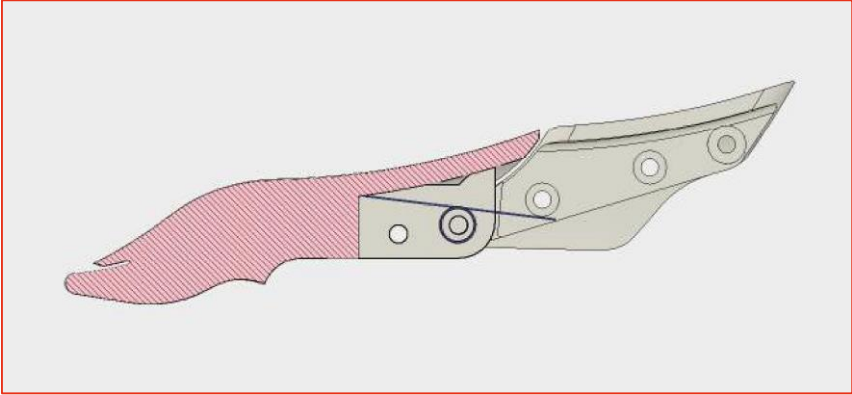


[3] Install R-I-02 and R-I-03. The R-I-03 is mounted via the shaft (10mm) as shown in the figure.

***Please pay attention to the position of the torsion spring at**



this time.



Cross-section view of torsion spring position

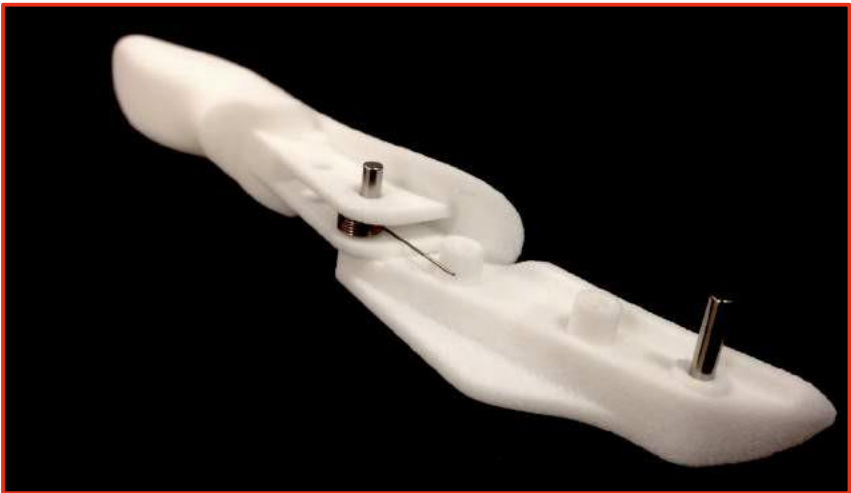
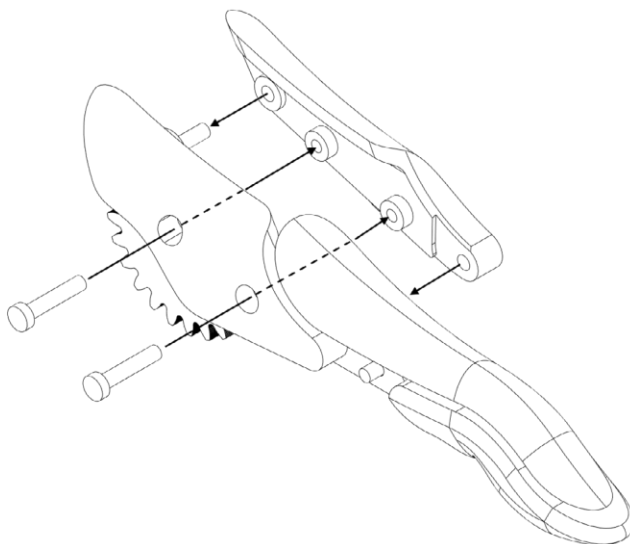
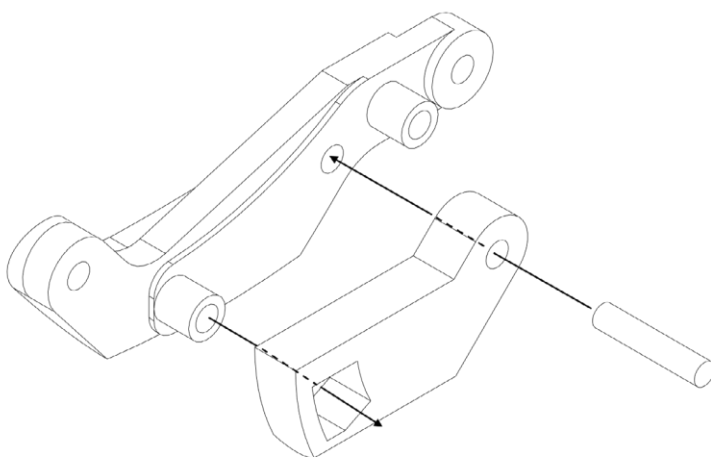


Photo of the torsion spring position

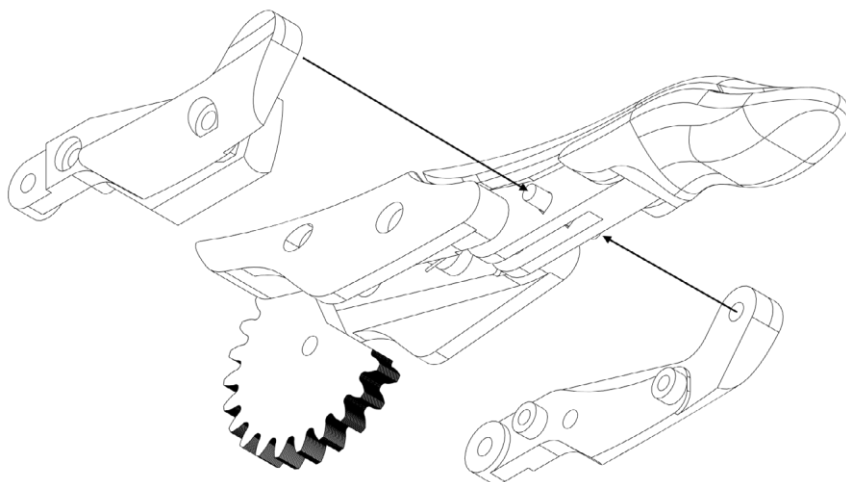
【4】 Cover the R-I-04 and fix it in two places with screws



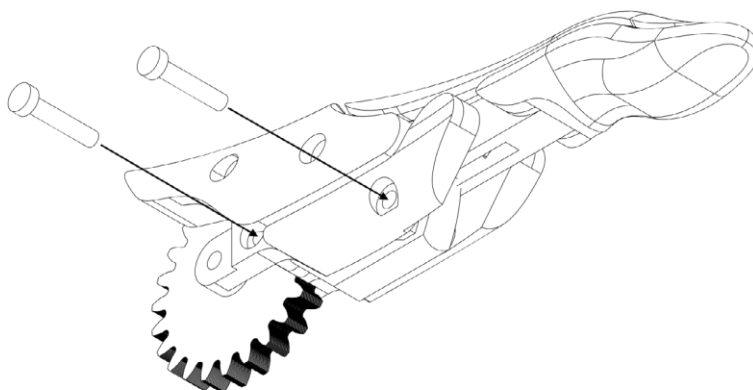
【5】 Attach it to the R-I-06 after passing the shaft (10mm) through the R-I-05.



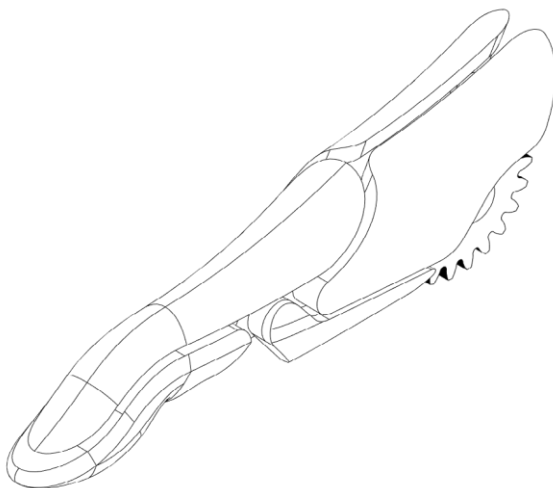
【6】 Using the parts assembled in R-l-07 and [5], the parts assembled in [4] are placed between both is.



【7】 Fix the combined parts with [6] in two places with screws (M2L10) as shown in the figure.



[8] Now the index finger is complete.



Mounting

09 Assembling Three Fingers

**Assemble the middle, ring and pinky fingers. They use the same pieces.*



-Used parts—

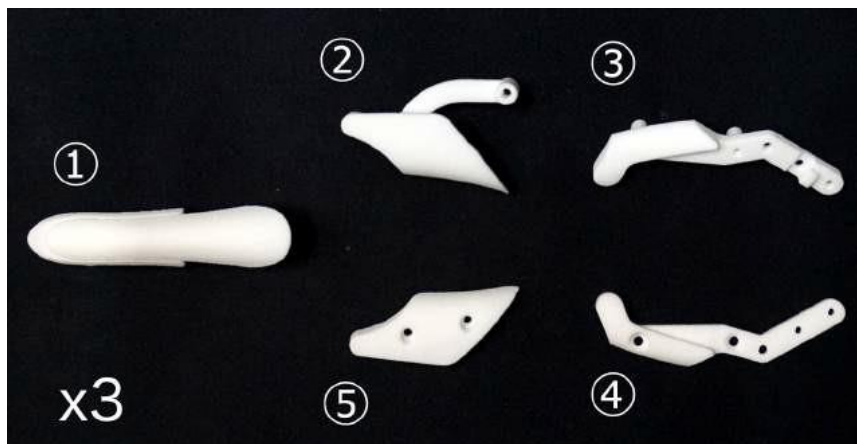
1. R-O-01

2. R-O-02

3. R-O-03

4. R-O-04

5. R-O-05



-General parts used-

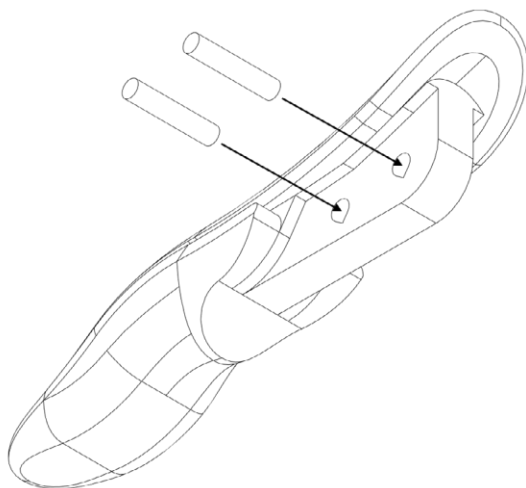


x12

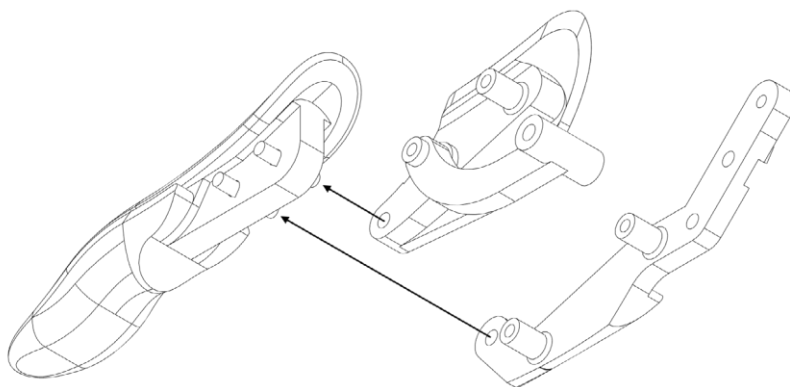


x6

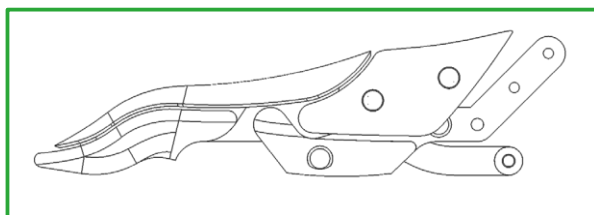
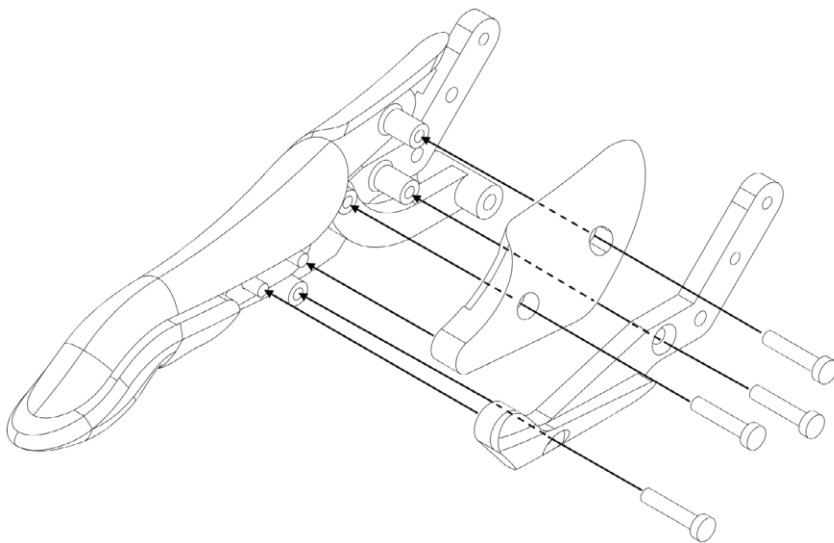
【1】 Connects two shafts (10mm) to the R-O-01.



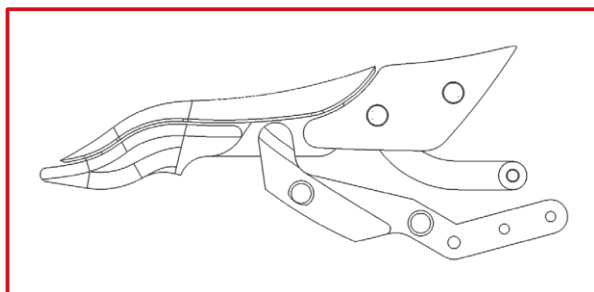
【2】 Install the R-O-02 and R-O-03 as shown in the figure below.



【3】 Fix the R-O-04 and R-O-05 with 4 screws (M2L10).

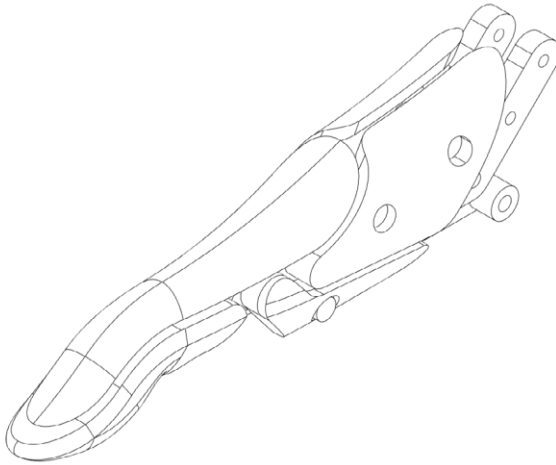


**Good example*



**Bad example*

[4] Gather three equal things. Your middle finger, ring finger, and pinky finger

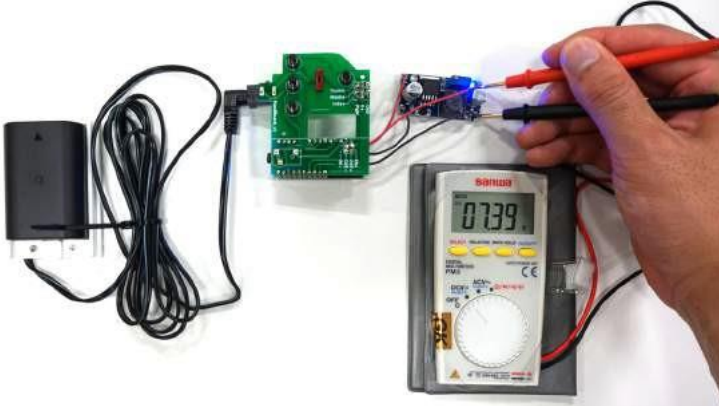


Mounting

10 Adjusting the Voltage and Adjusting the

Adjust the angle of the servo motor before mounting the hand. The H ACKberry Mk2 handboard does not need [1] and [2].

【1】 Insert the DC plug into the ACKberry Mk1 H manual plate and push the slide switch up to turn it on. Measure voltage by applying the tester to the OUT+ and OUT- of the CDC converter. In the unregulated state, the battery voltage is emitted as is.



【2】 To reduce this output voltage to 5V, turn the gold part of the CD D converter counterclockwise using a flathead screwdriver. The voltage gradually decreases after 5 or 6 turns when you start spinning. Turn off the power once you're done adjusting.



[3] Refers to the servo motor connector on the board pin connector. **Make sure the black (or brown) wire is facing out.** Insert a large servo motor connector into the pin connector marked Index, and a small servo motor connector into the pin connector marked Editdle and Thumb.

For Mk1 boards, black (or brown) leads to GND, red to V+, and white (or yellow) to PWM.

For Mk2 plates, black (or brown) leads to GND, red to VCC, and white (or yellow) to SIG.



After connection, the plate lights up and the angle of the motor changes, completing the adjustment. Mark servo motors of the same type in the middle and thumb to avoid errors when reconnecting later.

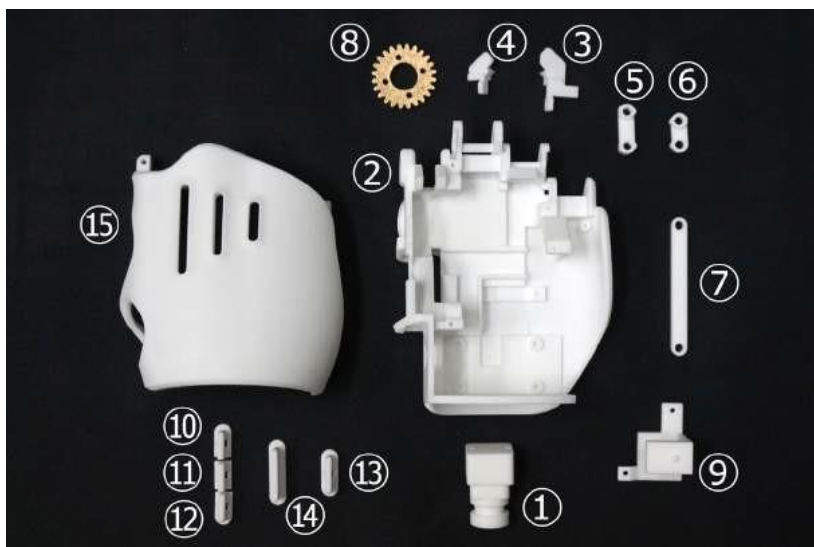
Mounting

11 Assembly of Palmas



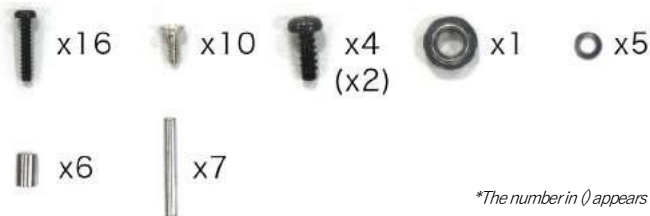
-Used parts—

- | | | |
|------------|------------|------------|
| 1. R-H-01 | 2. R-H-02 | 3. R-H-03 |
| 4. R-H-04 | 5. R-H-05 | 6. R-H-06 |
| 7. R-H-07 | 8. R-H-08 | 9. R-H-09 |
| 10. R-H-10 | 11. R-H-11 | 12. R-H-12 |
| 13. R-H-13 | 14. R-H-14 | 15. R-H-15 |



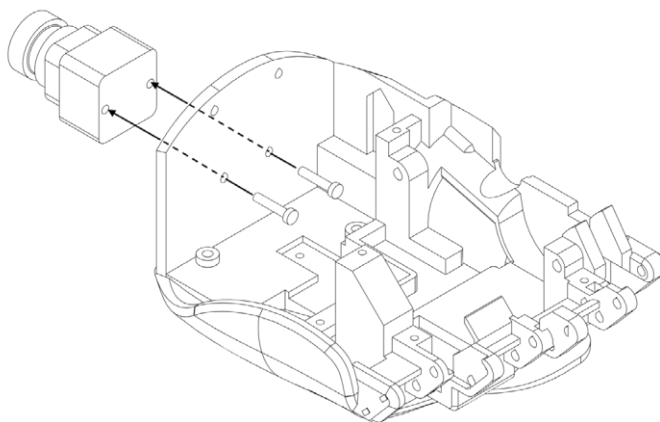
**The R-I-03 in the photo is 3D printed using metal materials to increase strength. Although it is not as strong or durable, it is possible to use resin materials like other parts.*

-General parts used-



**The number in () appears when using the Mk2*

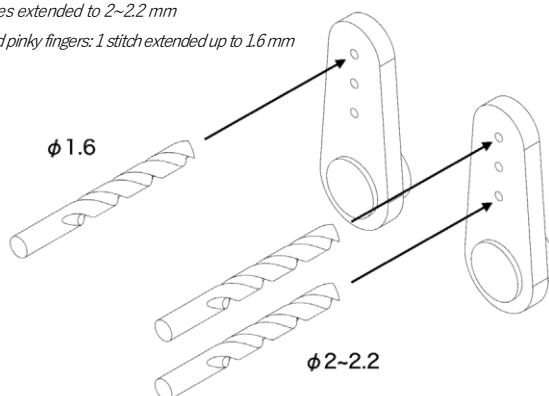
[1] Secure the R-H-01 to the R-H-02 with two screws (M2L10).



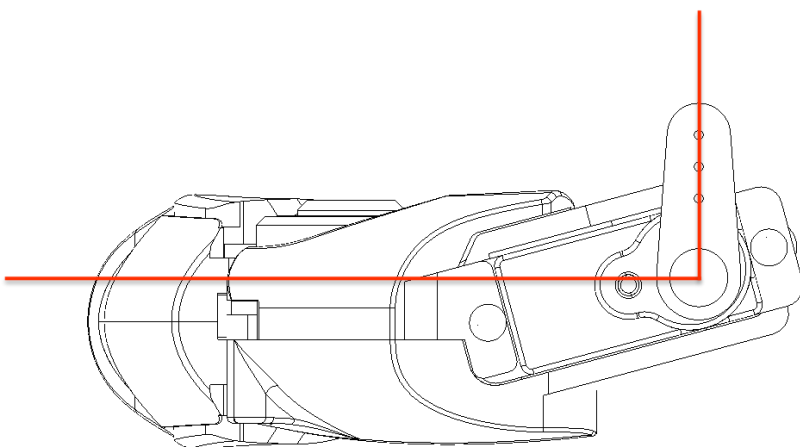
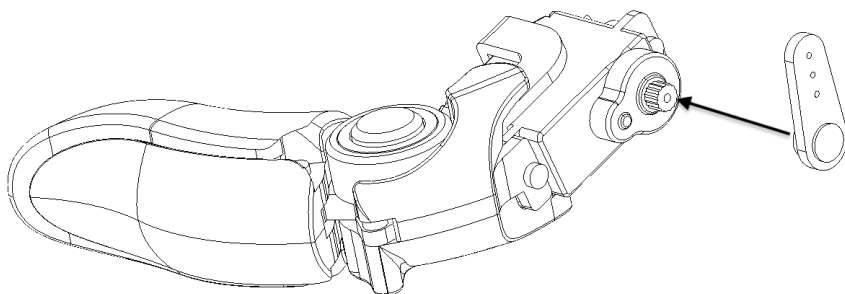
[2] Expand the servo horn hole (the one shown below) connected to the servo motor using a press screw. **Be careful not to confuse the location of the hole or the diameter.**

For thumb: 2 places extended to 2~2.2 mm

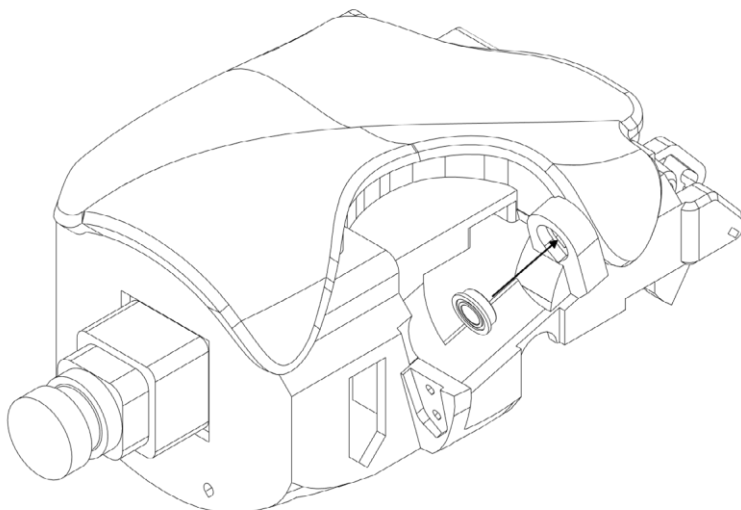
For middle, ring and pinky fingers: 1 stitch extended up to 1.6 mm



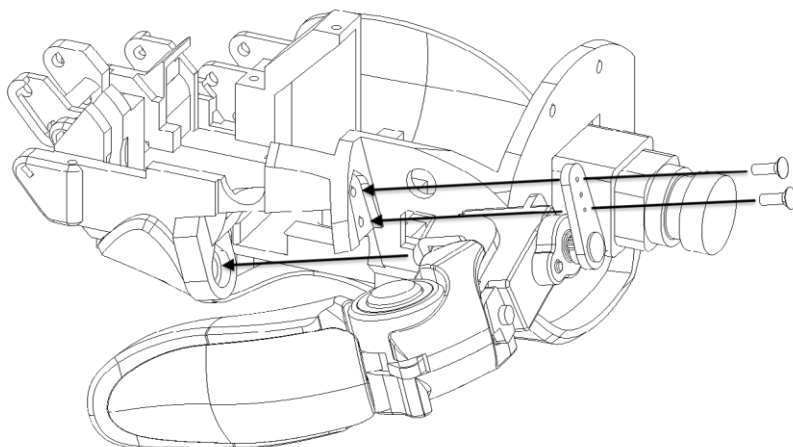
[3] The servo motor shaft is set to "10 Adjusting Voltage and Starting Position of the Servo Motor" (page 92)
Place the servo thumb and horn unit at right angles, being careful not to deviate from position.



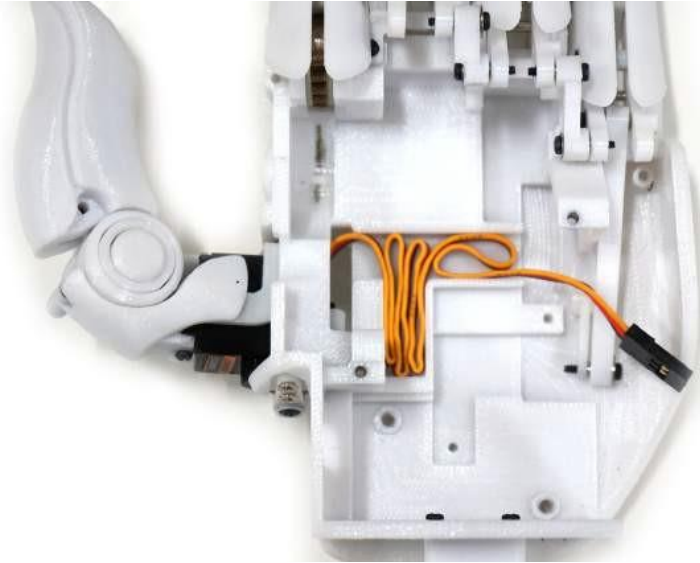
[4] Push the bearing towards the R-H-02.



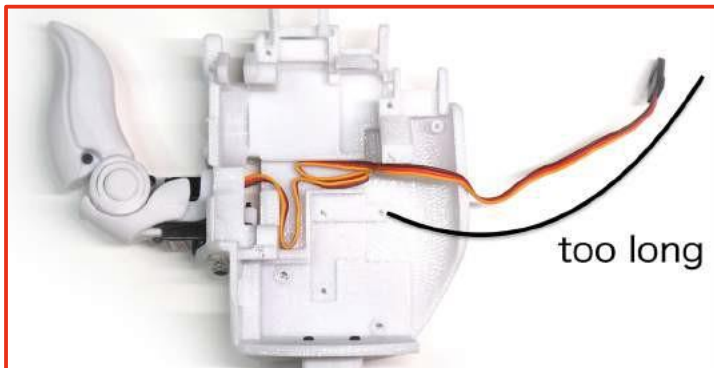
[5] Fix the thumb using two screws (M2L6).



[6] Insert the servo motor cable into the R-H-02 as shown in the figure below, then

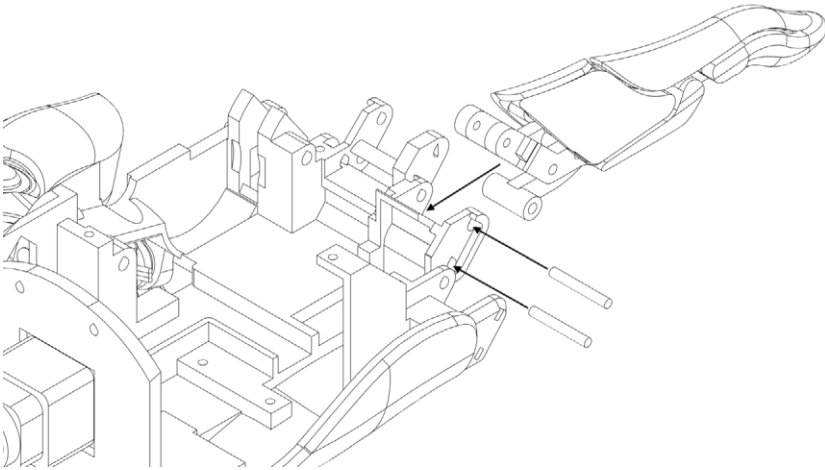


**Good example*

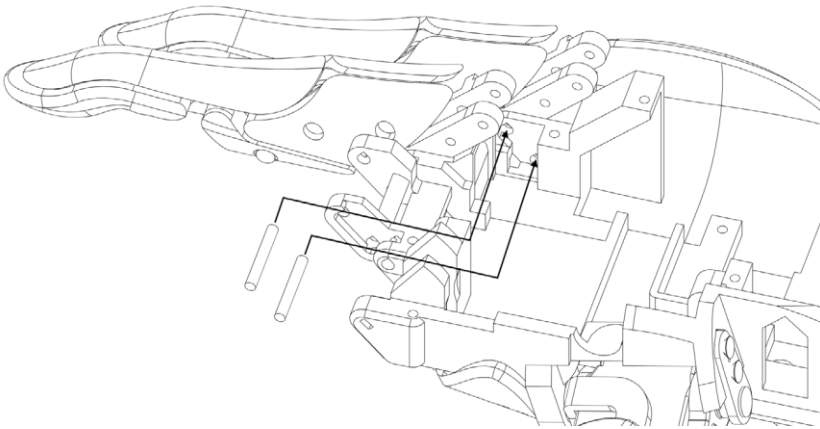


**Bad example*

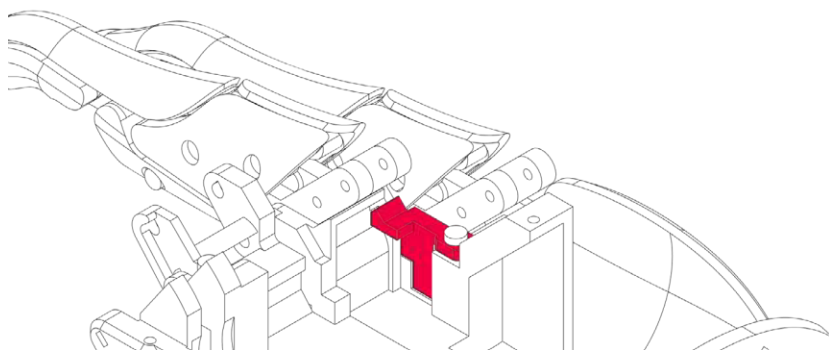
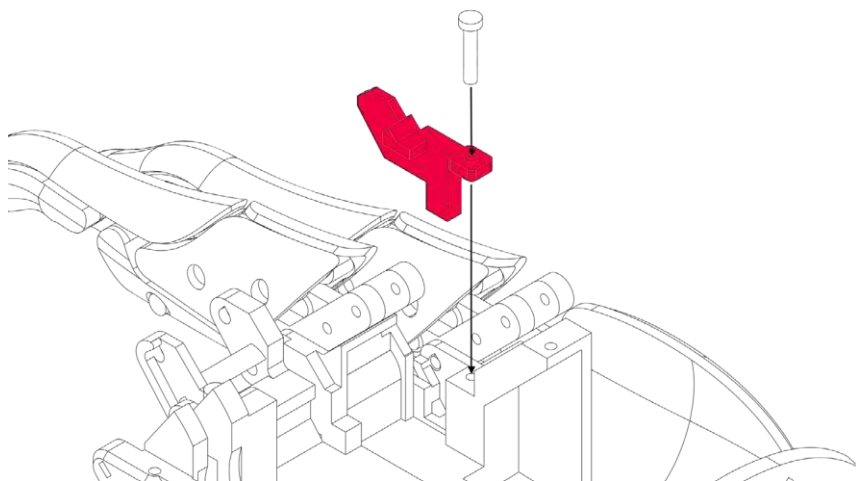
[7] Next, connect the ring finger unit via two axes (15mm) as shown in the figure



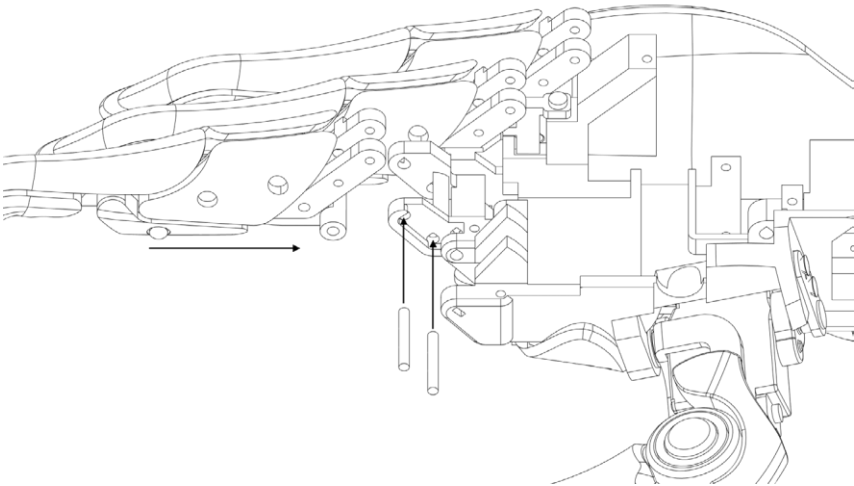
[8] In the same way, the pinky finger unit is also fixed by two axes (15 mm).



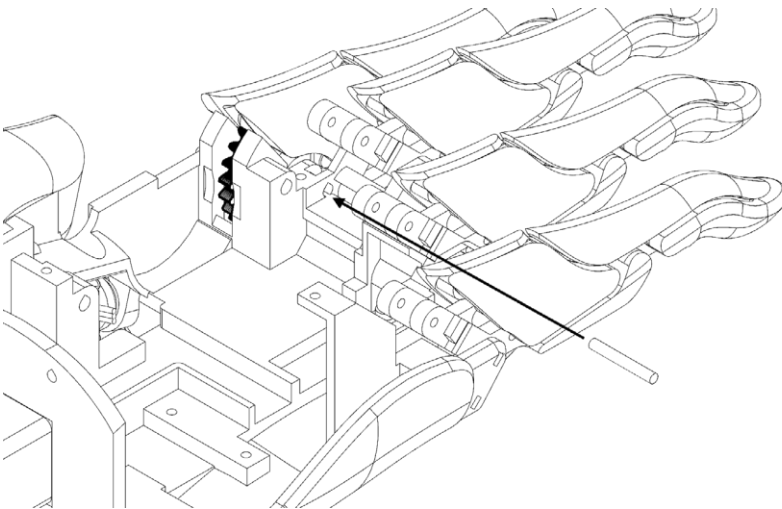
【9】 Install the R-H-03 with the screws (M2L10) as shown



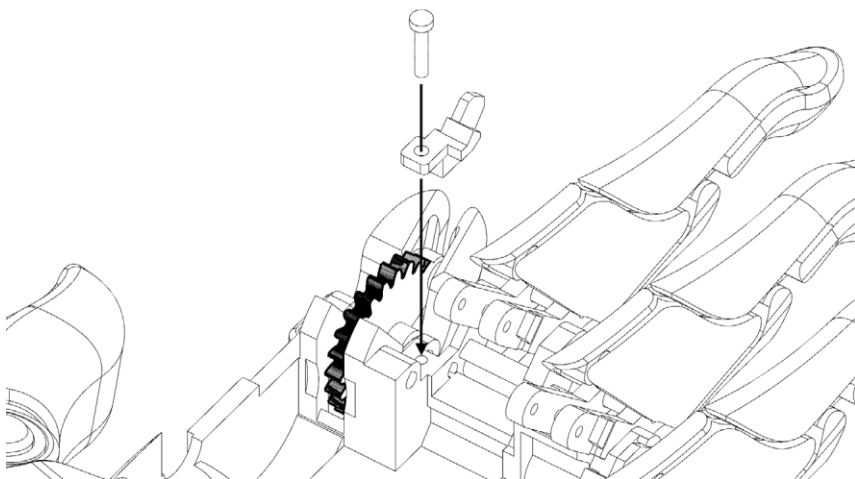
[10] Place your middle finger using two rods (15 mm) as shown in the



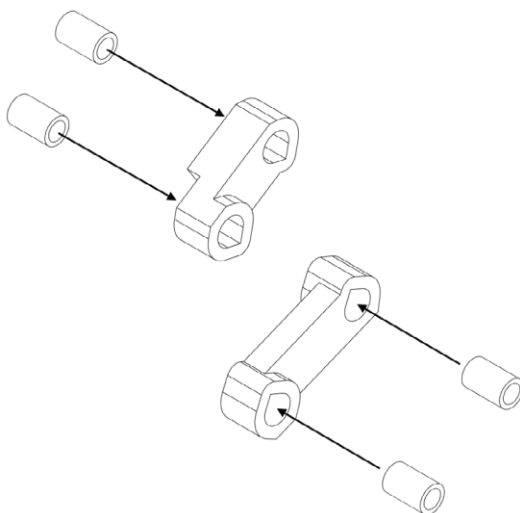
[11] Connect the index finger unit using an axis (15 mm) as shown in the figure below. At that point, the shaft should pass through the gear hole.



【1 2】 Fix the R-H-04 using screws (M2L10) as shown in the figure

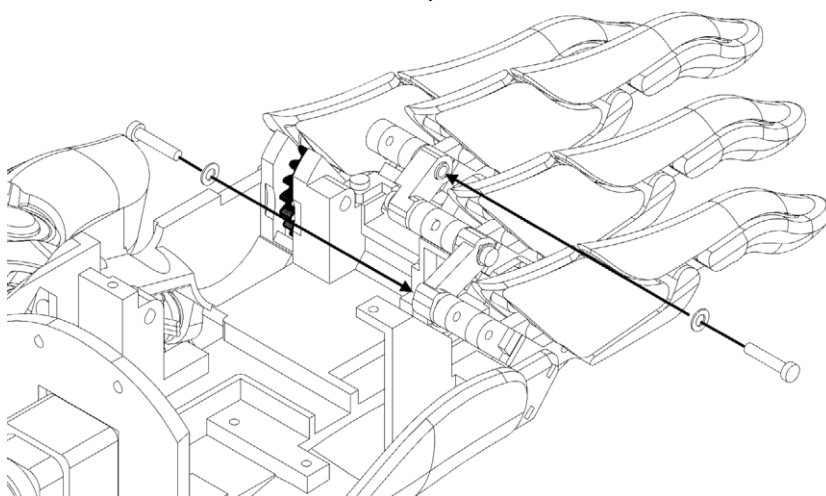
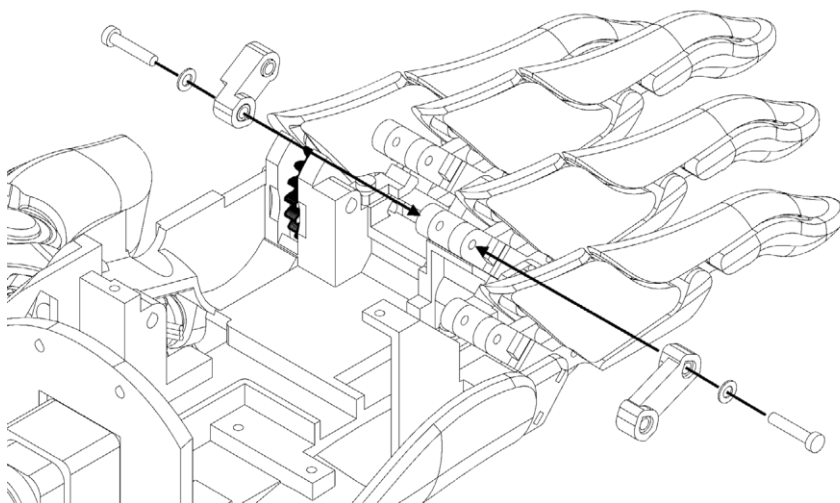


【1 3】 Insert two spacers into the R-H-05 and R-H-06.

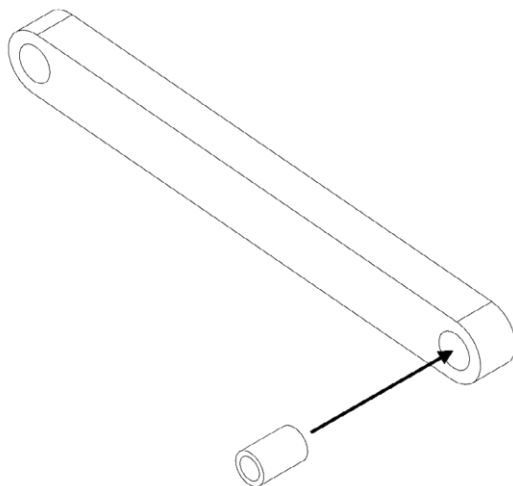


[14] R-H-05 and R-H-06 are screwed through the washer in 2 places each (4 in total) with screws (M2L10).

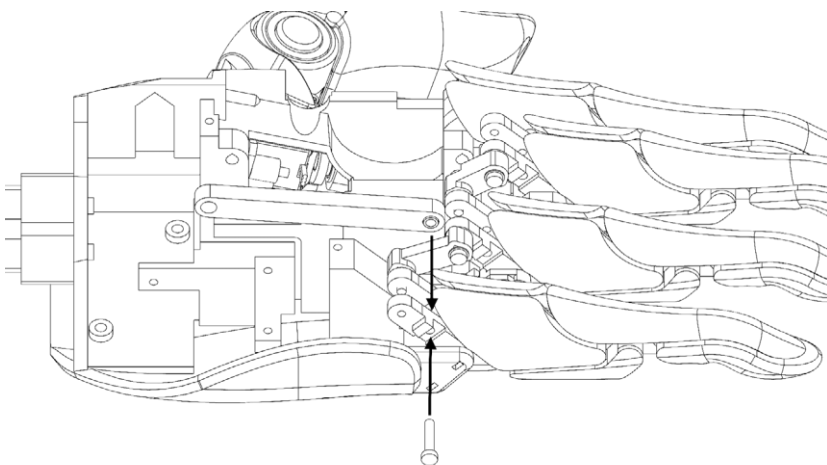
**Be careful to hold the R-H-05 (longest part) between your pinky and ring fingers.*



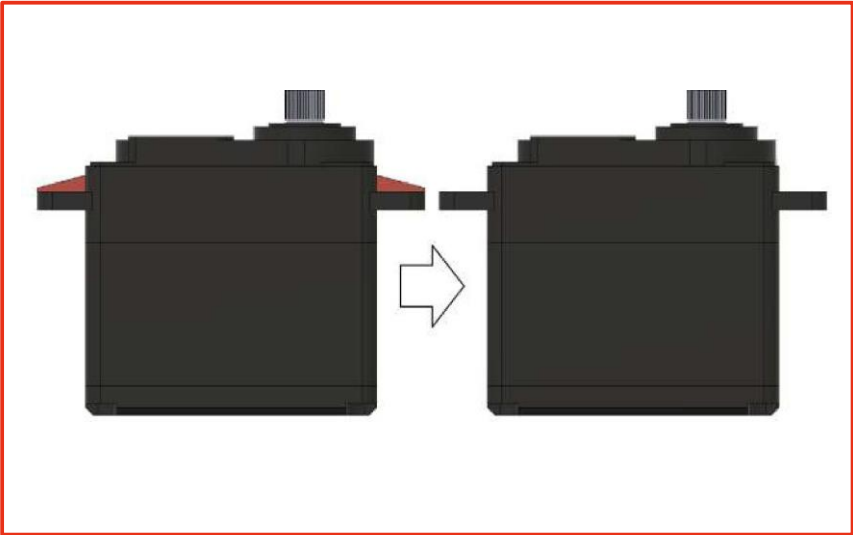
【15】 Insert the spacer into any of the holes in the R-H-07.



【16】 Attach the R-H-07 to the pinky unit with the M screw (2L10).

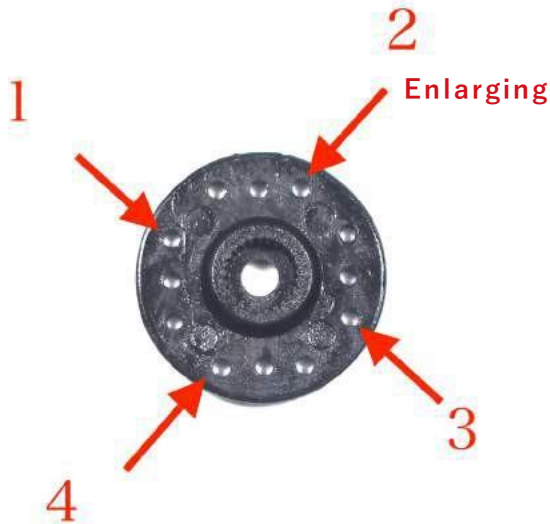


[17] Cut the part that appears in red in the bottom diagram of the size of the

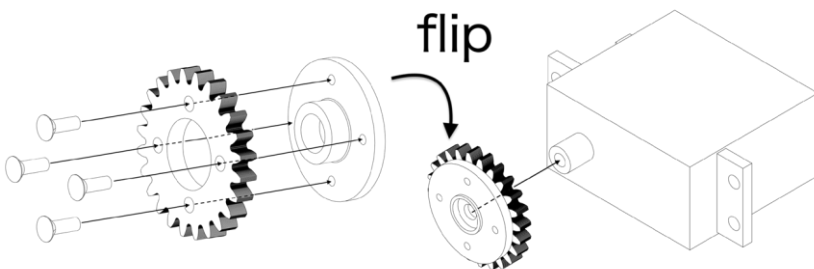


Side view

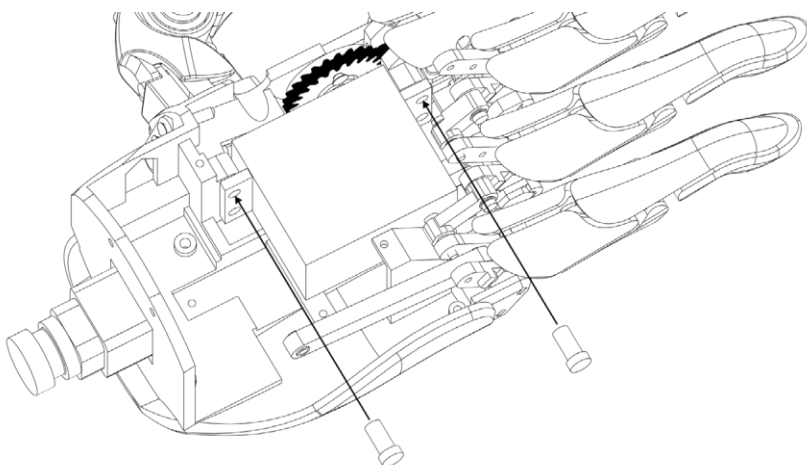
[18] The servo horn hole (the right end of the triple hole) **that is included in the servo motor size is** enlarged to 1.6 mm using four-pin screws.



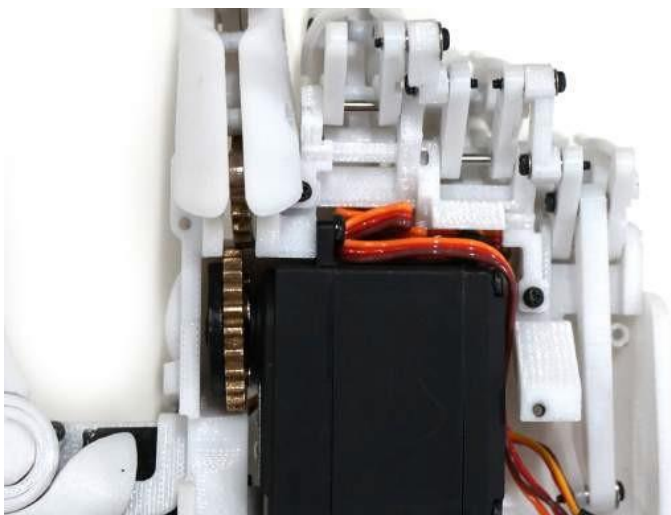
【1 9】 Connect the R-H-08 to the servo horn using 4 screws (M2L6).



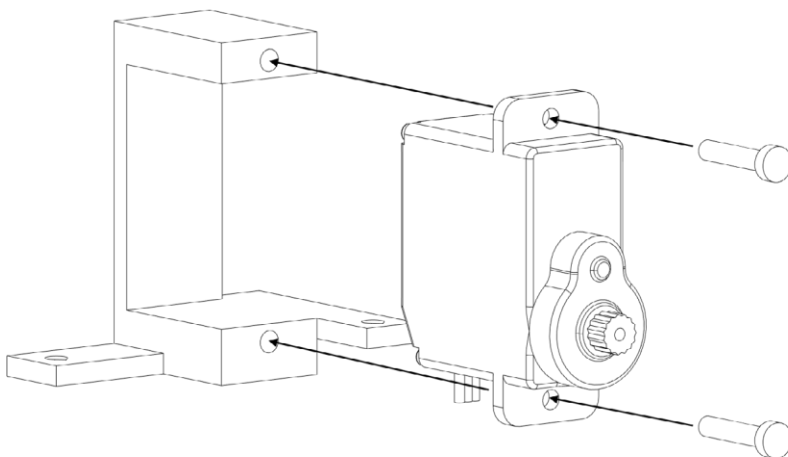
[2 0] Attach the large servo motor to the R-H-02 with two screws (M3L8) with your **index finger extended**, **being careful not to rotate the servo motor shaft**.



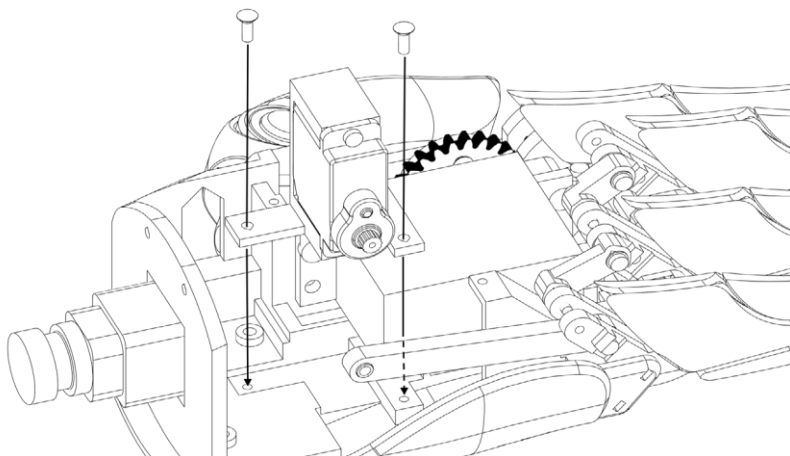
[2 1] Bring the wires together as shown in the figure below.

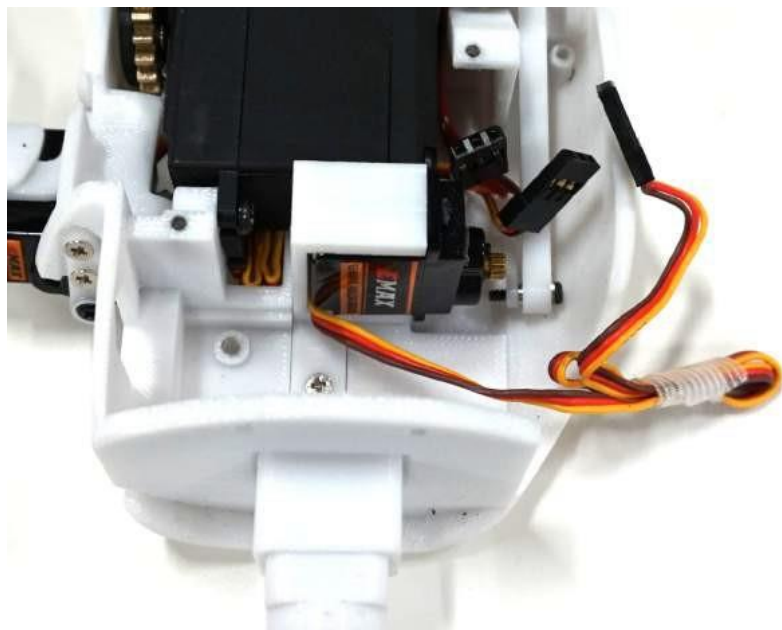


【2 2】 Attach the small servo motor to R-H-09 with 2 screws (M2L10). **At this point, the cable is**
Please be careful.

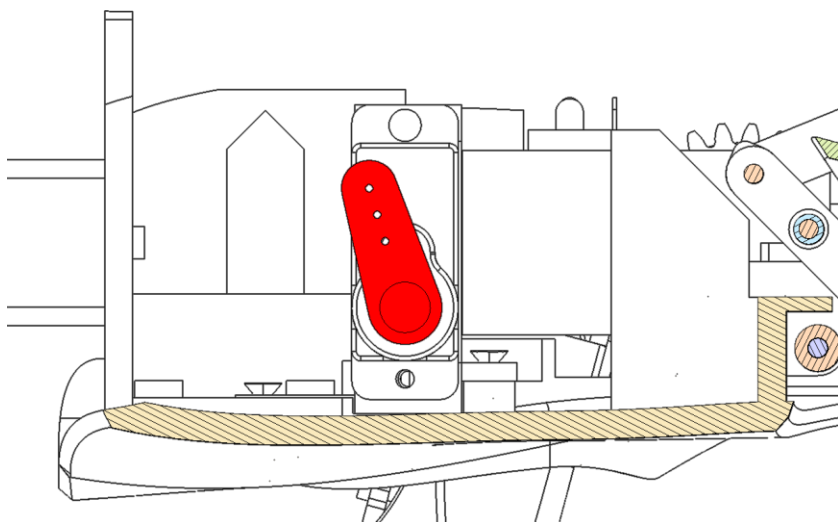
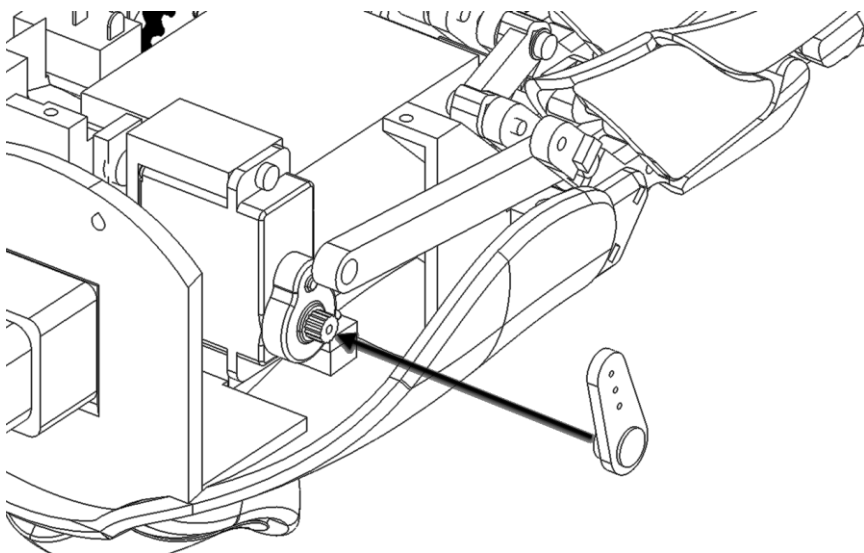


【2 3】 Secure this with two screws (M2L6) and wind the cable as shown in the second picture. **If you**
accidentally use the screw (M2L10) here, the screw will come out of the palm of your hand, so be
careful.

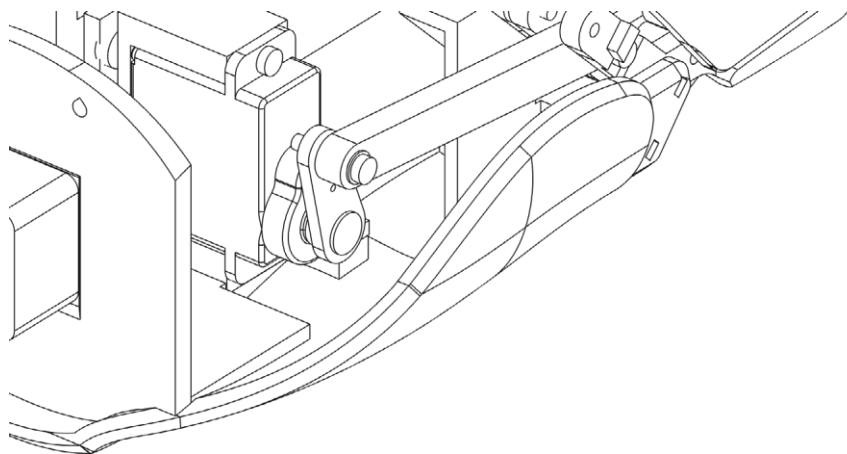
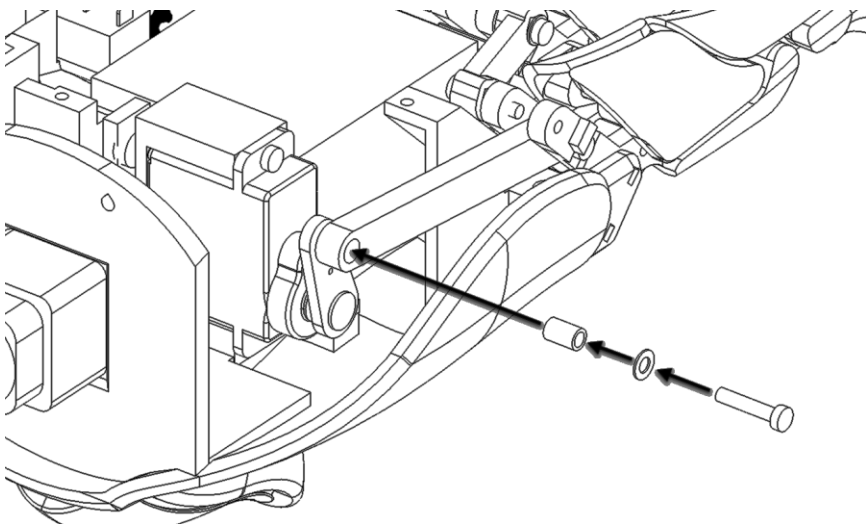




[2 4] The servo motor shaft is set to "10 Adjusting the voltage and setting the initial position of the servo motor"
Attach the servo horn to the servo motor at the angle shown in the diagram, being careful not to rotate from the state.



【2 5】 Attaches the R-H-07 to the servo horn with a thread (M2L10) through the washer and



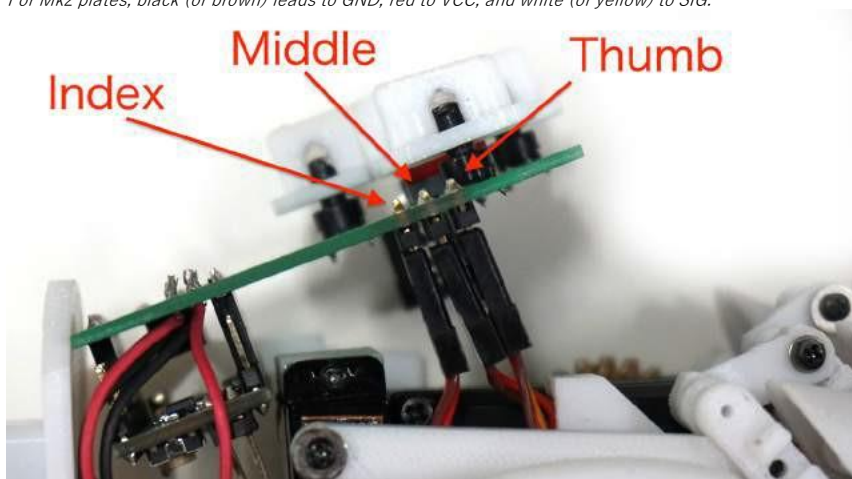
【2 6】 The servo motor cable is threaded as shown in the



【2 7】 Refers to the servo motor connector on the pin connector of the board. **Make sure the black (or brown) wire is facing out. The** pass connector marked Index has a large connector for servo motor, and the pin connector labeled Middle and Thumb has a small connector for servo motor.

For Mk1 boards, black (or brown) leads to GND, red to V+, and white (or yellow) to PWM.

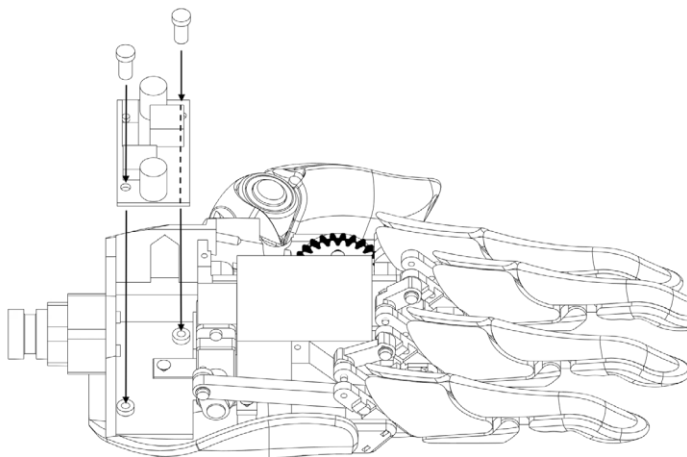
For Mk2 plates, black (or brown) leads to GND, red to VCC, and white (or yellow) to SIG.



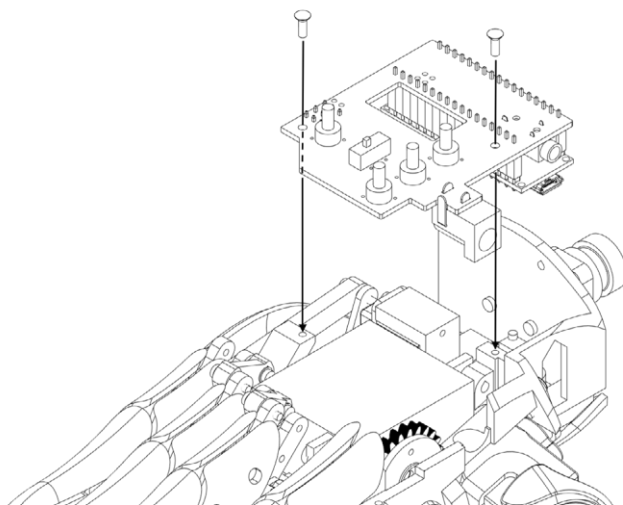
» next is the installation of the board.

For Mk1, see [28] and for M k2 boards, see [33] and later.

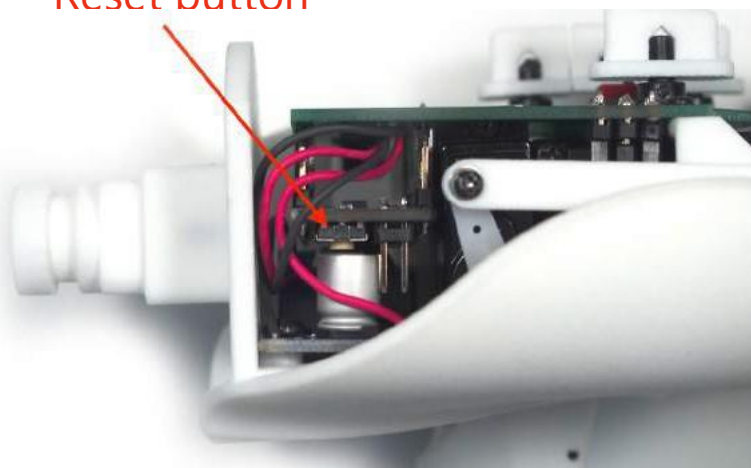
[2 8] Secure the CDC converter with two screws (M3L8).



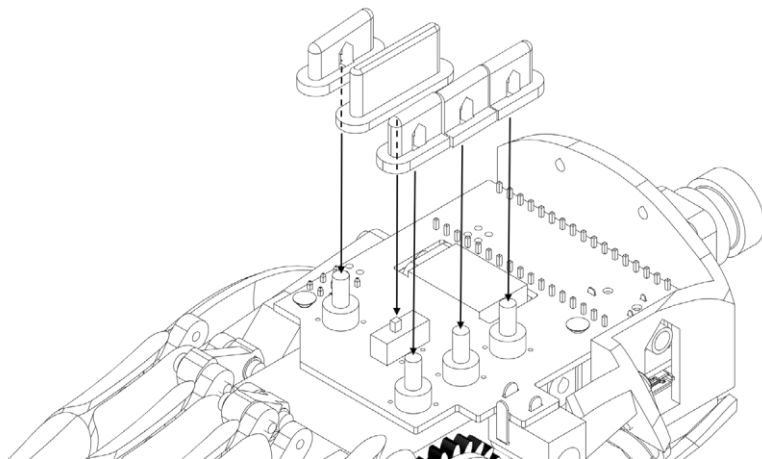
[2 9] Fix the plate using two screws (M2L6). At this point, if the rduino Micro is not fully inserted, or if the screws to fix the board are too tight, the reset button on the rduino Micro will still be pressed and HACKberry will not work properly, so be careful.



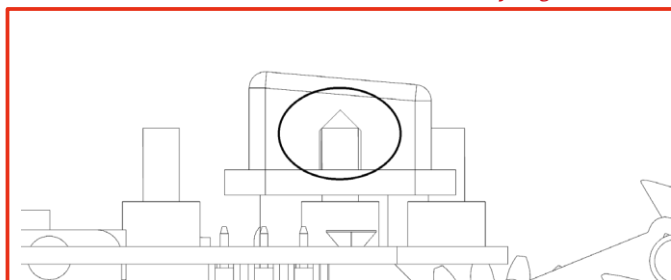
Reset button



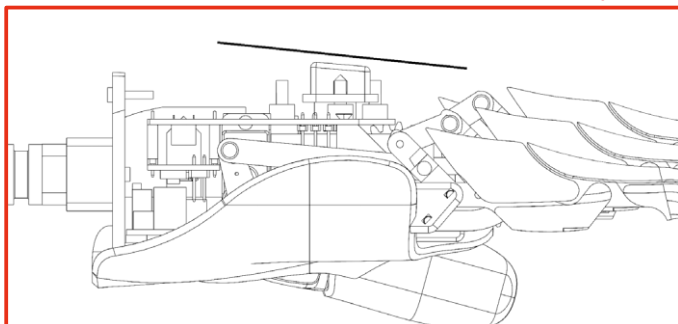
【30】 Install R-H-10, R-H-11, R-H-012, R-H-13, and R-H-14 as shown below.



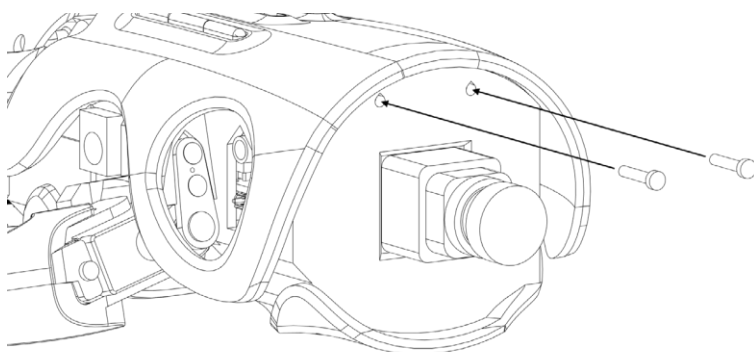
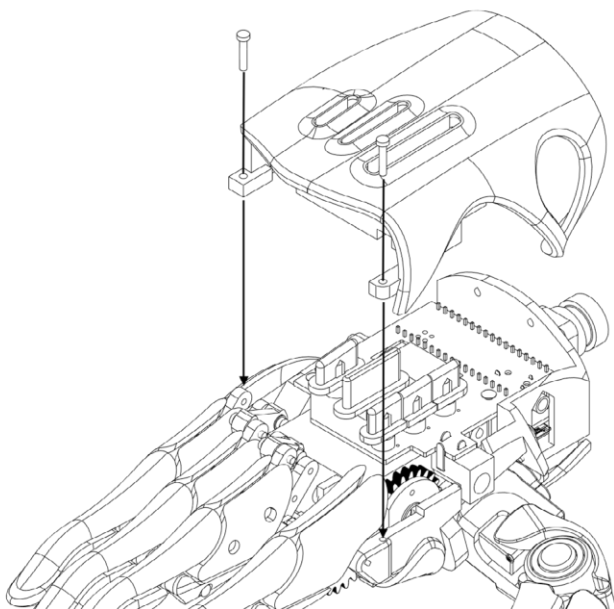
※ Please insert everything well into the back.



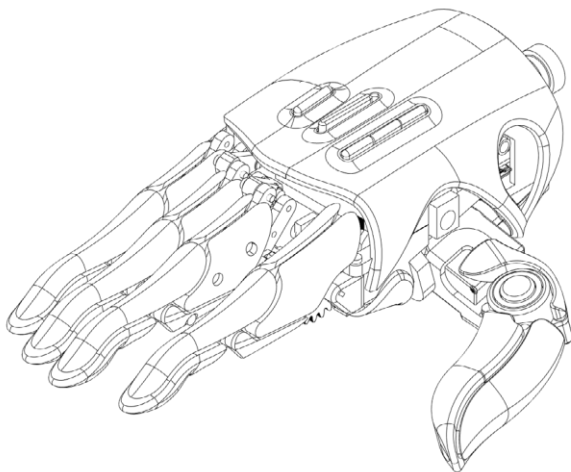
**For R-H-13, the short side should point toward your fingertips.*



[3 1] The R-H-15 is fixed using 4 screws (M2L10). It's easier to install if you bend your

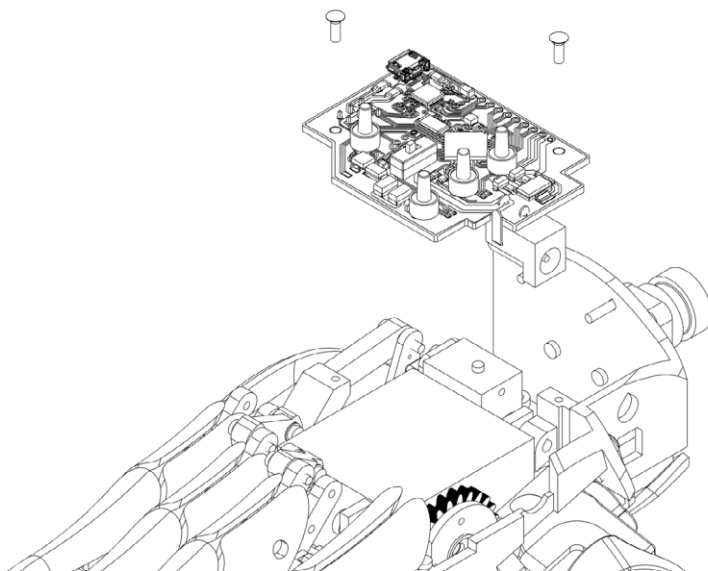


[3 2] Now the palm assembly is complete. Move on to the next chapter.

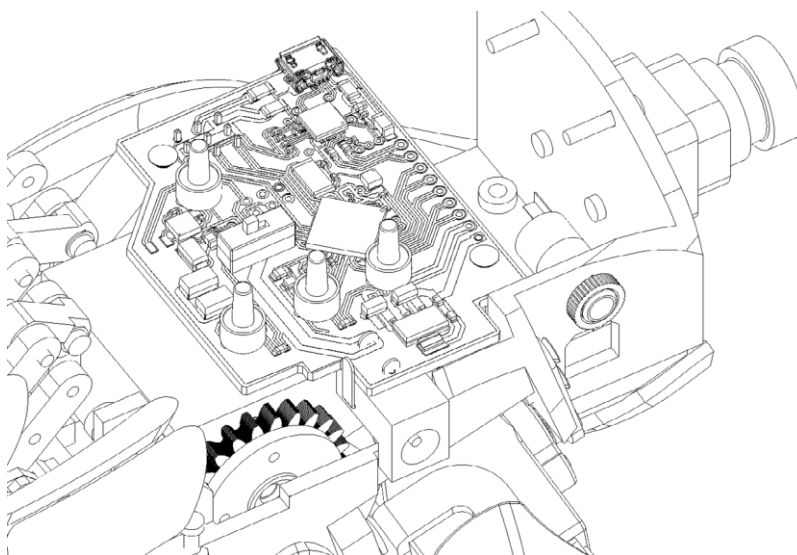
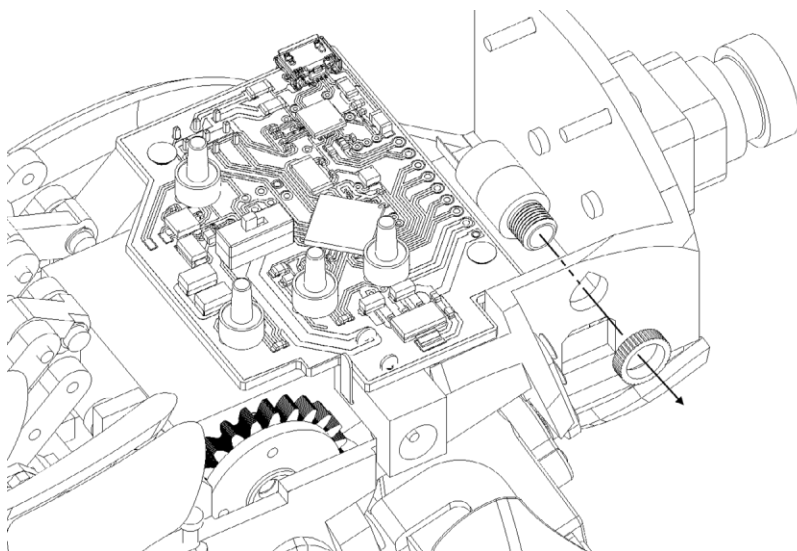


» for Mk2 boards, this is the case.

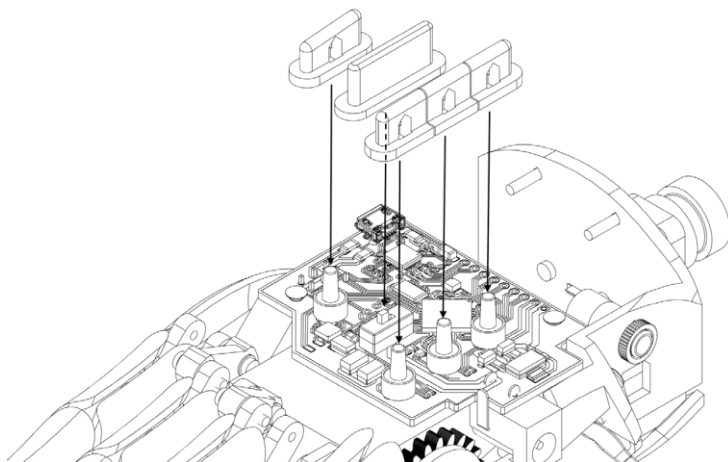
[3 3] Fix the ACKberry Mk2 handboard using two screws (M2L6).



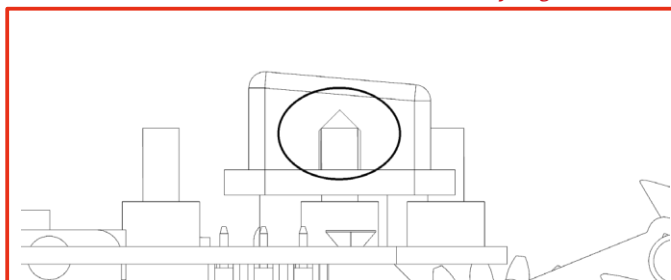
[3 4] Remove the nut from the stereo jack once, and then reinstall it by pinching the R-H-01.



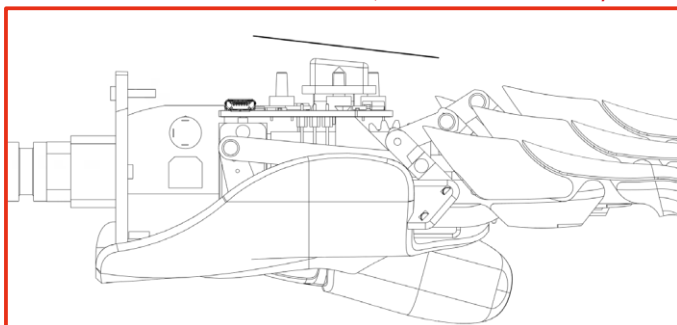
【3 5】 Install R-H-10, R-H-11, R-H-012, R-H-13, R-H-14 as shown in the figure below.



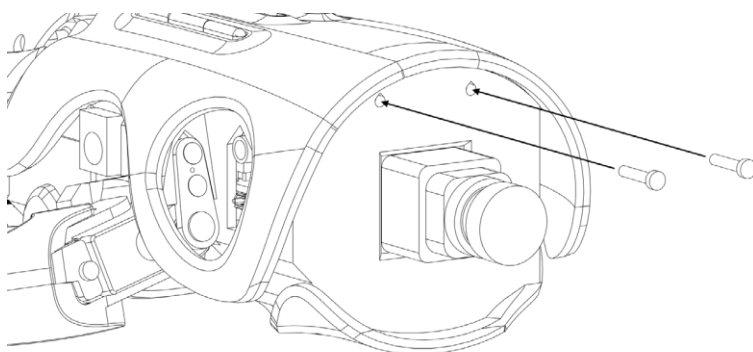
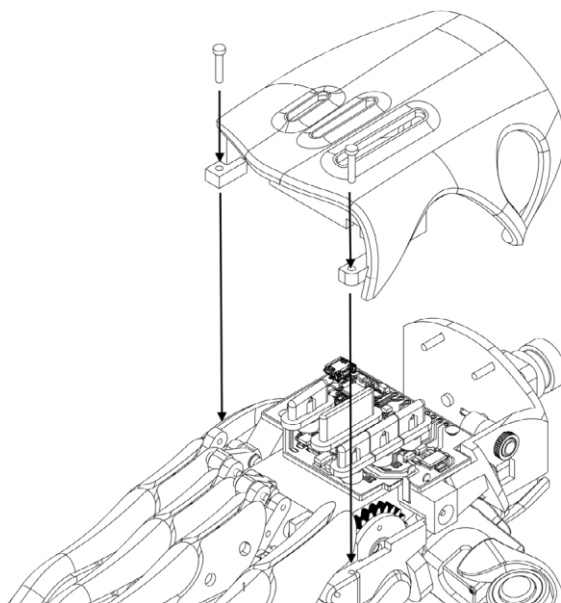
※ Please insert everything well into the back.



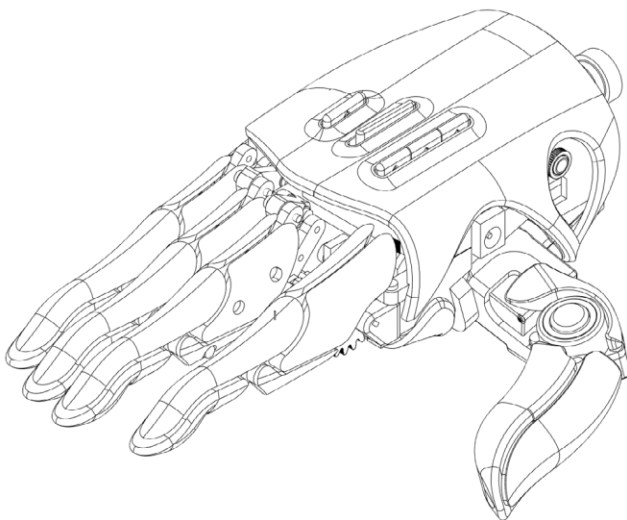
**For R-H-13, the short side should point toward your fingertips.*



【3 6】 The R-H-15 is fixed using 4 screws (M2L10). It's easier to install if you bend



[4 7] This completes the palm assembly.



Mounting

11 Wrist Mount

**In this section, we will mount the wrist unit. Note that the parts connected to the wrist unit are different when there is no dummy plug and when there is a dummy connector.*

No dummy bushing



With dummy sockets

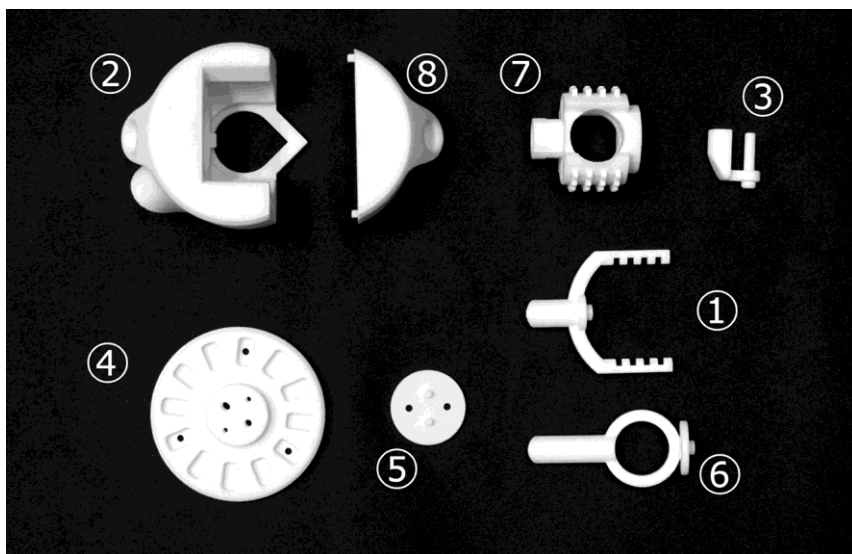


-Used parts—

1. R-W-01
4. R-W-04
7. R-W-07

2. R-W-02
5. R-W-05
8. R-W-08

3. R-W-03
6. R-W-06



-General parts used-

 x3
(x0)

 x2

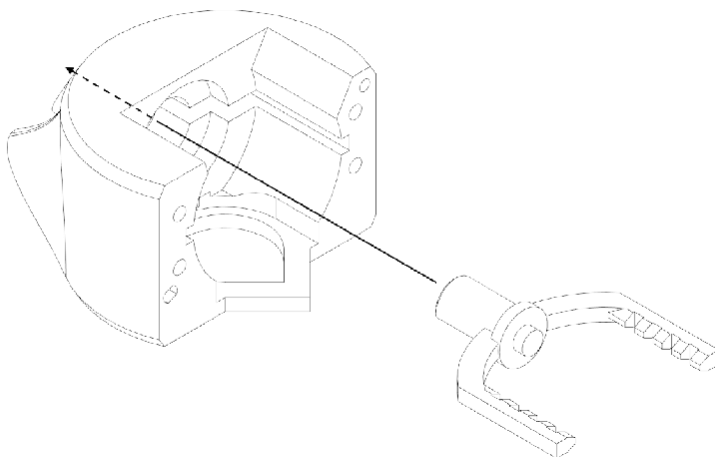
 x2

 x4

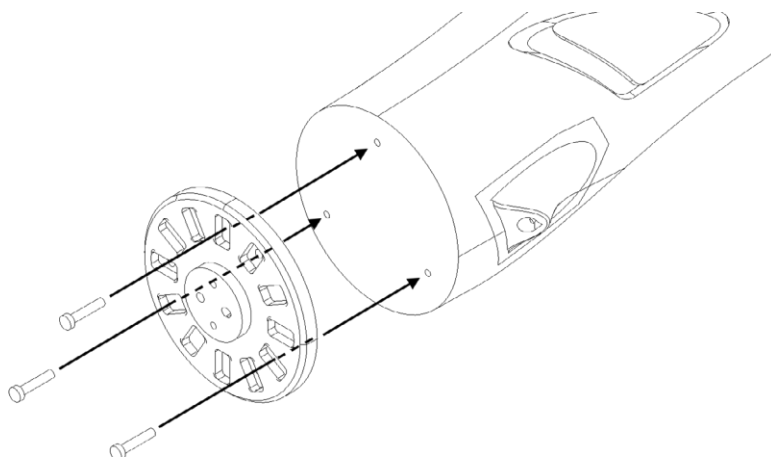
 x2

**The quantity in () is the quantity without a dummy socket.*

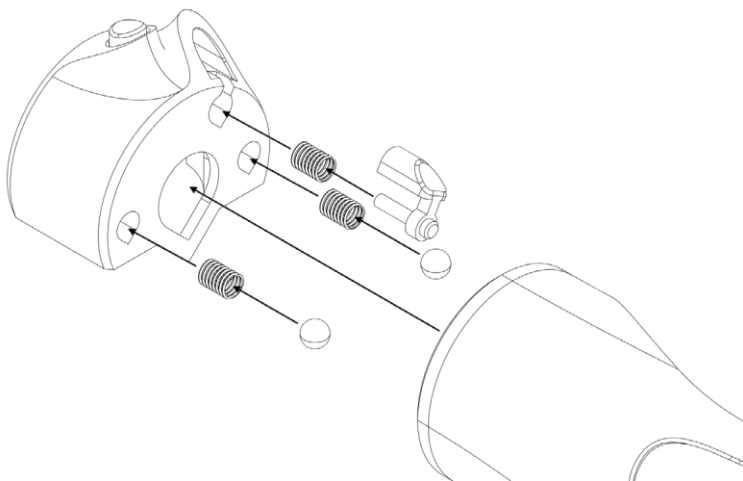
【1】 Connect R-W-01 into R-W-02. **The button face of the R-W-01 is diagonal, but the longer side is down.**



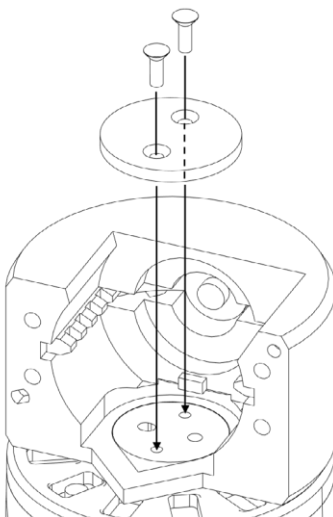
【2】 *(If there is no dummy tumbler, this process is not necessary)* Install the dummy tumbler into the R-W-04 using 3 screws (M2L10).



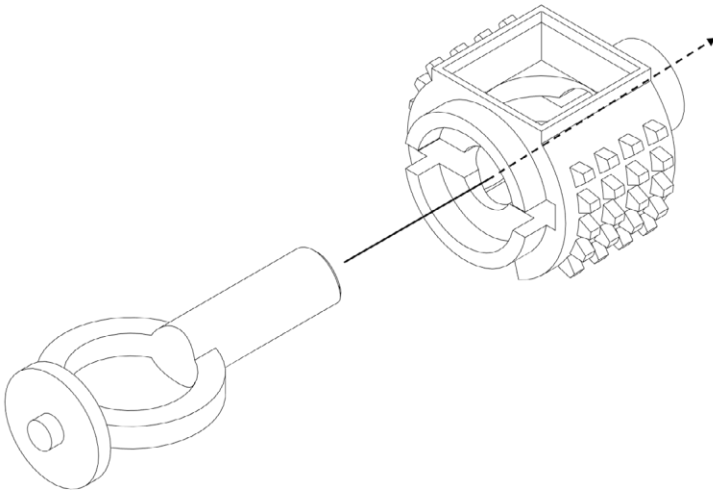
【3】 R-W-02 and R-W-04 have three coil springs and two steel balls, R-W-03.



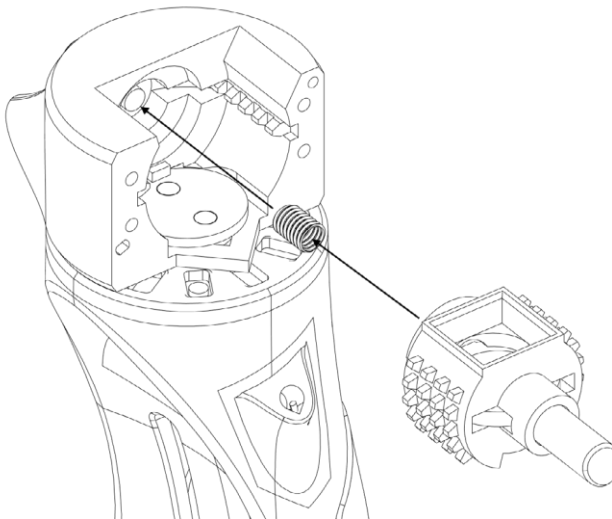
【4】 Attach the R-W-04 and R-W-05 with two screws (M2L6).



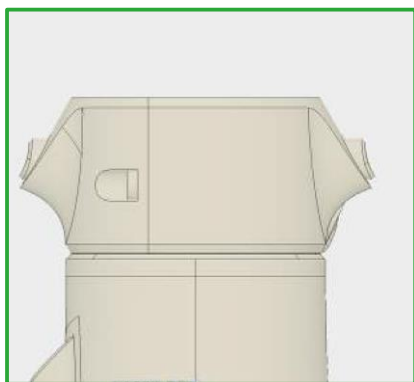
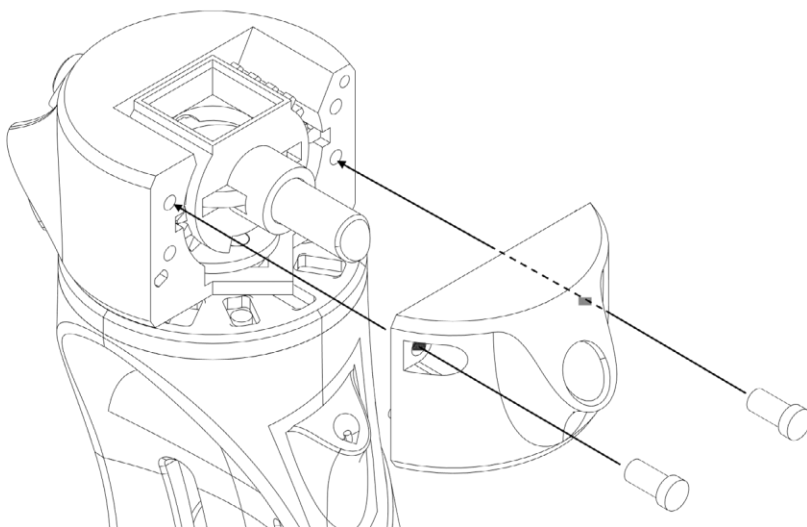
[5] Connect R-W-06 to R-W-07. **The face of the R-W-0 6 button is diagonal, but the longer side is down.**



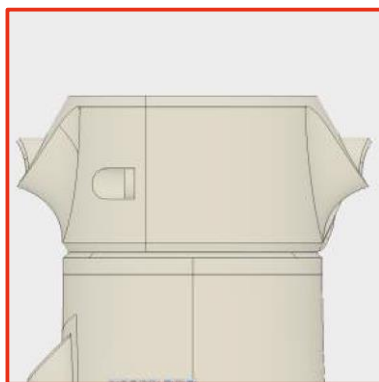
[6] Insert the coil spring and the parts you assembled earlier, as shown in the figure.



【7】 Install the R-W-08 and secure it with two screws (M3L8).

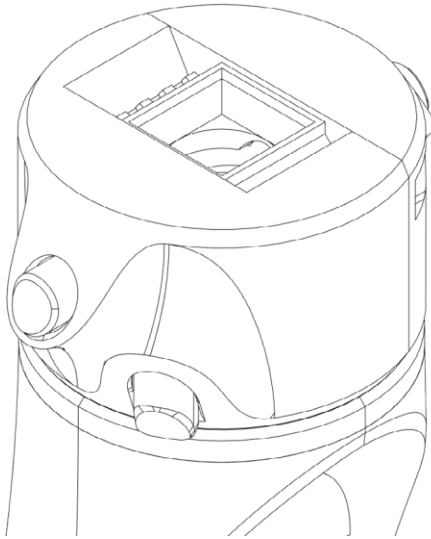


**Good example*



**Bad example*

[8] This completes the assembly of the wrist



Mounting

13 Mounting the Sensor



-Used parts—

1. C-S E-01
4. Velcro strap

2. C-SE-02
5. Stereo Cable

3. Sensor Board
6. Sensor Cushion



-General parts used-



【1】 Mount the S-Soring Plate to C-SE-01 in the orientation shown in the photo and fix it with two



【2】 Remove the release paper from the sensor cushion and stick it onto the sensor plate as shown in the photo.



[3] Remove the release paper on the other side and stick the C-SE-02 to the sensor



[2] Thread the Velcro strap through the C-SE-01 as shown in the photo. This is the end of the sensor assembly.



Mounting

14 How to Set the Starting

**In this section, the initial position of the engine is established. Due to individual differences in the motors or misalignments during installation, the angle of the fingers may be slightly out of adjustment, and this can be adjusted by changing the value in the program.*

If you are on your right hand, please proceed from this page, and if you are on your left hand, proceed from page 139.

For the right hand

[1] Insert the DC plug from the battery housing or the dummy plug containing the battery into the DC socket of the palm unit.



[2] Press the middle button on the back of your hand to activate the H ACKberry. Immediately after lighting, the thumb opens to one side of the hand and the other fingers are open, as shown in the figure. **However, due to individual differences in the motors and misalignment during installation, the fingers may not extend or, on the contrary, the "hehehe" sound may continue to sound when trying to overextend. If so, turn off the power immediately.**

**The Mk2 board takes a little longer from the time the servo motor current is turned on to start moving than the Mk1 board.*



As an example in this section, let's consider a case where the index finger is too bent and the thumb too open, as shown in the figure below.



[3] Open the program you used in "Upload 5 Programs" (5 pages 2). The rhythm of the finger angle is done by changing the six numbers shown in the figure.

```
Hackberry3.1
1) *
2) * Arduino micro code for HACKberry.
3) * Originally created by ext11 Inc.
4) * edited by Genta Kondo on 2017/6/11
5) */
6) #include <Servo.h>
7)
8) //Settings
9) const boolean isRight = 0; //right:1, left:0
10)
11) //For right hand, find optimal values of ThumbMin, IndexMax and OtherMax first.
12) //For left hand, find optimal values of ThumbMax, IndexMin and OtherMin first.
13) //Then, calculate the remaining values by following rules.
14) //Difference of ThumbMin and ThumbMax is 86
15) //Difference of IndexMin and IndexMax is 117
16) //Difference of OtherMin and OtherMax is 55
17)
18) const int outThumbMax = 150; //right:open, left:close
19) const int outIndexMax = 150; //right:open, left:close
20) const int outOtherMax = 150; //right:open, left:close
21)
22) const int outThumbMin = 30; //right:close, left:open
23) const int outIndexMin = 30; //right:close, left:open
24) const int outOtherMin = 30; //right:close, left:open
25)
26) const int speedMax = 6;
27) const int speedMin = 0;
28) const int speedReverse = -3;
29) const int thSpeedReverse = 15; //0-100
```

【4】 The value marked with MAX indicates the angle at which the finger is opened, and the value marked with M in indicates the angle at which the finger is closed. This time, my index finger was bent when it was supposed to be open, so I'll gradually increase the value of Max to open it wider.

In this case,

```
const int outIndexMax = 150;
```

What was

```
const int outIndexMax = 161;
```

When I uploaded the program, my finger was at the right angle.

```
17 |
18 | const int outThumbMax = 150; //right:open, left:close
19 | const int outIndexMax = 161; //right:open, left:close
20 | const int outOtherMax = 150; //right:open, left:close
21 |
22 | const int outThumbMin = 30; //right:close, left:open
23 | const int outIndexMin = 30; //right:close, left:open
24 | const int outOtherMin = 30; //right:close, left:open
25 |
```



[5] Next is the thumb adjustment. Because the thumb is too open, the value of the M-axis is gradually reduced.

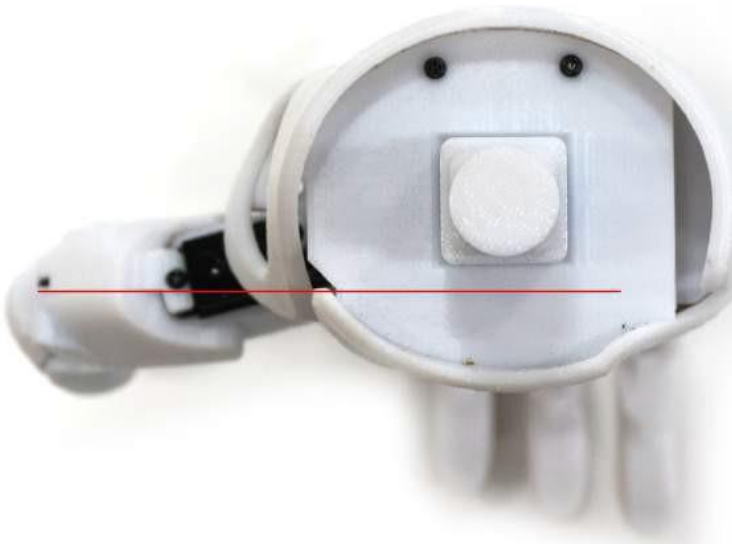
```
const int outThumbMax = 150;
```

What was

```
const int outThumbMax = 136;
```

When I uploaded the program, my finger was at the right angle.

```
17  
18 const int outThumbMax = 136; //right:open, left:close  
19 const int outIndexMax = 161; //right:open, left:close  
20 const int outOtherMax = 150; //right:open, left:close  
21  
22 const int outThumbMin = 30; //right:close, left:open  
23 const int outIndexMin = 30; //right:close, left:open  
24 const int outOtherMin = 30; //right:close, left:open  
25
```



[6] Next, adjust the value of the closed finger. For the angle difference between the closed and open states, refer to the following values for each motor.

```
-thumb:86  
-index:117  
-other:55
```

For example, since you only need to set the thumb aperture angle to 136, type 136-86. Set the other values as well and upload the program.

```
21  
22 const int outThumbMin = 136-86;//right:close, left:open  
23 const int outIndexMin = 161-117;//right:close, left:open  
24 const int outOtherMin = 150-55;//right:close, left:open  
25
```

For the left hand

[1] Insert the DC plug from the housing or battery plug containing the battery into the DC socket of the palm unit.



[2] Press the middle button on the back of your hand to activate the H ACKberry. Immediately after lighting, the thumb opens to one side of the hand and the other fingers are open, as shown in the figure. **However, due to individual differences in the motors and misalignment during installation, the fingers may not extend or, on the contrary, the "hehehe" sound may continue to sound when trying to overextend. If so, turn off the power immediately.**



As an example in this section, let's consider a case where the index finger is too bent and the thumb too open, as shown in the figure below.



[3] Open the program you used in "Upload 5 Programs" (5 pages 2). The rhythm of the finger angle is done by changing the six numbers shown in the figure.

```
Hackberry3.1
1) *
2) * Arduino micro code for HACKberry.
3) * Originally created by ext11 Inc.
4) * edited by Genta Kondo on 2017/6/11
5) */
6) #include <Servo.h>
7)
8) //Settings
9) const boolean isRight = 0; //right:1, left:0
10)
11) //For right hand, find optimal values of ThumbMin, IndexMax and OtherMax first.
12) //For left hand, find optimal values of ThumbMax, IndexMin and OtherMin first.
13) //Then, calculate the remaining values by following rules.
14) //Difference of ThumbMin and ThumbMax is 86
15) //Difference of IndexMin and IndexMax is 117
16) //Difference of OtherMin and OtherMax is 55
17)
18) const int outThumbMax = 150; //right:open, left:close
19) const int outIndexMax = 150; //right:open, left:close
20) const int outOtherMax = 150; //right:open, left:close
21)
22) const int outThumbMin = 30; //right:close, left:open
23) const int outIndexMin = 30; //right:close, left:open
24) const int outOtherMin = 30; //right:close, left:open
25)
26) const int speedMax = 6;
27) const int speedMin = 0;
28) const int speedReverse = -3;
29) const int thSpeedReverse = 15; //0-100
```

【4】 The value marked with M in indicates the angle at which the finger is opened, and the value marked with M ax indicates the angle at which the finger is closed. This time, my index finger was bent when it was supposed to be open, so I'll gradually reduce the minimum value to make it more open.

In this case,

```
const int outIndexMin = 30;
```

What was

```
const int outIndexMin = 12;
```

When I uploaded the program, my finger was at the right angle.

```
17  
18 const int outThumbMax = 150; //right:open, left:close  
19 const int outIndexMax = 150; //right:open, left:close  
20 const int outOtherMax = 150; //right:open, left:close  
21  
22 const int outThumbMin = 30; //right:close, left:open  
23 const int outIndexMin = 12; //right:close, left:open  
24 const int outOtherMin = 30; //right:close, left:open  
25
```



[5] Next is the thumb adjustment. Since the thumb is too open, the value of M in should be gradually increased.

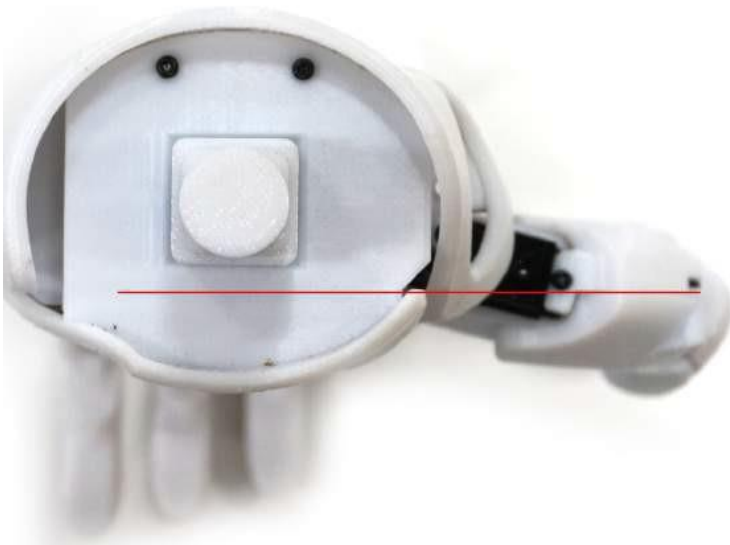
```
const int outThumbMin = 30;
```

What was

```
const int outThumbMin = 55;
```

When I uploaded the program, my finger was at the right angle.

```
17
18 const int outThumbMax = 150; //right:open, left:close
19 const int outIndexMax = 150; //right:open, left:close
20 const int outOtherMax = 150; //right:open, left:close
21
22 const int outThumbMin = 55; //right:close, left:open
23 const int outIndexMin = 12; //right:close, left:open
24 const int outOtherMin = 30; //right:close, left:open
25
```



[6] Next, adjust the value of the closed finger. For the angle difference between the closed and open states, refer to the following values for each motor.

```
- thumb:86  
- index:117  
- other:55
```

For example, since you just set the open angle of the thumb to 55, type 5 5+86. Set the other values as well and upload the program.

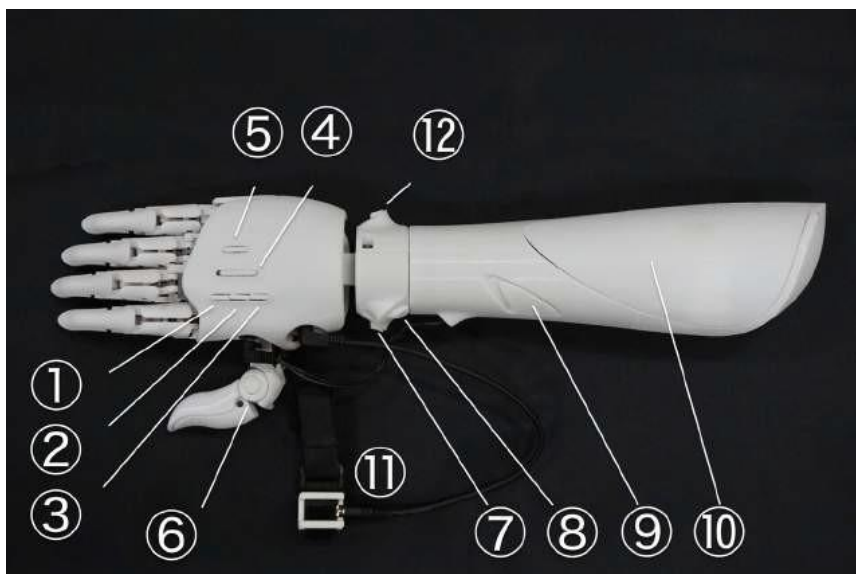
```
17  
18 const int outThumbMax = 55+86;//right:open, left:close  
19 const int outIndexMax = 12+117;//right:open, left:close  
20 const int outOtherMax = 30+55;//right:open, left:close  
21
```


3. Come on, move

- 1. **Name of each department** P146
- 2. **Installing batteries** P147
- 3. **Calibration** P148
- 4. **Manual assembly** P150
- 5. **Wrist flexion** P151
- 6. **Movement with myoelectric sensors** P151

How to use HACKberr

01 Names



1. Calibration button

4. Power switch

7. Wrist flexion and palmar
dorsiflexion buttons

10. Battery Door

2. Expand button

5. Fixed three-finger button

8. Wrist Rotate Button

11. Simple Pressure
Sensor

3. Thumb Rotate Button

6. Thumb lock button

9. Battery Door Lever

12. Wrist attachment
and removal button

How to use HACKberr and **Battery Installation**

Slide the battery door lever to open the battery door and install the battery. Slide it through the slot and make sure it clicks.

(When removing, press C-B-02 and slide it in the opposite direction.))



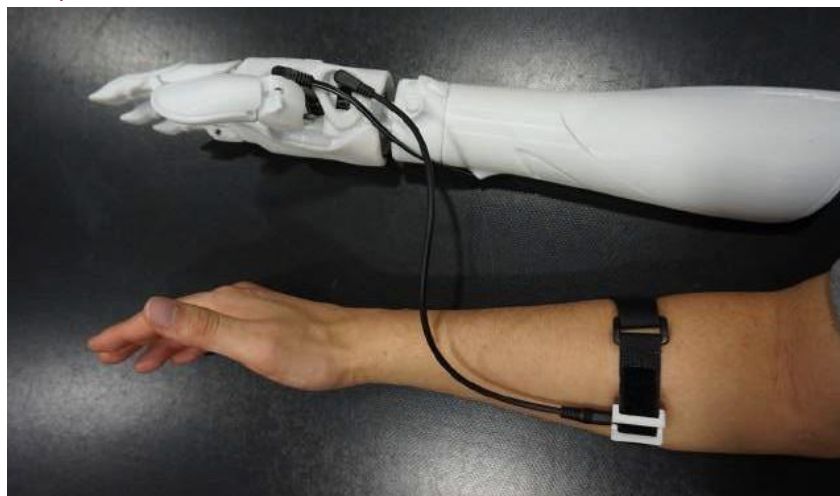
How to use HACKberry and

03 Calibration

Insert the DC plug into the DC connector as shown.



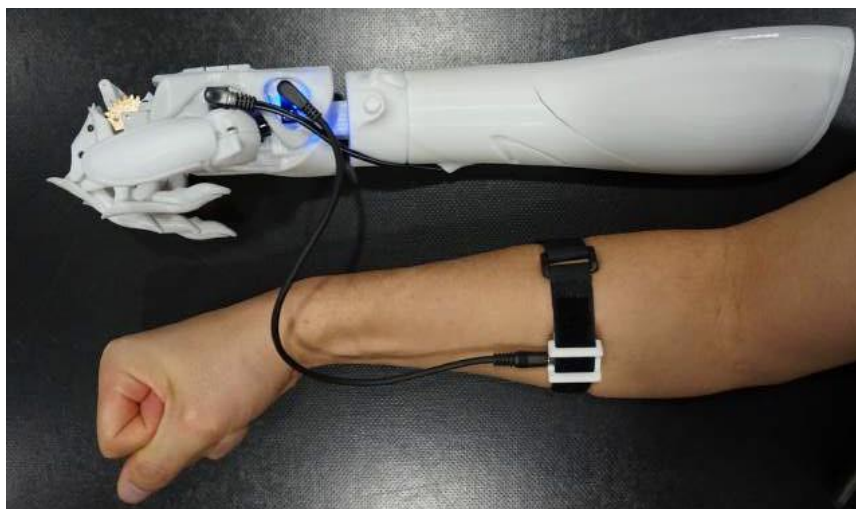
Place the flat part of the sensor against the skin (the part with muscles as much as possible) and attach it to the arm with a Velcro strap that tightens it gently. After that, connect the HACKberry and the sensor with an audio cable. **Be careful to insert it firmly into the depths.**



Then, calibrate it.

Calibration tells HACKberry how much muscle lumps vary from user to user.

While you are powered on, press the calibration button and repeat the "squeeze" → "release" motions several times for 4 seconds.



When the calibration is finished, the index finger to the pinky finger moves in unison.

How to use HACKberr and

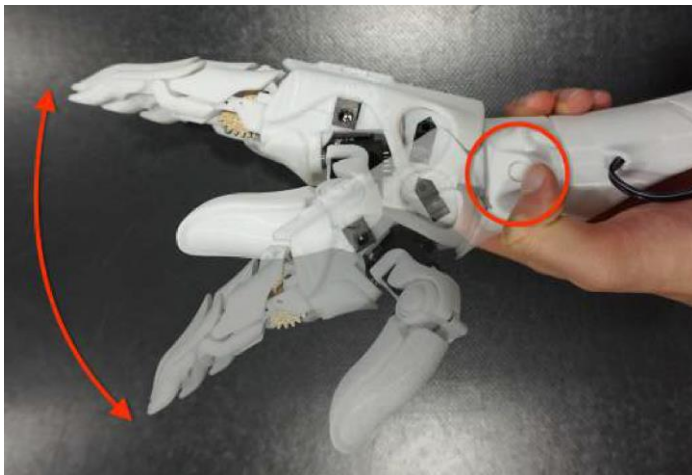
04 Putting on and taking off your

Press the hand join button and the detach button to attach and detach your hand and wrist. The alignment of hands and wrists can be adjusted every 90 degrees.



How to use HACKberr and **05 Palmar dorsiflexion and radial-**

When you press the wrist flexion and palmar dorsiflexion buttons, you can tilt your wrist in the direction of palmar dorsiflexion or radial-ulnar flexion.



Press the wrist induction button to rotate your hand in the direction of the figure.

How to use HACKberr and **06 Wrist Inward and Absorption**



4. Have more fun with HACKberry

1. **Introduction** P154
2. **HACKberry handboard Mk2 仕様** P155
3. **Manipulation of HACKberry by myoelectric sensor** · P158
4. **Acetone surface treatment** P162
5. **Nylon dyed color** P168

This chapter presents examples of H A C K B E R R

R R CUSTOMIZATION customization. In verse 1,

customization was improved compared to the Mk 1

plate.

H A C K b e r r y h a n d b o a r d M k 2 基 板 の 特 徴 や 仕 様 に つ い て 説 明 し ま す 。

In Section 2, we will introduce how to operate H A C K b e r r y as an electromyoelectric prosthetic hand using an MK 2 plate and a commercially available myoelectric sensor (2 ch).

In Section 3, we'll discuss how to treat acetone surface to improve the visual quality of 3D printed parts for the home.

In section 4, the external 3D printing service was used to print

We'll show you how to dye 3D printed parts from nylon materials.

Customize

0 2 HACKberry handboard Mk2 仕

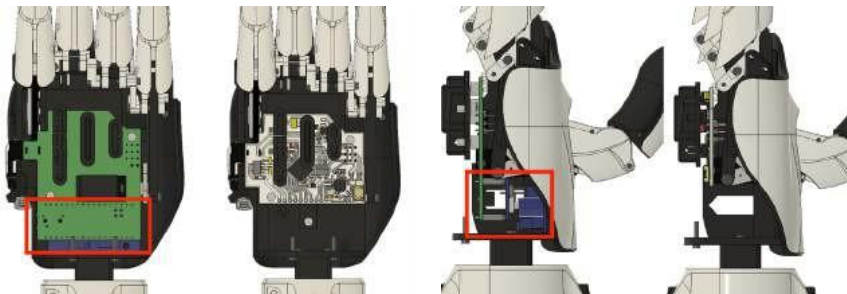
HACKberry handboard Mk2 was developed primarily with the aim of improving customization over the MK 1. The next two are the main improvements.

1. Space security through motherboard integration
2. Simplified customization with additional expansion ports



[1] Ensure space by integrating tables

The functions of the Arduino Micro and D C D C converters that were separated on the MK 1 were installed on a single motherboard. This has reduced the space dedicated to electrical circuits in the enclosure and, in turn, increased the rate of installation of additional sensors and modules.



**Top view of spatial comparison (Left Mk1, Right Mk2)*

**Spatial comparison side view (Left Mk1, Right Mk2)*

[2] Simplified customization by adding expansion ports

Custom power and communication port on the bottom of the base (analog force, I2C communication, Syrian communication)
We have added . This makes it easy to interface with other sensors and modules.



**Input and communication ports for customization on the bottom of the board*

The method is detailed below. The M k 2 board is a compatible machine, and the Processa has a
Since A T m e g a 3 2 8 P - A U is used, the allocation of pines is the same as in A T m e g a 3 2 8 P .

Pins	Inputs	Calibration Button	A6
		Displacement Origin Button	A0
		Negative Ternary Button	D10
		Positive Ternary Button	A7
		Potentiometer	A1
	Outputs	Manual Advance Button	D5
		Negative Ternary Button	D11
		Positive Ternary Button	D9
	Pines expansion	5V	5V
		Ground	GND
Input voltage	Programming configuration	Arduino Nano	
		ATMega328P-AU	
	Microprocessor VS	AVRISP MKII	
		ICSP	
Input voltage	Min. Input voltage	7V~12V	
	Max. input voltage	20V	
	Communication speed	115200	
	- DDC capacitors convert any interpeak to at least 12V before 12V. - Regular capacitor with three terminals at less than 20V. - PTC fuses are not–desigted–Index & Middle hest up to 300mA.		
Servo pins	Index & Middle pins can deliver up to 500mA.		
Servo pins	Index & Middle pins can be up to 500mA.		

※ HACKberry Mk2 Hand Board 詳細仕様

Customize

03 Operation of the HA CKb err and myoelectric sensor

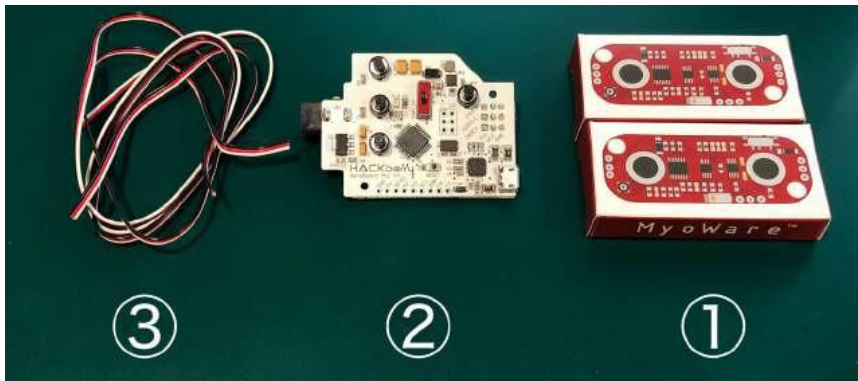
[1] Introduction

While many prosthetic hands on the market use myoelectric sensors, the HACKberry uses a simple pressure sensor. The characteristics of simple pressure sensors are low noise, ease of operation, and extremely low cost. On the other hand, it is difficult to multi-channel and can only measure a single signal in one channel. If you want to operate the HACKberry using a myoelectric sensor like a commercial prosthetic hand, check out this section to customize it.

[2] What is myoelectricity?

When a person moves a muscle, the electrical signals emitted by the brain are transmitted to the muscles through the nerves. This electrical signal leaks slightly to the surface of the skin. The signal observed by electrodes attached to the surface of the skin is myoelectricity.

[3] Parts used



1. MyoWare 2 PCs

Myoelectric sensors manufactured by Advancer Technologies. It has built-in myopotential amplification and smoothing circuitry, which makes it relatively easy to measure myoelectric signals.

2. HACKberry Mk2 Hand Table

Use the audio jack without plugging it in.

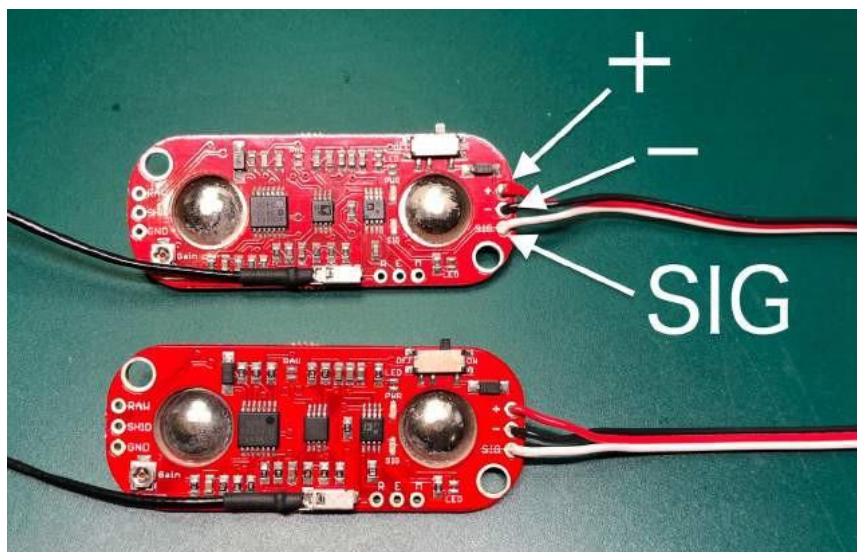
3. 2 x 3 row flat cables of 50 cm

In this section, we'll use two types of M and or W to r, each of which is palm flexion (wrist flexion to the palm side) and dorsiflexion.

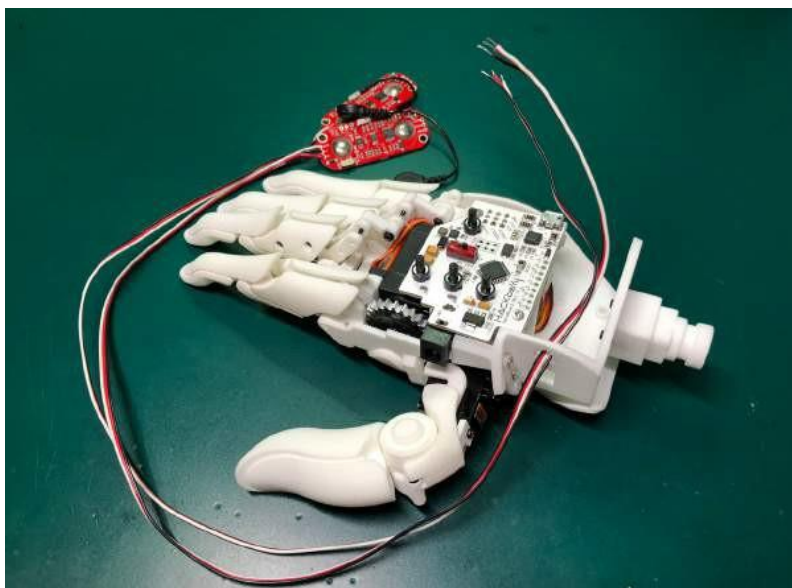
(bending the wrist towards the back of the hand) to detect the movement of the hand and perform the operation of HACKBERRY through the two channels.

[4] Method of implementation

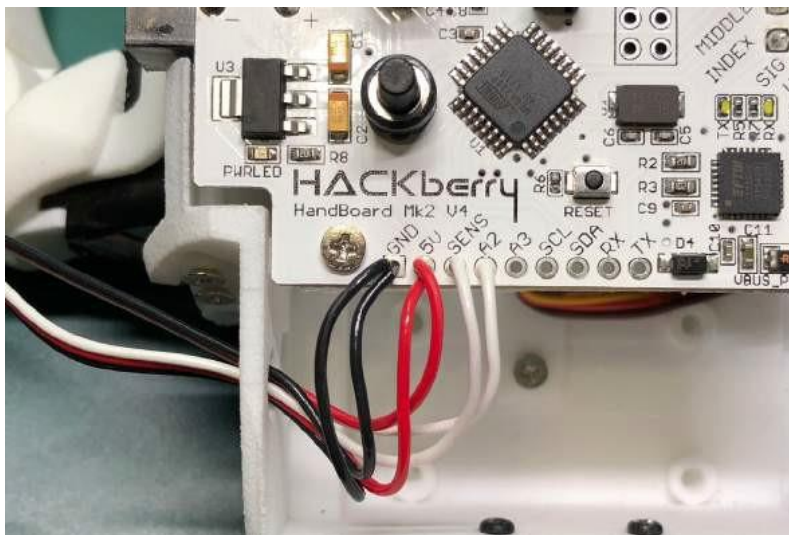
Solder the red, black, and white lines at the three locations of MyoWare's +, E, and S IG tracks, respectively.



Connect the HACKberry's Mk2 handboard to the HACKberry and thread the ribbon cable through the hole in the audio jack as shown.



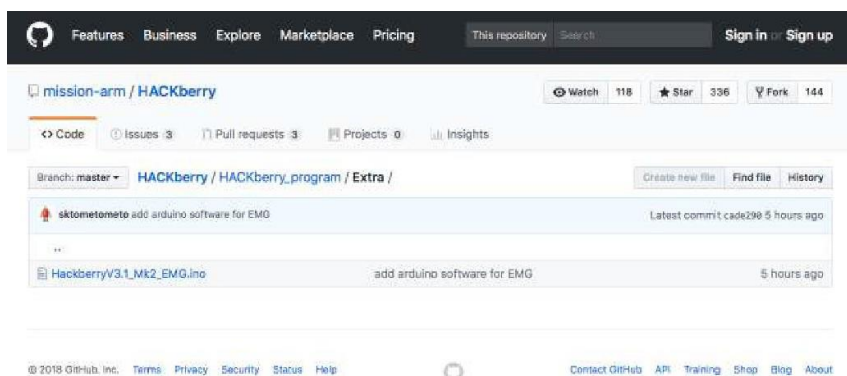
Solder the ribbon cable to the expansion port of the HACKberry Mk2 handboard. As shown in the figure, connect two black wires to GND, two red wires to 5V, one white wire to SENS, and one white wire to A2.



This is the end of the hardware work.

Next, download the program for myoelectric manipulation from Github.

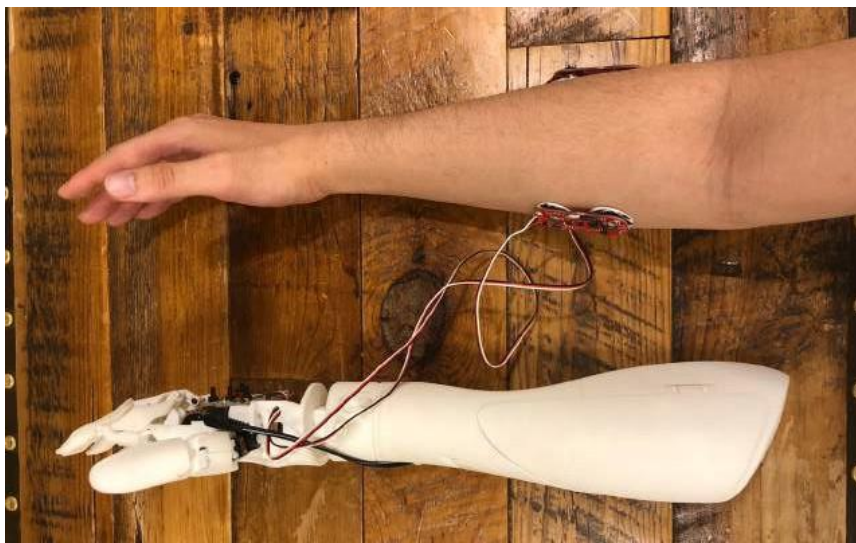
Please refer to [02 How to get parts] (page 19) to visit HACKberry's Github page and download the "HackberryV3.1_Mk2_EMG_180412" under the chart.



Then, refer to [0 5 Upload Program] (page 52) and type the program you downloaded earlier into the HACKberry Mk2 handheld table.

[5] How to trade

MyoWare White Cable Flexors Connected to the SENS on the External HACKberry Handboard Mk2 Port (the muscle that rises during palm flexion) and attaches the other MyoWare to the extensor muscle (the muscle that rises during dorsiflexion). For detailed sensor mounting positions, please refer to the



MyoWare instruction manual.

Turn on the HACKberry and press the calibration button. After pressing the button, the calibration will be performed for 4 seconds, during which you will alternate forcing the flexors and extensors to preset the width of the EMG signal in HACKberry. Once the calibration is complete, your fingers flex when you press on your flexors and extend when you press on your extensors.



YouTube筋電操作動画

<https://youtu.be/pDvJqjNLdBo>

03 Surface treatment with vaporized

1] Introduction

FD M (hot melt stacking), which is commonly used in home 3D printers, can print three-dimensional shapes at low cost, but due to the nature of stacking layers one by one, the roughness of the surface layer is noticeable. In that case, surface treatment with vaporized acetone can be used to smooth the surface. The advantages and disadvantages of this method are listed below.

Pros

- The time required is short (about 1/10 compared to the method of grinding and smoothing the surface with sandpaper, etc.)
- You can manage hard-to-archive details with it.
- Although the stacking step is large during 3D printing, it is effective, so the printing time can be greatly reduced.
- As the surface is molten and uniform, it becomes a kind of melting state, which reduces the weak bond between layers and improves the overall strength.

Disadvantages

- Organic solvents are used, so you have to be careful when using them.
- Due to the nature of melting and smoothing the surface as a whole, the details may be crushed, and it is not suitable for adjustment and pressure that require precision.

As a countermeasure against these disadvantages, there are methods such as separating the exposed parts of the housing from other structural and mechanical parts and treating only them, or placing the parts to be treated superficially in an acetone-prone position (described below).

2] Precautions

Acetone is an organic solvent. It is designated as a Class 4 hazardous substance (first water-soluble petroleum class 2) by the Fire Service Act. The characteristics of acetone that should be paid special attention to are **"strong flammability (ignition at room temperature)"** and **"high volatility"**. Please note:

- To avoid inflammation and excessive inhalation, **drive in a well-ventilated** and ventilated place to **avoid the accumulation of vapor**.
- If it spills from the container while you're working, wipe it off quickly with a cloth.
- If acetone vapor gets into your eyes, wash them quickly with plenty of water. Even if there is no pain or abnormalities, it is recommended to seek medical attention at a specialized medical center immediately.
- It is also advisable to seek medical attention when swallowing by mistake or inhaling gas.
- After using acetone, be sure to wash your hands and gargle well.

3] Equipment

- Sealed container (material that does not dissolve in acetone. A 5L polyethylene container is used here.)
- Aluminum foil
- Paper towels
- アセトン (ホートク 純アセトン1 L <http://amzn.asia/4U79i9l>)
- Pipettes

4] Procedure

[1] Line the airtight bottom of the container with aluminum foil. Cut the foil into large pieces so that it sticks out of the outside of the container. This is to avoid touching the pieces with the foil rim when you take them out of the container later.



[2] Place the pieces on the foil. At this point, the **surface you want to treat is at the top**. This is because the vaporized acetone falls on the parts from above, so the top is more likely to dissolve.



[3] Fold 3 sheets of paper towels in half, leave them as shown in the photo, and use a pipette to soak them in 15 ml of acetone.





[4] Immediately after, cover and start the timer. If printing with a 0.3mm pitch, please take about 1 hour.



【5】 After the set time, open the lid and take out the parts. Right now, the pieces are soft, **Grab the edge of the foil without directly touching the pieces and pull them out.**



【6】 The surface can be seen to be melted and smoothed. Leave it in a well-ventilated place for 4-5 hours.



[7] If surface treatment is the goal, at this point it's more or less smooth, but if you're
When blowing a matte spray, etc., the texture will return to the original.

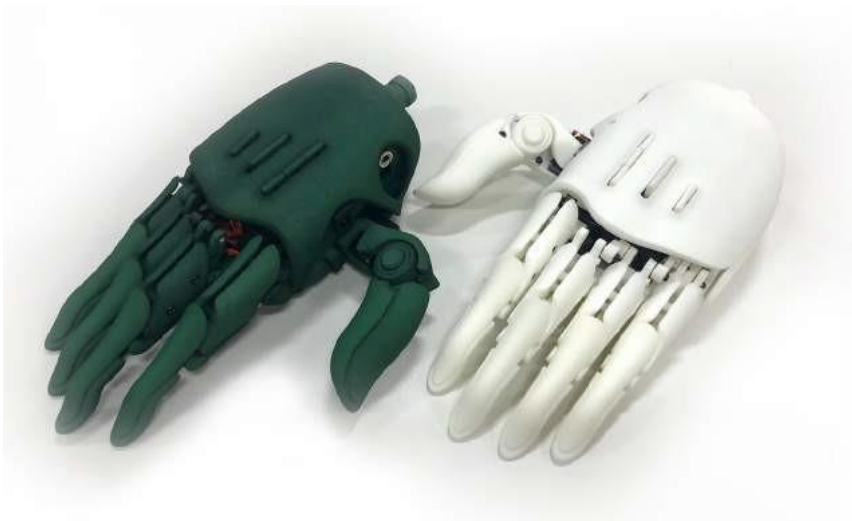


**Diagram before the plant*



**Illustration after nuance*

04 Nylon dyeing



[1] Introduction

In this section, one of the ways to color HACKberry is to dye 3D printed parts made of nylon using commercially available resin dyes.

3D printed parts made of nylon material are printed in white, making them easy to dye. (Not only that, nylon material is cheap and high-strength, so we use it as standard material for HACKberry.) Please note that nylon is not compatible with FDM (Hot Stacked Fuse) 3D printers for home use. (The resin in 3D printers for home use is usually A, B, S, and P, PLA.) A, B, S, and PLA are not suitable for dyeing.)

Nylon is compatible with 3D printers, which use the SLS (powder sintering) method, a method in which the powder is baked and laminated layer by layer with a laser. As SLS 3D printers are not widely used for home use, it is recommended to use a third-party 3D printing service (using an external service allows you to choose from a variety of materials other than nylon). We use DMM.com 3D printing services.

DMM.com 3Dプリントサービス:<https://make.dmm.com/print/>

3D printing services that handle nylon may also offer dyeing services. Here I want to adjust my favorite color to my liking, so I'll dye it myself using commercial resin dyes.

[2] Equipment



- | | | |
|--|-----------------------------|-----------|
| 1. Diluted dyeing solution with resin dye | 2. Stainless Steel Strainer | 3. |
| Paper towels | | |
| 4. Rubber gloves | 5. Measuring cup | 6. Funnel |
| 7. Vegetable Chopsticks | 8. Thermometer | |
| 9. Induction heater (to heat the dyeing solution, gas cooker is also acceptable) | | |
| 10. Stainless steel pot (size that can hold a strainer) | | |

[3] Dyeing Flow

[1] First, prepare a dye solution. This time, we used the "Dye SDN Resin" from Osaka Chemical Products Co., Ltd. Dilute the dye 20 times using a measuring cup to determine the number of parts you want to dye for soaking. Keep in mind that the dilution concentration varies depending on the dye.



[2] Heat the dyeing solution in a stainless steel pan.



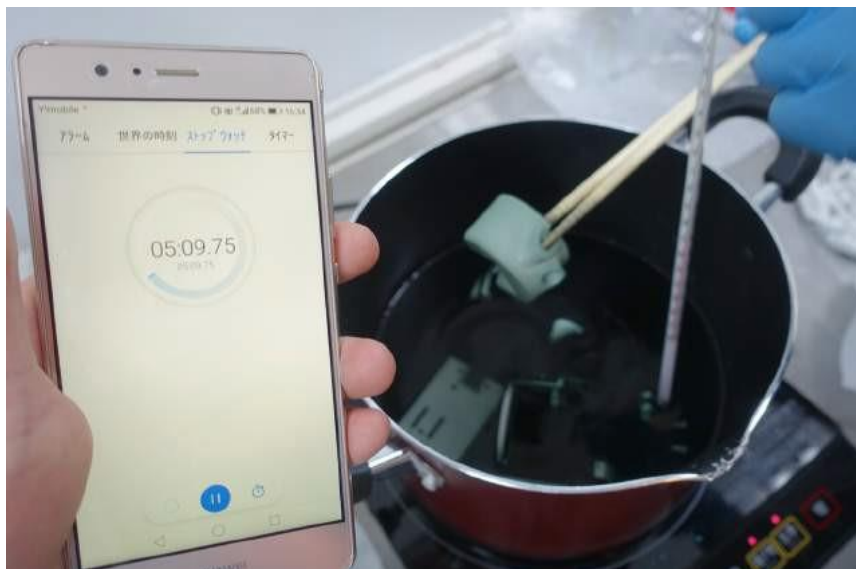
[3] The standard dyeing conditions for "resin dye SDN" nylon are $70 \pm 10^{\circ} \text{C}$ for 3~5 minutes. The temperature of the dyeing solution is 70

When the 10°C range $\pm$ reached, dip the 3D-printed sculpture into the dye solution and start the timer. Keep it at $70 \pm 10^{\circ} \text{C}$ of the specified temperature while dyeing.

Also, remove dyeing fluid sparingly.



[4] The manufacturer's recommended dyeing time is 3-5 minutes, but in our environment it didn't seem to dye much even though I took it apart after 5 minutes, so I dyed it for about 20 minutes.



※ After 5 minutes



※ After 20 minutes

[5] Once the dye is dyed your favorite color, remove the 3D printed sculpture from the dye solution and place it on paper towels to dry. This completes the dyeing of the 3D



[6] Dyeing fluid contains substances that are harmful to the human body, so it **cannot be disposed of in the sink even after dyeing has finished**. The dye liquid can be reused, so use a funnel to store it in a plastic bottle. If you want to dispose of it, dilute the dye solution more than 200 times and pour it into the sink.



5. Is it a failure? If you believe

Troubleshooting

Is it a failure? When I thought

Won't turn on

- ◆ Make sure the battery is properly installed.
- ◆ Make sure the DC plug is securely inserted into the DC connector on the board.
- ◆ Please check if the welded position is correct.
 - ➔ Are DC plug wires soldered correctly to the positive and negative sides?
 - ➔ Are the VC C (or V+) and GND on the board shorted?
- ◆ Place a tester on the battery terminal to check if the voltage is displayed correctly.
 - ➔ If the circuit is shorted, the protection circuit in the battery may be working and the voltage may not be generated. Please contact each battery manufacturer for recovery methods.

Cannot be calibrated

- ◆ When you press the calibration button, check for a clicking sensation.
 - ➔ The button may remain pressed for reasons such as tilt and welding. Redo the solder or scrape the parts of the box to repair it.
- ◆ Make sure you have written the correct version of the program (left or right, m k1 or mk2).
- ◆ Please check if the initial position of the servo motor is correct (see page 134 for more details.))
- ◆ Make sure the servo motor pin connectors are connected to the board in the correct orientation. (See page 112 for details.))
- ◆ Check that the sensor is connected correctly.
 - ➔ Is the audio cable securely connected from the rear?
 - ➔ If there is no response even after connecting successfully, the sensor or the audio cable itself may be damaged.

Fingers don't bend well

- ◆ Check if the initial position of the servo motor is correct. (See page 134 for details.))

Noise from the servo motor

- ◆ Check if the initial position of the servo motor is correct. When the servo motor tries to move in a direction that it cannot structurally rotate, it generates abnormal noise. (See page 134 for details.))
- ◆ Please check if the voltage setting of the DCDC converter is correct. If the voltage is not adjusted, excessive voltage will be applied to the servo motor, which will damage the servo motor. (See page 92 for details.))

Suddenly, the servo motor stopped working

- ◆ (In the case of MK1) Check if the polyswitch is inserted correctly.
 - ➔ If not inserted, the servo motor may become hot and damaged. In that case, change the motor, insert the poly switch, and reassemble it.
 - ➔ Even if inserted, the circuit in the servo motor may have been shorted due to poor soldering or lack of insulation. In that case, change the engine.
- ◆ Please check if the voltage setting of the DCDC converter is correct. If the voltage is not adjusted, excessive voltage will be applied to the servo motor, which will damage the servo motor. (See page 92 for details.))

Sensor does not respond when pressed

- ◆ Check if the initial position of the servo motor is correct. (See page 134 for details.))
- ◆ Check that the sensor is connected correctly.
 - ➔ Is the audio cable securely connected from the rear?
 - ➔ If there is no response even after connecting successfully, the sensor or the audio cable itself may be damaged.
- ◆ Try adjusting the position where you want to mount the sensor.
 - ➔ Place him in a position where the muscles are easy to lift as much as possible.

Unable to write a program

- ◆ Make sure to use a USB cable with a communication function. (Communication cannot be done with a dedicated USB cable for charging.))
- ◆ Check if the COM port is recognized correctly. (See page 54.))
- ◆ Make sure you select the correct board. (See page 56.))
- ◆ Make sure the driver is installed correctly.
 - ➔ Drivers are basically installed automatically, but may not be available depending on the environment. Please refer to the following site for driver installation.
https://synapse.kyoto/tips/FTDI_driver/page001.html

Fingers get out of control

- ◆ Check that the sensor is connected correctly.
 - ➔ Is the audio cable properly connected from the back?
 - ➔ If there is no response even after connecting successfully, the sensor or the audio cable itself may be damaged.

HACKberry MANUAL

— Production Works —

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